

D.C. CIRCUITS

10 D.C. circuits

10.1 Practical circuits

Candidates should be able to:

- 1 recall and use the circuit symbols shown in section 6 of this syllabus
- 2 draw and interpret circuit diagrams containing the circuit symbols shown in section 6 of this syllabus
- 3 define and use the electromotive force (e.m.f.) of a source as energy transferred per unit charge in driving charge around a complete circuit
- 4 distinguish between e.m.f. and potential difference (p.d.) in terms of energy considerations
- 5 understand the effects of the internal resistance of a source of e.m.f. on the terminal potential difference

10.2 Kirchhoff's laws

Candidates should be able to:

- 1 recall Kirchhoff's first law and understand that it is a consequence of conservation of charge
- 2 recall Kirchhoff's second law and understand that it is a consequence of conservation of energy
- 3 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in series
- 4 use the formula for the combined resistance of two or more resistors in series
- 5 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in parallel
- 6 use the formula for the combined resistance of two or more resistors in parallel
- 7 use Kirchhoff's laws to solve simple circuit problems

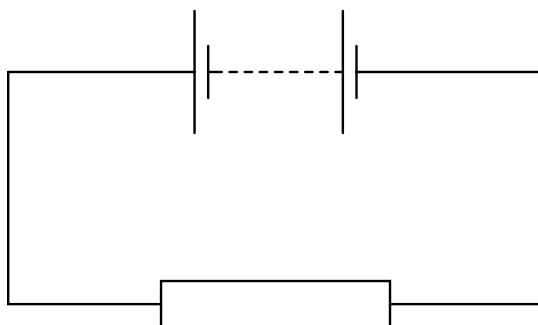
10.3 Potential dividers

Candidates should be able to:

- 1 understand the principle of a potential divider circuit
- 2 recall and use the principle of the potentiometer as a means of comparing potential differences
- 3 understand the use of a galvanometer in null methods
- 4 explain the use of thermistors and light-dependent resistors in potential dividers to provide a potential difference that is dependent on temperature and light intensity

1 9702/1/M/J/02/Q31*

In the circuit below, the battery converts an amount E of chemical energy to electrical energy when charge Q passes through the resistor in time t .



Which expressions give the e.m.f. of the battery and the current in the resistor?

	e.m.f.	current
A	EQ	Q/t
B	EQ	Qt
C	E/Q	Q/t
D	E/Q	Qt

2 9702/1/M/J/02/Q32

The filament of a 240 V, 100 W electric lamp heats up from room temperature to its operating temperature. As it heats up, its resistance increases by a factor of 16.

What is the resistance of this lamp at room temperature?

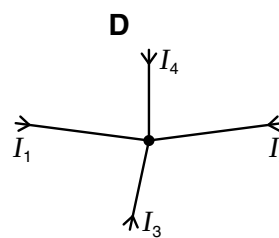
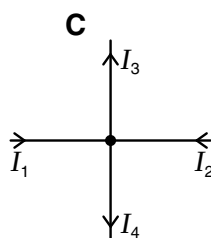
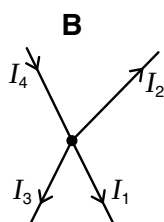
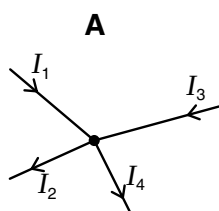
- A** 36 Ω **B** 580 Ω **C** 1.5 k Ω **D** 9.2 k Ω

3 9702/1/M/J/02/Q33

The diagrams show connected wires which carry currents I_1 , I_2 , I_3 and I_4 .

The currents are related by the equation $I_1 + I_2 = I_3 + I_4$.

To which diagram does this equation apply?



4 9702/1/M/J/02/Q34

When four identical lamps P, Q, R and S are connected as shown in diagram 1, they have normal brightness.

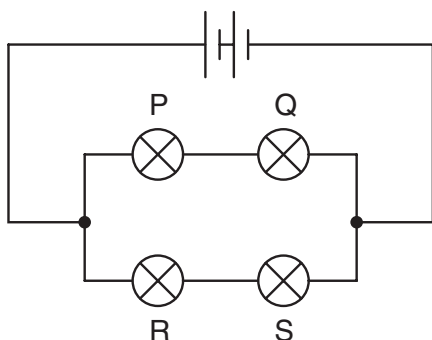


diagram 1

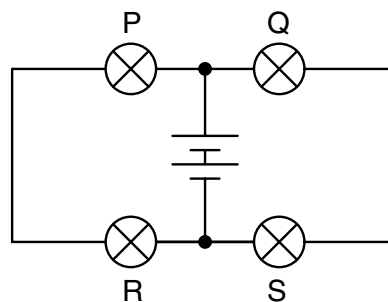


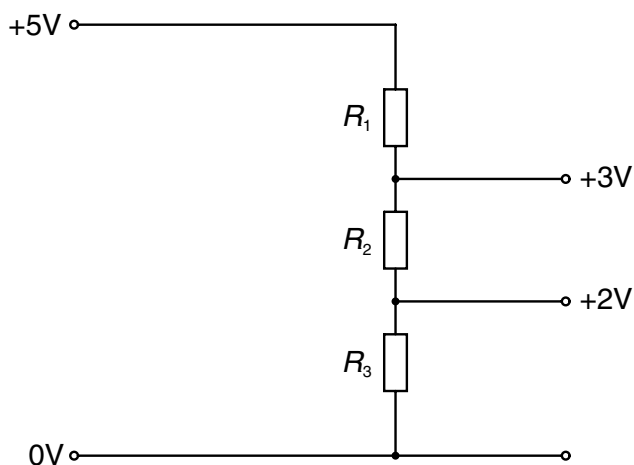
diagram 2

When the four lamps are connected as shown in diagram 2, which statement is correct?

- A The lamps do not light.
- B The lamps are less bright than normal.
- C The lamps have normal brightness.
- D The lamps are brighter than normal.

5 9702/1/M/J/02/Q35

A potential divider is used to give outputs of 2 V and 3 V from a 5 V source, as shown.

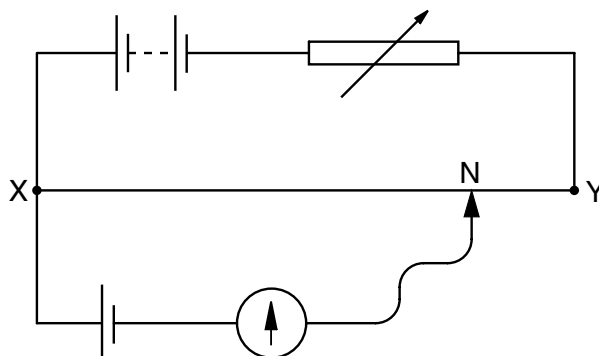


What are possible values for the resistances R_1 , R_2 and R_3 ?

	$R_1/\text{k}\Omega$	$R_2/\text{k}\Omega$	$R_3/\text{k}\Omega$
A	2	1	5
B	3	2	2
C	4	2	4
D	4	6	10

6 9702/1/O/N/02/Q35

In the potentiometer circuit below, the moveable contact is placed at N on the bare wire XY, such that the galvanometer shows zero deflection.



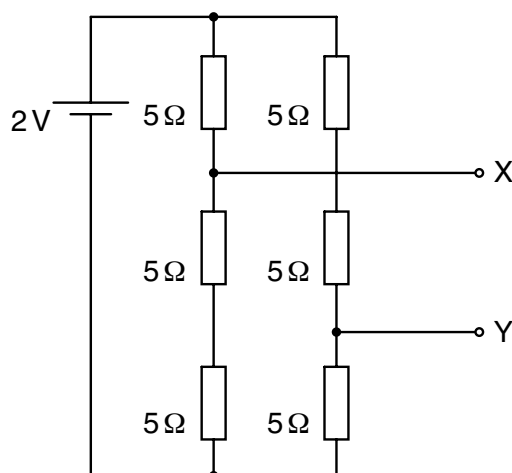
The resistance of the variable resistor is now increased.

What is the effect of this increase on the potential difference across the wire XY and on the position of the moveable contact for zero deflection?

	potential difference across XY	position of moveable contact
A	increases	nearer to X
B	increases	nearer to Y
C	decreases	nearer to X
D	decreases	nearer to Y

7 9702/1/O/N/02/Q36

Six resistors, each of resistance $5\ \Omega$, are connected to a $2\ \text{V}$ cell of negligible internal resistance.



What is the potential difference between terminals X and Y?

- A** $\frac{2}{3}\ \text{V}$ **B** $\frac{8}{9}\ \text{V}$ **C** $\frac{4}{3}\ \text{V}$ **D** $2\ \text{V}$

8 9702/01/M/J/03/Q31

The sum of the electrical currents into a point in a circuit is equal to the sum of the currents out of the point.

Which of the following is correct?

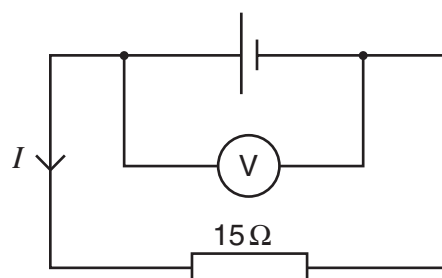
- A** This is Kirchhoff's first law, which results from the conservation of charge.
- B** This is Kirchhoff's first law, which results from the conservation of energy.
- C** This is Kirchhoff's second law, which results from the conservation of charge.
- D** This is Kirchhoff's second law, which results from the conservation of energy.

9 9702/01/M/J/03/Q32

The e.m.f. of the cell in the following circuit is 9.0 V. The reading on the high-resistance voltmeter is 7.5 V.

What is the current I ?

- A** 0.1 A
- B** 0.5 A
- C** 0.6 A
- D** 2.0 A

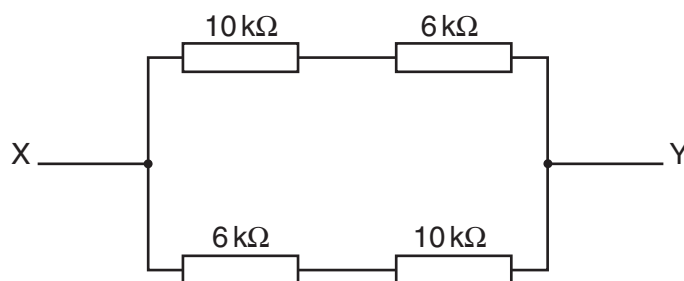


10 9702/01/M/J/03/Q33

The diagram shows an arrangement of four resistors.

What is the resistance between X and Y?

- A** 4 k Ω
- B** 8 k Ω
- C** 16 k Ω
- D** 32 k Ω

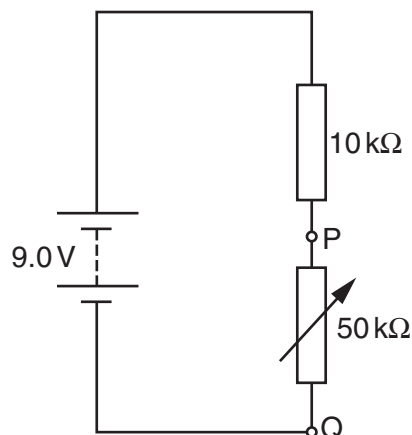


11 9702/01/M/J/03/Q34

The diagram shows a potential divider connected to a 9.0 V supply of negligible internal resistance.

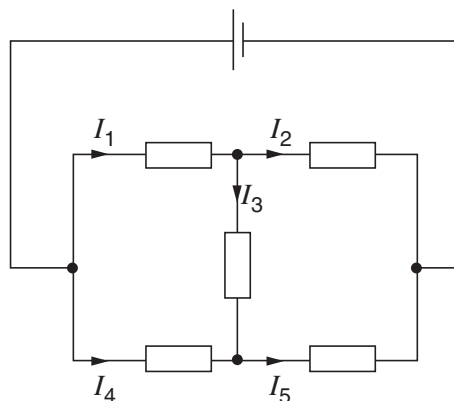
What range of voltages can be obtained between P and Q?

- A** zero to 1.5 V
- B** zero to 7.5 V
- C** 1.5 V to 7.5 V
- D** 1.5 V to 9.0 V



12 9702/11/O/N/16/Q37

The diagram shows currents I_1 , I_2 , I_3 , I_4 and I_5 in different branches of a circuit.

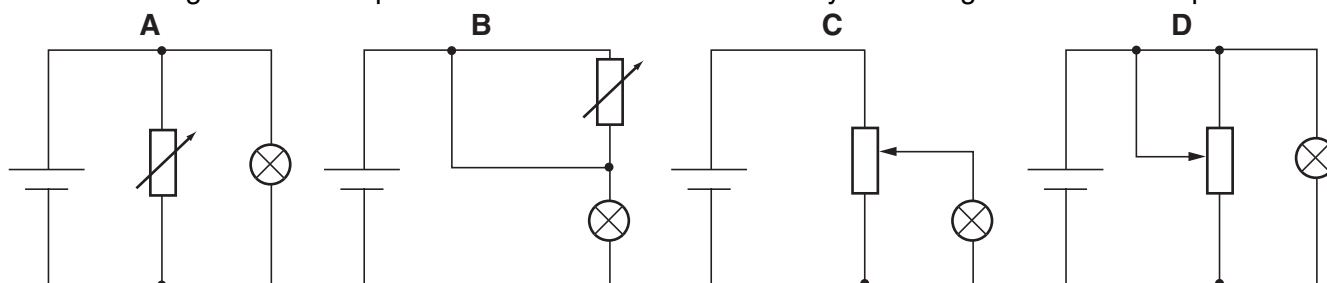


Which one of the following is correct?

- A $I_1 = I_2 + I_3$
- B $I_2 = I_1 + I_3$
- C $I_3 = I_4 + I_5$
- D $I_4 = I_5 + I_3$

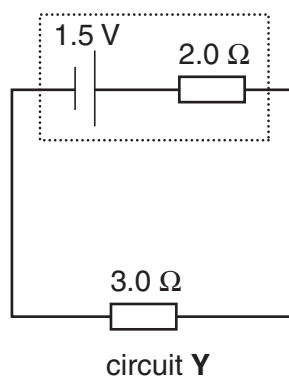
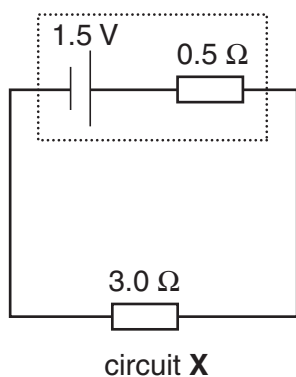
13 9702/01/O/N/03/Q33

Which diagram shows a potential divider circuit that can vary the voltage across the lamp?



14 9702/01/O/N/03/Q34

The diagram shows two circuits. In these circuits, only the internal resistances differ.



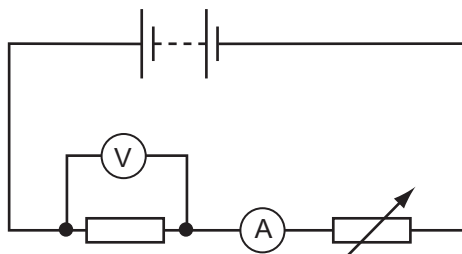
Which line in the table is correct?

	potential difference across $3.0\ \Omega$ resistor	power dissipated in $3.0\ \Omega$ resistor
A	greater in X than in Y	less in X than in Y
B	greater in X than in Y	greater in X than in Y
C	less in X than in Y	less in X than in Y
D	less in X than in Y	greater in X than in Y

15 9702/01/M/J/04/Q35*

The diagram shows a battery, a fixed resistor, an ammeter and a variable resistor connected in series.

A voltmeter is connected across the fixed resistor.



The value of the variable resistor is reduced.

Which correctly describes the changes in the readings of the ammeter and of the voltmeter?

	ammeter	voltmeter
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

16 9702/01/M/J/04/Q36

Kirchhoff's two laws for electric circuits can be derived by using conservation laws.

On which conservation laws do Kirchhoff's laws depend?

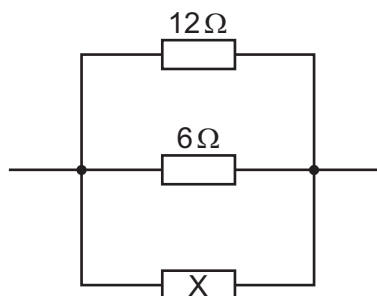
	Kirchhoff's first law	Kirchhoff's second law
A	charge	current
B	charge	energy
C	current	mass
D	energy	current

17 9702/01/M/J/04/Q37

The diagram shows a parallel combination of three resistors. The total resistance of the combination is 3Ω .

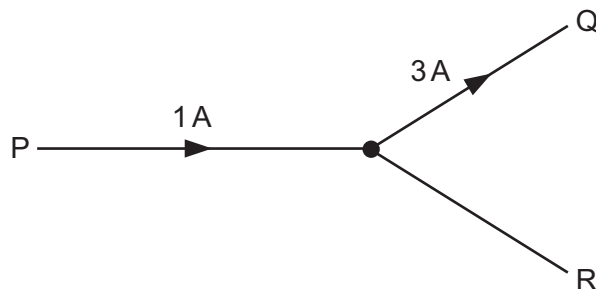
What is the resistance of resistor X?

- A** 2Ω
- B** 3Ω
- C** 6Ω
- D** 12Ω



18 9702/01/O/N/04/Q35

The diagram shows a junction in a circuit where three wires P, Q and R meet. The currents in P and Q are 1 A and 3 A respectively, in the directions shown.



How many coulombs of charge pass a given point in wire R in 5 seconds?

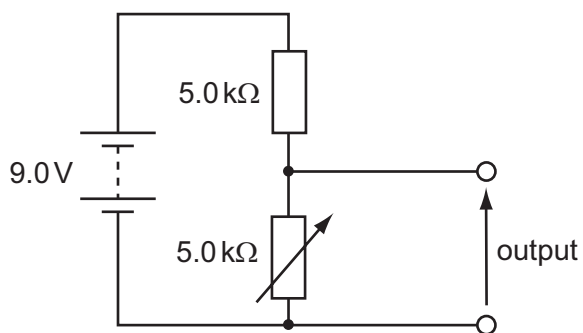
- A 0.4 B 0.8 C 2 D 10

19 9702/01/O/N/04/Q36

The diagram shows a potential divider circuit designed to provide a variable output p.d.

Which gives the available range of output p.d?

	maximum output	minimum output
A	3.0 V	0
B	4.5 V	0
C	9.0 V	0
D	9.0 V	4.5 V



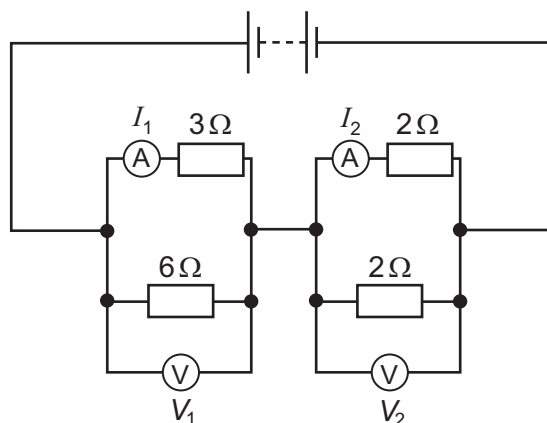
20 9702/01/O/N/04/Q37*

In the circuit shown, the ammeters have negligible resistance and the voltmeters have infinite resistance.

The readings on the meters are I_1 , I_2 , V_1 and V_2 , as labelled on the diagram.

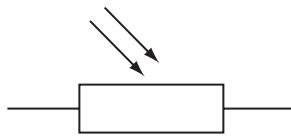
Which is correct?

- A $I_1 > I_2$ and $V_1 > V_2$
 C $I_1 < I_2$ and $V_1 > V_2$
 B $I_1 > I_2$ and $V_1 < V_2$
 D $I_1 < I_2$ and $V_1 < V_2$



21 9702/01/M/J/05/Q35

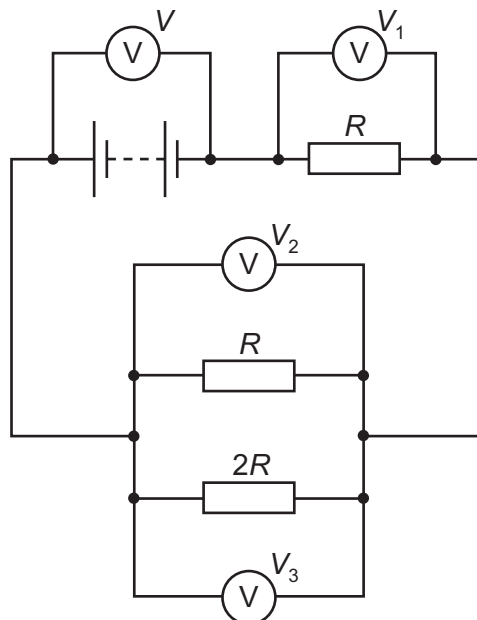
Which electrical component is represented by the following symbol?



- A a diode
- B a light-dependent resistor
- C a resistor
- D a thermistor

22 9702/01/M/J/05/Q36

The diagram shows a circuit with four voltmeter readings V , V_1 , V_2 and V_3 .

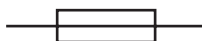


Which equation relating the voltmeter readings must be true?

- | | |
|-------------------------|-------------------|
| A $V = V_1 + V_2 + V_3$ | C $V_3 = 2(V_2)$ |
| B $V + V_1 = V_2 + V_3$ | D $V - V_1 = V_3$ |

23 9702/01/O/N/05/Q35

An electrical component has the following circuit symbol.

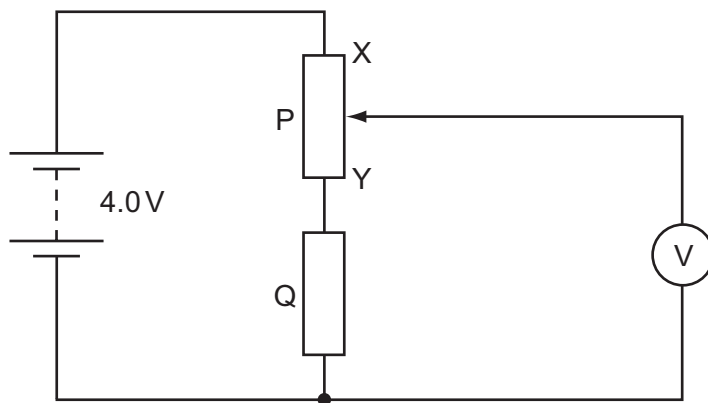


What does this symbol represent?

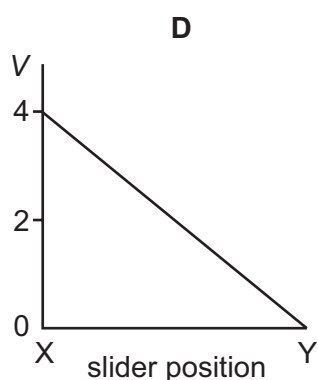
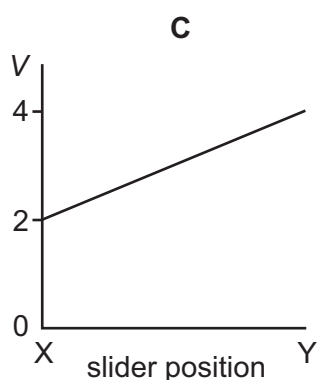
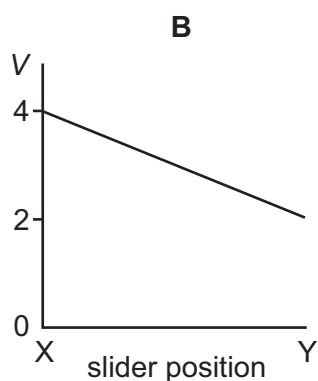
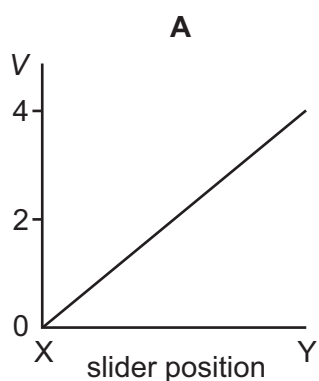
- | | |
|--------------------------------|--------------|
| A variable resistor (rheostat) | B fuse |
| C light-dependent resistor | D thermistor |

24 9702/13/O/N/12/Q36**

In the circuit below, P is a potentiometer of total resistance $10\ \Omega$ and Q is a fixed resistor of resistance $10\ \Omega$. The battery has an e.m.f. of $4.0\ \text{V}$ and negligible internal resistance. The voltmeter has a very high resistance. The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading V is plotted against slider position.



Which graph is obtained?

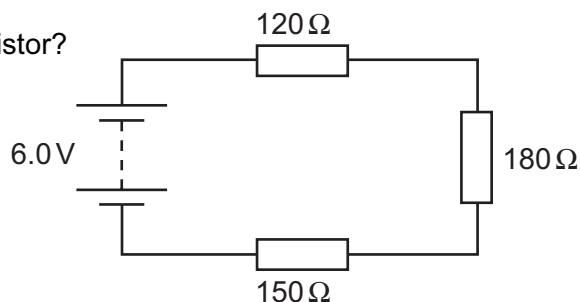


25 9702/01/O/N/05/Q36

Three resistors are connected in series with a battery as shown in the diagram. The battery has negligible internal resistance.

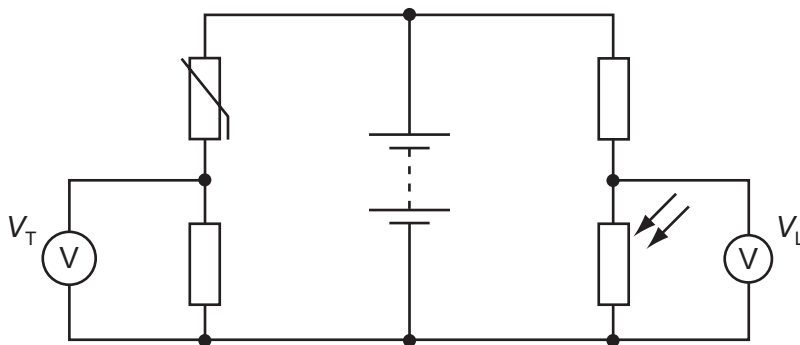
What is the potential difference across the $180\ \Omega$ resistor?

- A 1.6 V
- B 2.4 V
- C 3.6 V
- D 6.0 V



26 9702/12/M/J/13/Q38

In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



The readings on V_T and V_L are both high.

What are the conditions of temperature and light level?

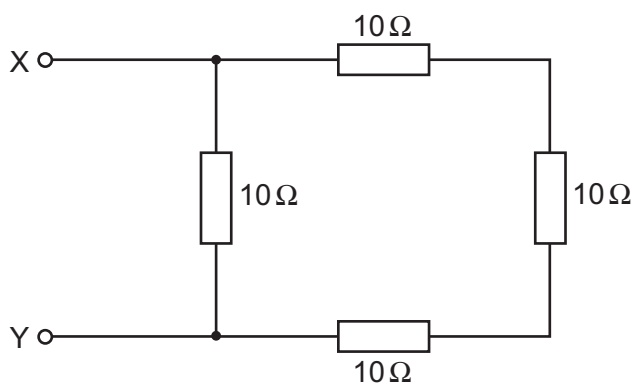
	temperature	light level
A	low	low
B	low	high
C	high	low
D	high	high

27 9702/12/O/N/15/Q36

The diagram shows an arrangement of resistors.

What is the total electrical resistance between X and Y?

- A less than $1\ \Omega$
- B between $1\ \Omega$ and $10\ \Omega$
- C between $10\ \Omega$ and $30\ \Omega$
- D $40\ \Omega$



28 9702/01/M/J/06/Q36

When four identical lamps P, Q, R and S are connected as shown in diagram 1, they have normal brightness.

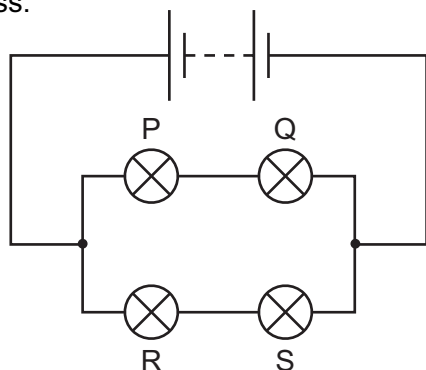


diagram 1

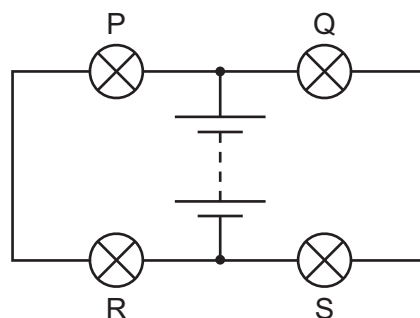


diagram 2

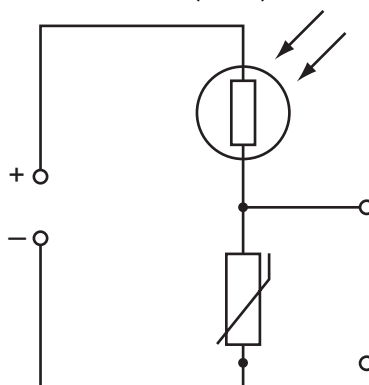
The four lamps and the battery are then connected as shown in diagram 2.

Which statement is correct?

- A** The lamps do not light. **C** The lamps have normal brightness.
B The lamps are less bright than normal. **D** The lamps are brighter than normal.

29 9702/01/M/J/06/Q37

The diagram shows a light-dependent resistor (LDR) and a thermistor forming a potential divider.



Under which set of conditions will the potential difference across the thermistor have the greatest value?

	illumination	temperature
A	low	low
B	high	low
C	low	high
D	high	high

30 9702/01/O/N/06/Q35

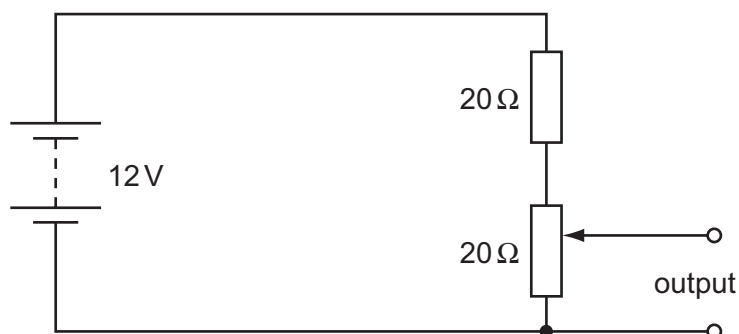
The resistance of a device is designed to change with temperature.

What is the device?

- A** a light-dependent resistor **C** a semiconductor diode
B a potential divider **D** a thermistor

31 9702/01/O/N/06/Q34

The diagram shows a potentiometer and a fixed resistor connected across a 12 V battery of negligible internal resistance.



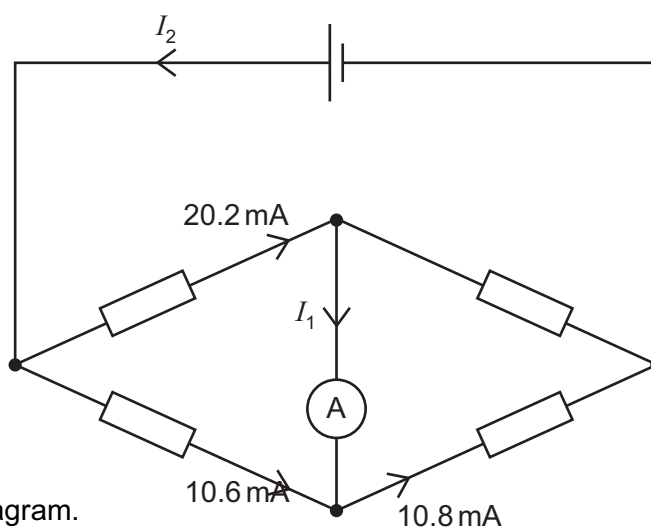
The fixed resistor and the potentiometer each have resistance 20 Ω. The circuit is designed to provide a variable output voltage.

What is the range of output voltages?

- A 0–6V B 0–12V
C 6–12V D 12–20V

32 9702/01/O/N/06/Q36

The diagram represents a circuit.



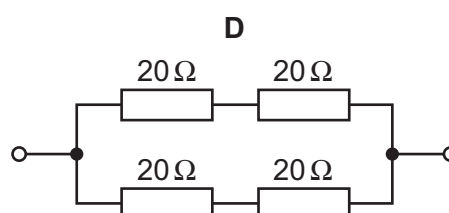
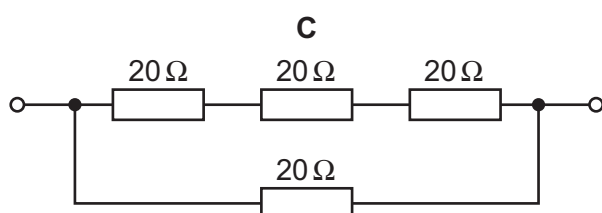
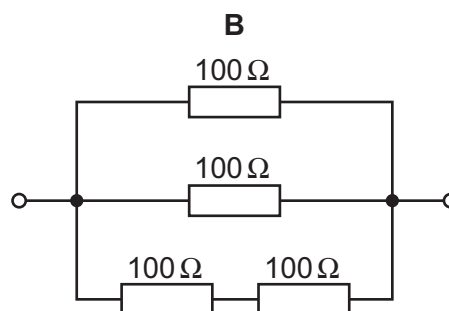
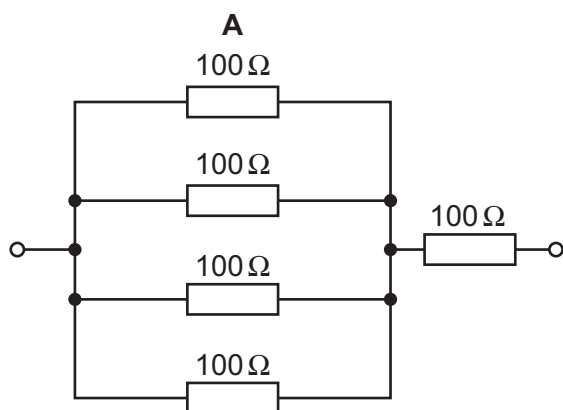
Some currents have been shown on the diagram.

What are the currents I_1 and I_2 ?

	I_1	I_2
A	0.2 mA	10.8 mA
B	0.2 mA	30.8 mA
C	–0.2 mA	20.0 mA
D	–0.2 mA	30.8 mA

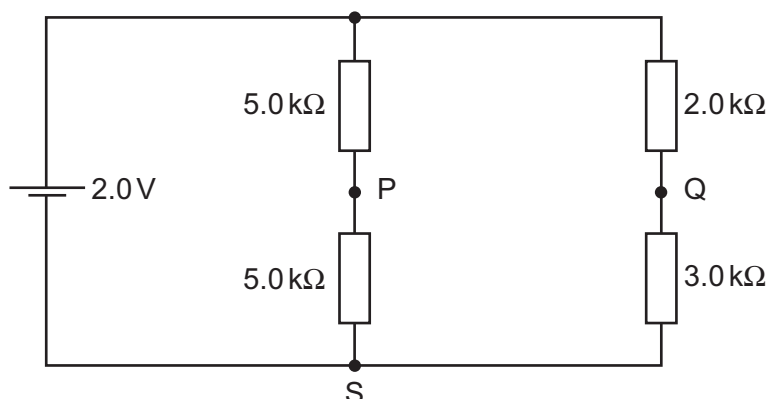
33 9702/01/O/N/06/Q37

Which circuit has a resistance of $40\ \Omega$ between the terminals?



34 9702/01/M/J/07/Q33

A cell of e.m.f. $2.0\ \text{V}$ and negligible internal resistance is connected to the network of resistors shown.



V_1 is the potential difference between S and P. V_2 is the potential difference between S and Q.

What is the value of $V_1 - V_2$?

- A** $+0.50\ \text{V}$ **B** $+0.20\ \text{V}$ **C** $-0.20\ \text{V}$ **D** $-0.50\ \text{V}$

35 9702/01/M/J/07/Q34

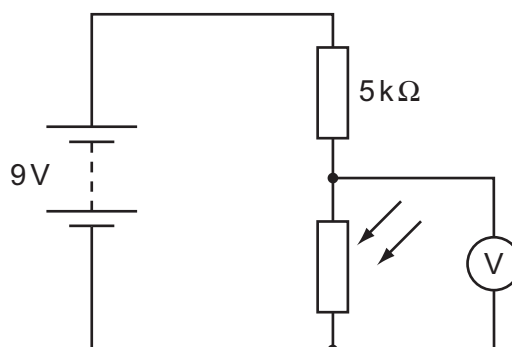
A circuit is set up with an LDR and a fixed resistor as shown.

The voltmeter reads $4\ \text{V}$.

The light intensity is increased.

What is a possible voltmeter reading?

- A** $3\ \text{V}$ **B** $4\ \text{V}$
C $6\ \text{V}$ **D** $8\ \text{V}$

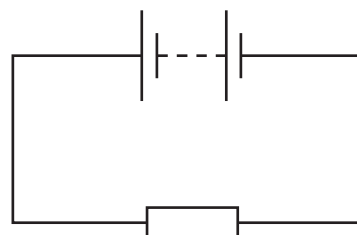


36 9702/01/M/J/07/Q35

In the circuit below, the battery converts an amount E of chemical energy to electrical energy when charge Q passes through the resistor in time t .

Which expressions give the e.m.f. of the battery and the current in the resistor?

	e.m.f.	current
A	EQ	Q/t
B	EQ	Qt
C	E/Q	Q/t
D	E/Q	Qt



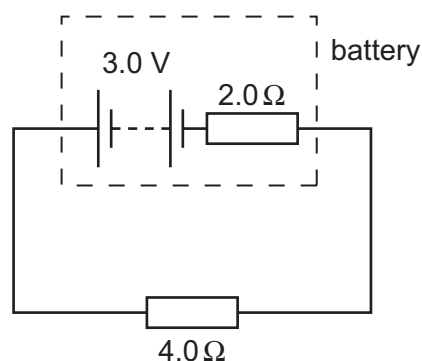
37 9702/01/M/J/07/Q36

A battery has an e.m.f. of 3.0 V and an internal resistance of $2.0\ \Omega$.

The battery is connected to a load of $4.0\ \Omega$.

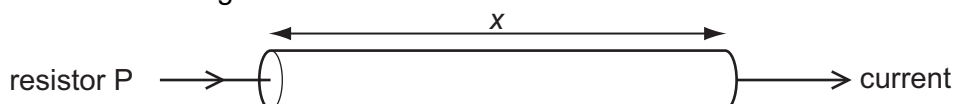
What are the terminal potential difference V and output power P

	V/V	P/W
A	1.0	0.50
B	1.0	1.5
C	2.0	1.0
D	2.0	1.5

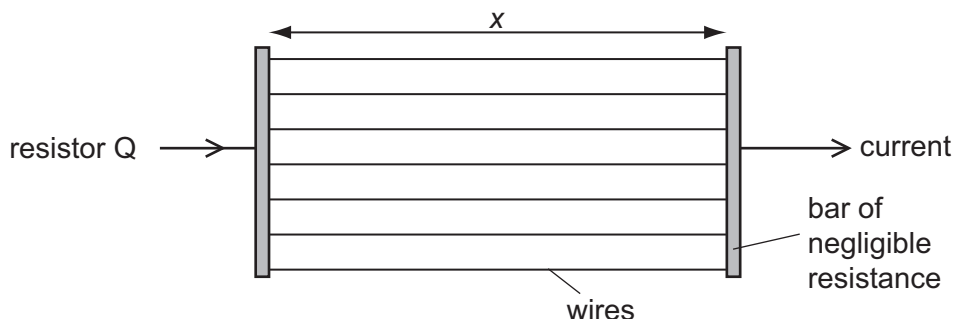


38 9702/01/M/J/07/Q37

A researcher has two pieces of copper of the same volume. All of the first piece is made into a cylindrical resistor P of length x .



All of the second piece is made into uniform wires each of the same length x which he connects between two bars of negligible resistance to form a resistor Q .

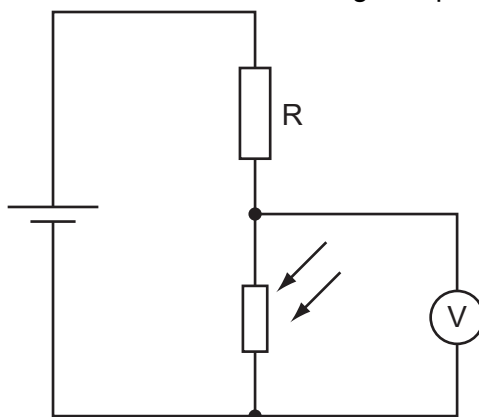


How do the electrical resistances of P and Q compare?

- A** P has a larger resistance than Q .
- B** Q has a larger resistance than P .
- C** P and Q have equal resistance.
- D** Q may have a larger or smaller resistance than P , depending on the number of wires made.

39 9702/01/O/N/07/Q32

A potential divider consists of a fixed resistor R and a light-dependent resistor (LDR).



What happens to the voltmeter reading, and why does it happen, when the intensity of light on the LDR increases?

- A The voltmeter reading decreases because the LDR resistance decreases.
- B The voltmeter reading decreases because the LDR resistance increases.
- C The voltmeter reading increases because the LDR resistance decreases.
- D The voltmeter reading increases because the LDR resistance increases.

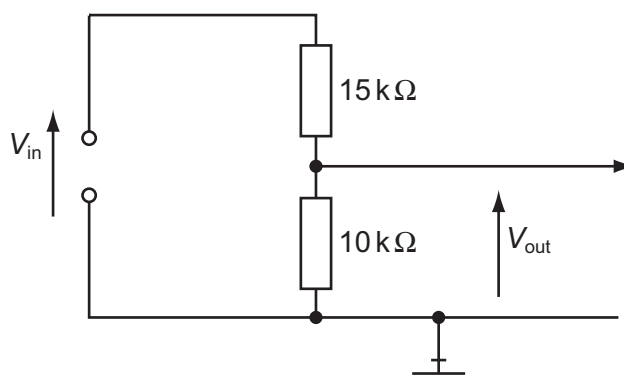
40 9702/01/O/N/07/Q33

The circuit is designed to trigger an alarm system when the input voltage exceeds some preset value. It does this by comparing V_{out} with a fixed reference voltage, which is set at 4.8 V.

V_{out} is equal to 4.8 V.

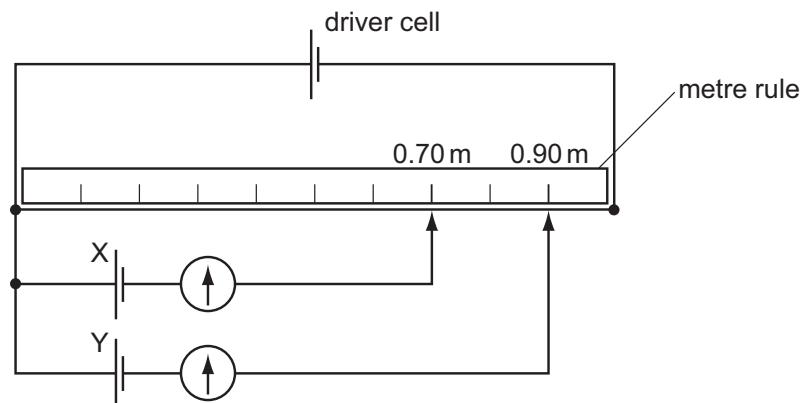
What is the input voltage V_{in} ?

- A 4.8 V
- B 7.2 V
- C 9.6 V
- D 12 V



41 9702/01/O/N/07/Q34

A potentiometer is used as shown to compare the e.m.f.s of two cells.



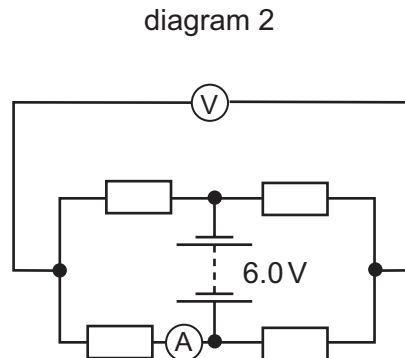
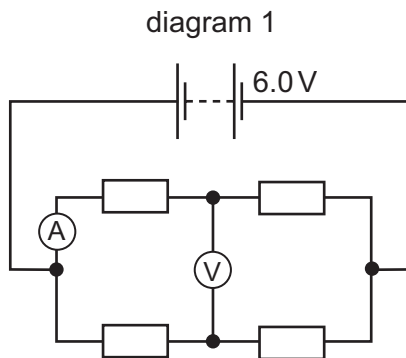
The balance points for cells X and Y are 0.70 m and 0.90 m respectively.

If the e.m.f. of cell X is 1.1 V, what is the e.m.f. of cell Y?

- A 0.69 V
- B 0.86 V
- C 0.99 V
- D 1.4 V

42 9702/01/O/N/07/Q35

When four identical resistors are connected as shown in diagram 1, the ammeter reads 1.0 A and the voltmeter reads zero.



The resistors and meters are reconnected to the supply as shown in diagram 2.

What are the meter readings in diagram 2?

	voltmeter reading / V	ammeter reading / A
A	0	1.0
B	3.0	0.5
C	3.0	1.0
D	6.0	0

43 9702/01/O/N/07/Q36

How is it possible to distinguish between the isotopes of uranium?

- A** Their nuclei have different charge and different mass, and they emit different particles when they decay.
- B** Their nuclei have different charge but the same mass.
- C** Their nuclei have the same charge but different mass.
- D** Their nuclei have the same charge and mass, but they emit different particles when they decay.

44 9702/01/O/N/07/Q37

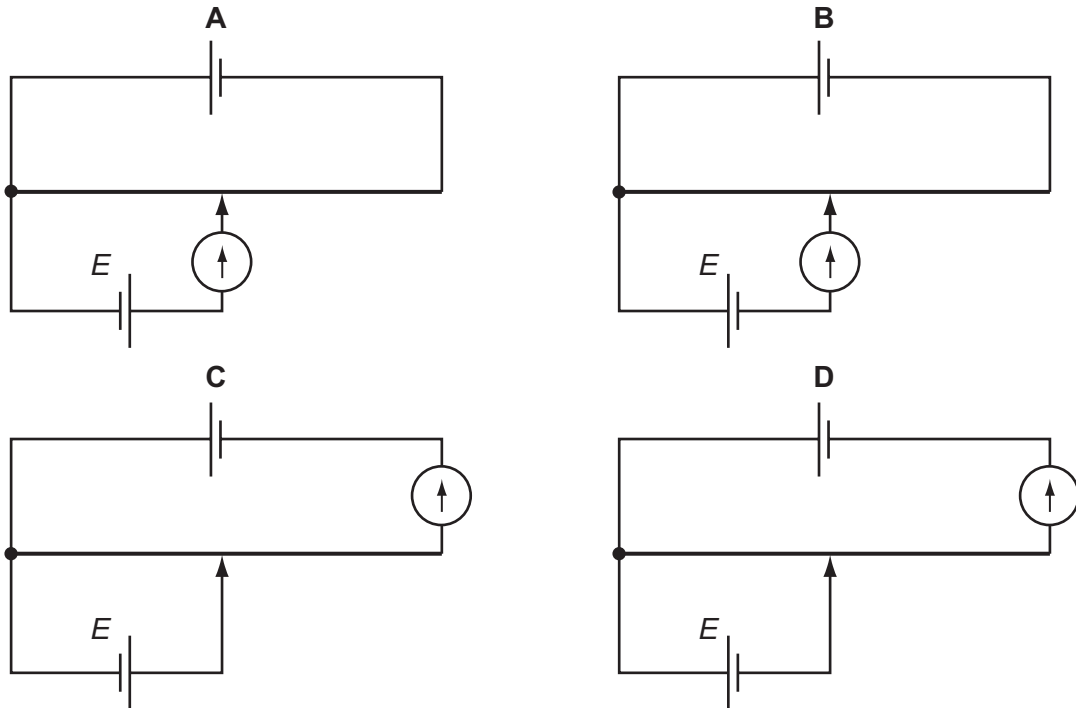
What is **not** conserved in nuclear processes?

- A** energy and mass together
- B** nucleon number
- C** neutron number
- D** charge

45 9702/01/M/J/08/Q38

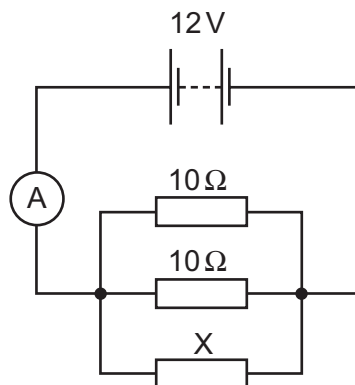
The unknown e.m.f. E of a cell is to be determined using a potentiometer circuit. The balance length is to be measured when the galvanometer records a null reading.

What is the correct circuit to use?



46 9702/01/O/N/08/Q35

The diagram shows a circuit containing three resistors in parallel.



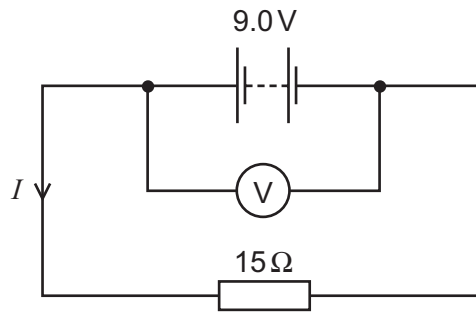
The battery has e.m.f. 12V and negligible internal resistance. The ammeter reading is 3.2A.

What is the resistance of X?

- A** 2.1Ω **B** 4.6Ω **C** 6.0Ω **D** 15Ω

47 9702/01/O/N/08/Q36

The e.m.f. of the battery is 9.0 V. The reading on the high-resistance voltmeter is 7.5 V.

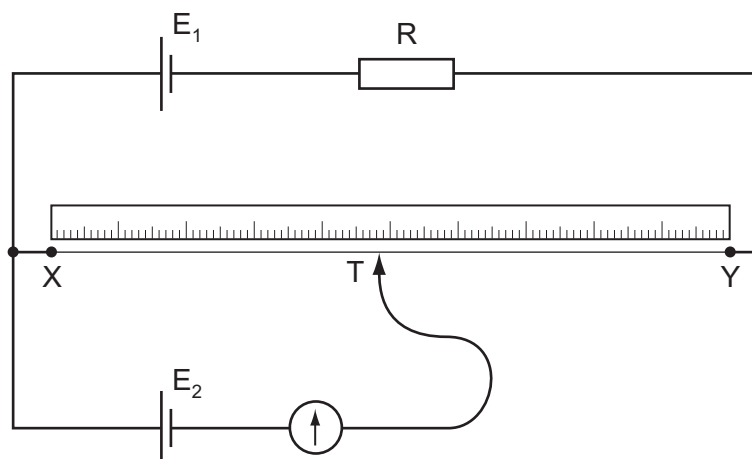


What is the current I ?

- A** 0.10 A **B** 0.50 A **C** 0.60 A **D** 2.0 A

48 9702/01/O/N/08/Q37

The diagram shows a potentiometer circuit.



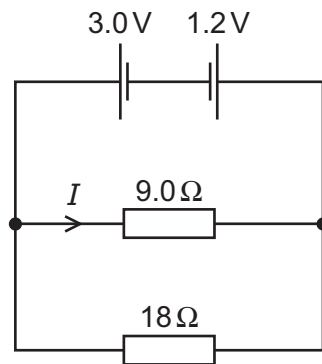
The contact T is placed on the wire and moved along the wire until the galvanometer reading is zero. The length XT is then noted.

In order to calculate the potential difference per unit length on the wire XY , which value must also be known?

- A** the e.m.f. of the cell E_1
B the e.m.f. of the cell E_2
C the resistance of resistor R
D the resistance of the wire XY

49 9702/01/M/J/09/Q33

Two cells of e.m.f. 3.0 V and 1.2 V and negligible internal resistance are connected to resistors of resistance 9.0Ω and 18Ω as shown.

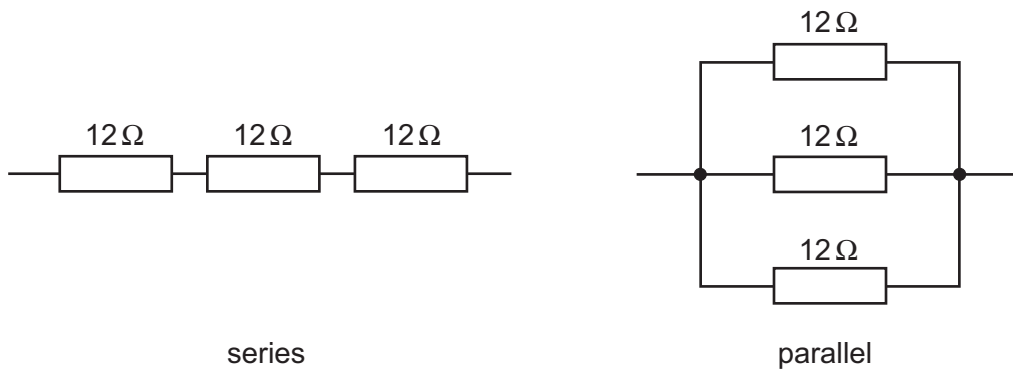


What is the value of the current I in the 9.0Ω resistor?

- A 0.10 A B 0.20 A C 0.30 A D 0.47 A

50 9702/01/M/J/09/Q34

Six identical 12Ω resistors are arranged in two groups, one with three in series and the other with three in parallel.

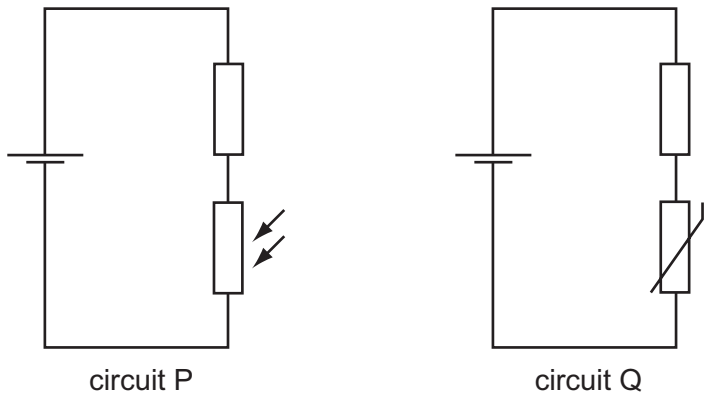


What are the combined resistances of each of these two arrangements?

	series	parallel
A	4.0Ω	0.25Ω
B	4.0Ω	36Ω
C	36Ω	0.25Ω
D	36Ω	4.0Ω

51 9702/01/M/J/09/Q35

The diagrams show a light-dependent resistor in circuit P, and a thermistor in circuit Q.



How does the potential difference across the fixed resistor in each circuit change when both the brightness of the light on the light-dependent resistor and the temperature of the thermistor are increased?

	circuit P	circuit Q
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

52 9702/12/O/N/09/Q32

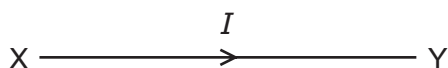
Each of Kirchhoff's two laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	the algebraic sum of currents into a junction is zero	charge
B	the algebraic sum of currents into a junction is zero	energy
C	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	charge
D	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	energy

53 9702/12/O/N/09/Q33

The diagram shows the symbol for a wire carrying a current I .

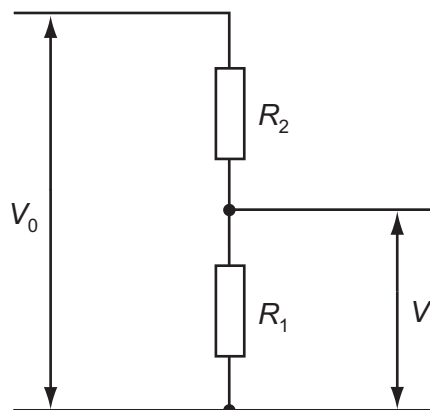


What does this current represent?

- A the amount of charge flowing past a point in XY per second
- B the number of electrons flowing past a point in XY per second
- C the number of positive ions flowing past a point in XY per second
- D the number of protons flowing past a point in XY per second

54 9702/12/O/N/09/Q34

A potential divider consisting of resistors of resistance R_1 and R_2 is connected to an input potential difference of V_0 and gives an output p.d. of V .

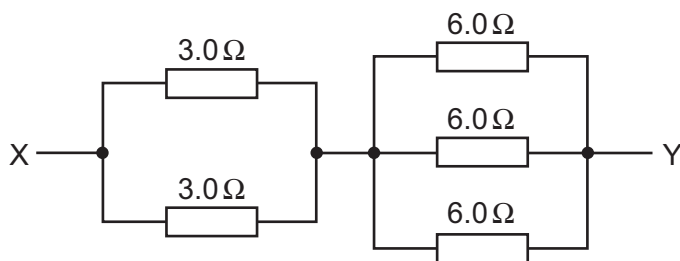


What is the value of V ?

- A $\frac{V_0 R_1}{R_2}$
- B $\frac{V_0 R_1}{R_1 + R_2}$
- C $\frac{V_0 R_2}{R_1 + R_2}$
- D $\frac{V_0 (R_1 + R_2)}{R_1}$

55 9702/12/O/N/09/Q35

A network of resistors consists of two 3.0Ω resistors and three 6.0Ω resistors.



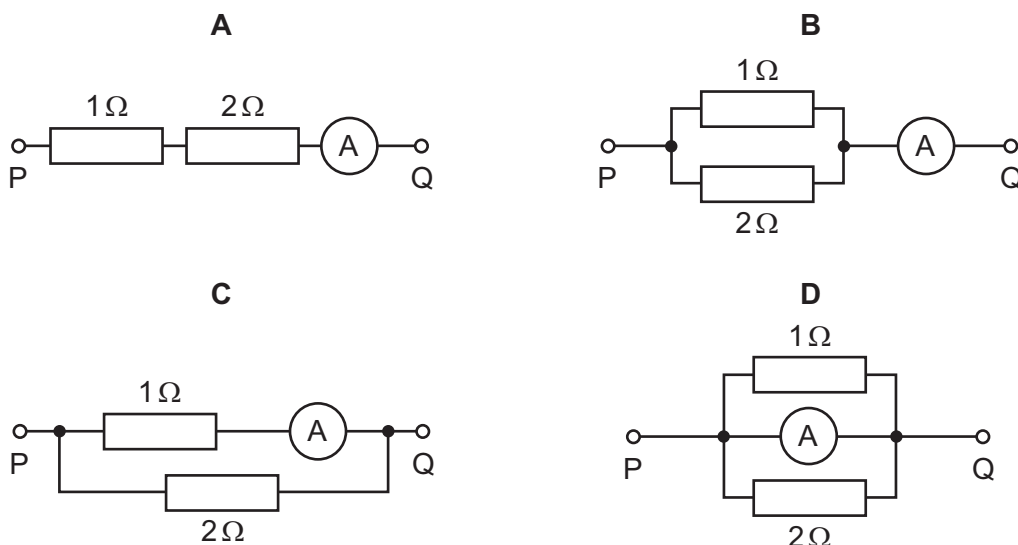
What is the combined resistance of this network between points X and Y?

- A 0.86Ω
- B 1.2Ω
- C 3.5Ω
- D 24Ω

56 9702/12/M/J/10/Q32*

In each arrangement of resistors, the ammeter has a resistance of 2Ω .

Which arrangement gives the largest reading on the ammeter when the same potential difference is applied between points P and Q?



57 9702/12/M/J/10/Q33

A source of e.m.f. of 9.0 mV has an internal resistance of 6.0Ω .

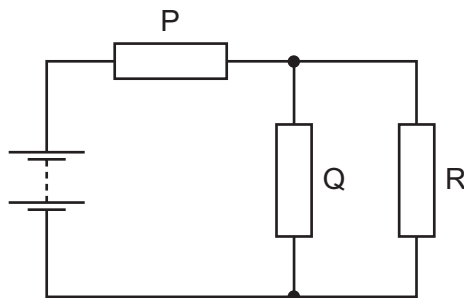
It is connected across a galvanometer of resistance 30Ω .

What will be the current in the galvanometer?

- A $250\mu\text{A}$ B $300\mu\text{A}$ C 1.5 mA D 2.5 mA

58 9702/12/M/J/10/Q34

The resistors P, Q and R in the circuit have equal resistance.



The battery, of negligible internal resistance, supplies a total power of 12 W .

What is the power dissipated by heating in resistor R?

- A 2 W B 3 W C 4 W D 6 W

59 9702/12/M/J/10/Q35

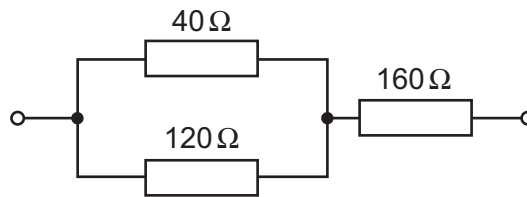
In deriving a formula for the combined resistance of three different resistors in series, Kirchhoff's laws are used.

Which physics principle is involved in this derivation?

- A the conservation of charge
 B the direction of the flow of charge is from negative to positive
 C the potential difference across each resistor is the same
 D the current varies in each resistor, in proportion to the resistor value

60 9702/11/O/N/10/Q35

The diagram shows part of a circuit.

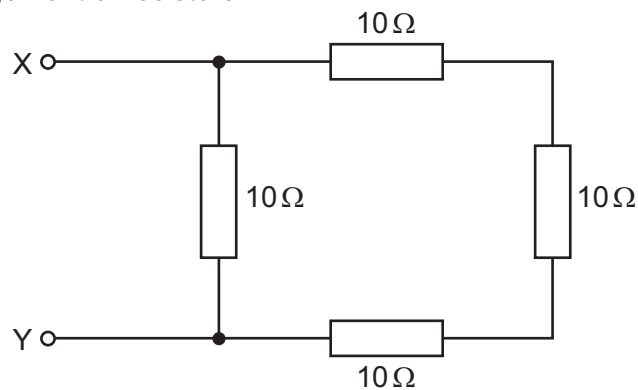


What is the total resistance of the combination of the three resistors?

- A** 320 Ω **B** 240 Ω **C** 190 Ω **D** 80 Ω

61 9702/11/O/N/10/Q36

The diagram shows an arrangement of resistors.

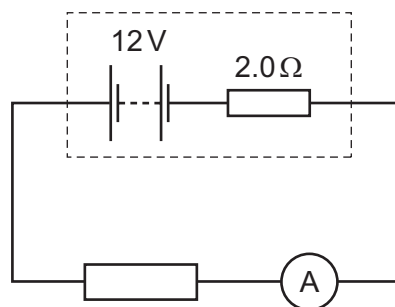


What is the total electrical resistance between X and Y?

- A** less than 1 Ω **C** between 10 Ω and 30 Ω
B between 1 Ω and 10 Ω **D** 40 Ω

62 9702/12/O/N/10/Q33

A battery of e.m.f. 12 V and internal resistance 2.0 Ω is connected in series with an ammeter of negligible resistance and an external resistor. External resistors of various different values are used.

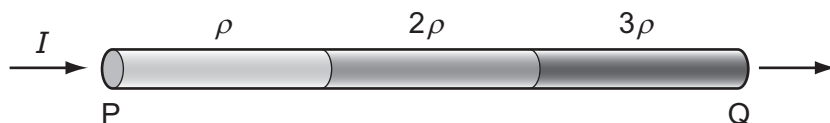


Which combination of current and resistor value is **not** correct?

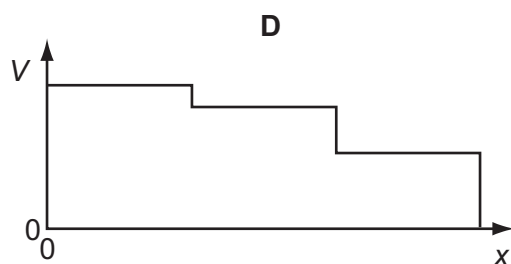
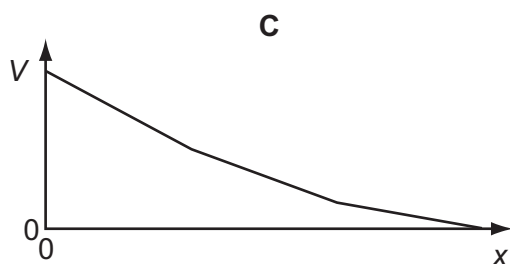
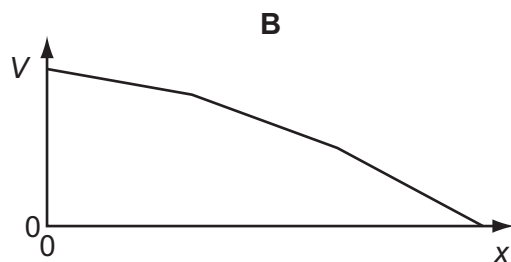
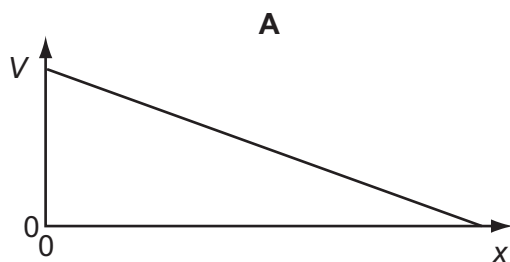
	current / A	external resistor value / Ω
A	1.0	10
B	1.2	8
C	1.5	6
D	1.8	4

63 9702/12/O/N/10/Q34

A wire PQ is made of three different materials, with resistivities ρ , 2ρ and 3ρ . There is a current I in this composite wire, as shown.

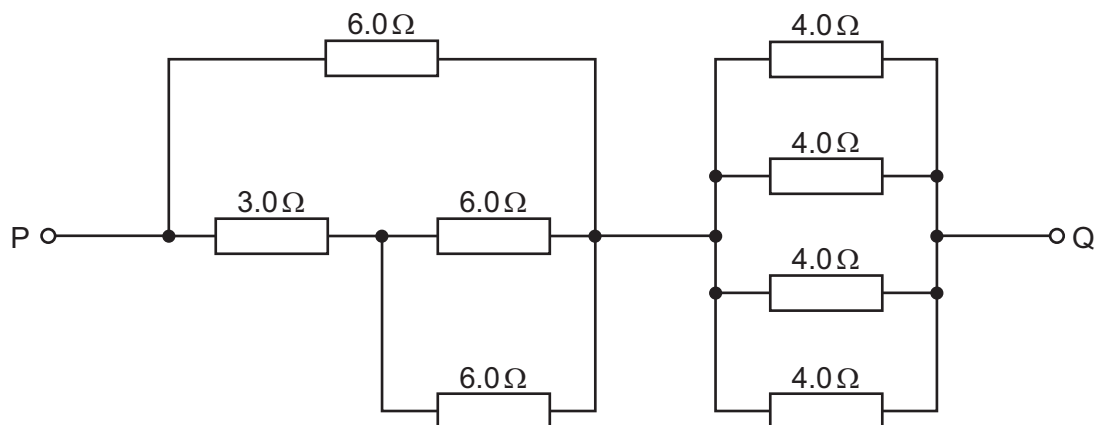


Which graph best shows how the potential V along the wire varies with distance x from P?



64 9702/12/O/N/10/Q35

The diagram shows part of a circuit.

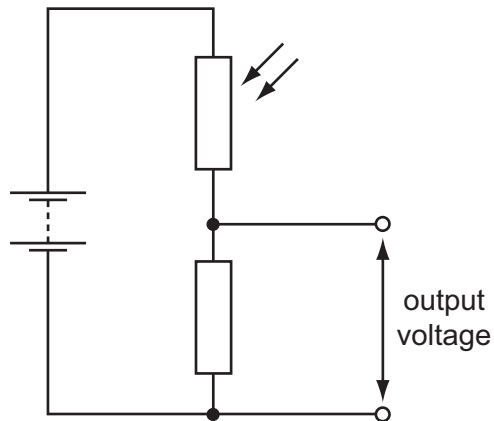


What is the resistance between the points P and Q due to the resistance network?

- A** $1.3\ \Omega$ **B** $4.0\ \Omega$ **C** $10\ \Omega$ **D** $37\ \Omega$

65 9702/12/O/N/10/Q36

The diagram shows a potential divider circuit.



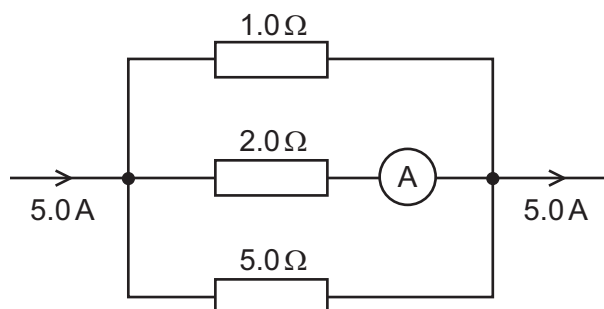
The light level increases.

What is the effect on the resistance of the light-dependent resistor (LDR) and on the output voltage?

	resistance of the LDR	output voltage
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

66 9702/11/M/J/11/Q35

The diagram shows part of a current-carrying circuit. The ammeter has negligible internal resistance.

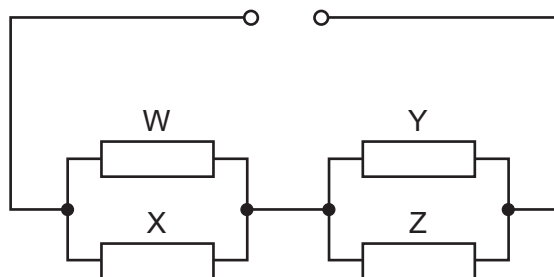


What is the reading on the ammeter?

- A** 0.7 A **B** 1.3 A **C** 1.5 A **D** 1.7 A

67 9702/11/M/J/11/Q36

Four resistors of equal value are connected as shown.

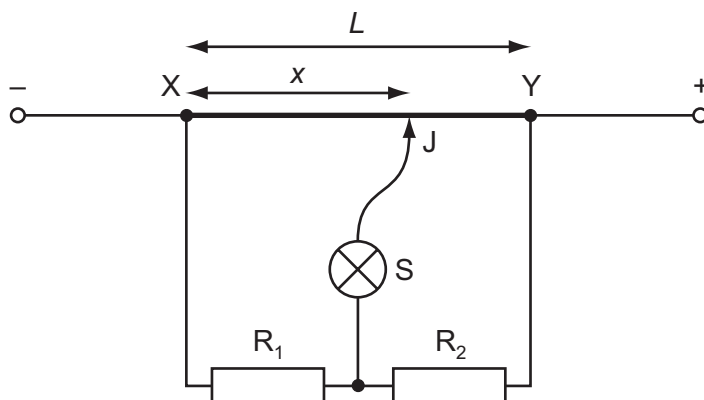


How will the powers to the resistors change when resistor W is removed?

- A The powers to X, Y and Z will all increase.
- B The power to X will decrease and the powers to Y and Z will increase.
- C The power to X will increase and the powers to Y and Z will decrease.
- D The power to X will increase and the powers to Y and Z will remain unaltered.

68 9702/11/M/J/11/Q37

In the circuit shown, XY is a length L of uniform resistance wire. R_1 and R_2 are unknown resistors. J is a sliding contact that joins the junction of R_1 and R_2 to points on XY through a small signal lamp S.



To determine the ratio $\frac{V_1}{V_2}$ of the potential differences across R_1 and R_2 , a point is found on XY at which the lamp is off. This point is at a distance x from X.

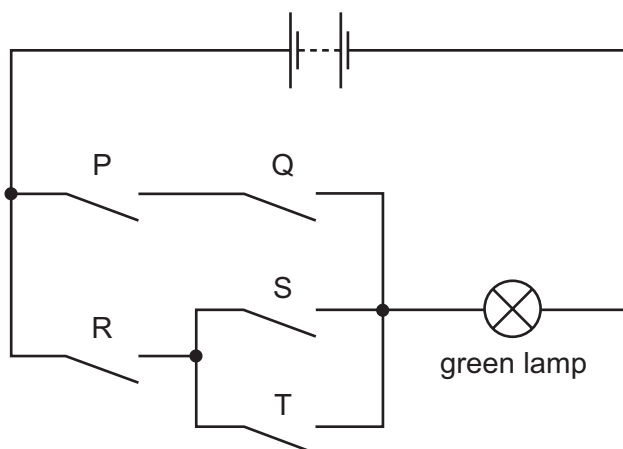
What is the value of the ratio $\frac{V_1}{V_2}$?

- A $\frac{L}{x}$
- B $\frac{x}{L}$
- C $\frac{L-x}{x}$
- D $\frac{x}{L-x}$

69 9702/12/M/J/11/Q36

Safety on railways is increased by using several electrical switches.

In the diagram, switches P, Q, R, S and T control the current through a green lamp.

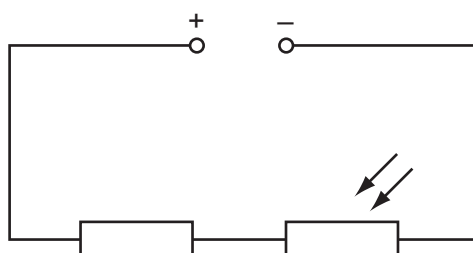


Which row does **not** allow the green lamp to light?

	P	Q	R	S	T
A	closed	closed	closed	open	closed
B	closed	open	closed	closed	open
C	closed	open	open	closed	closed
D	open	open	closed	open	closed

70 9702/12/M/J/11/Q37

The diagram shows a fixed resistor and a light-dependent resistor (LDR) in series with a constant low-voltage supply.



When the LDR is in the dark, the fixed resistor and the LDR have the same value of resistance.

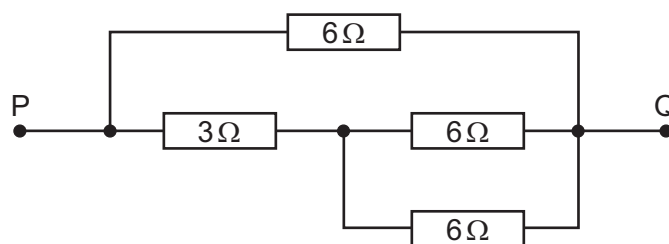
Light is shone on the LDR.

What happens to the potential differences across the two components?

	p.d. across resistor	p.d. across LDR
A	decreased	increased
B	increased	decreased
C	no change	increased
D	no change	decreased

71 9702/12/M/J/11/Q38

The diagram shows a d.c. circuit.



What is the resistance between the points P and Q due to the resistance network?

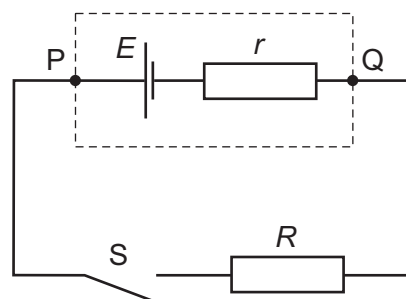
- A 0.47Ω B 2.1Ω C 3.0Ω D 21Ω

72 9702/11/O/N/11/Q36

A cell of e.m.f. E and internal resistance r is connected in series with a switch S and an external resistor of resistance R .

The p.d. between P and Q is V .

When S is closed,

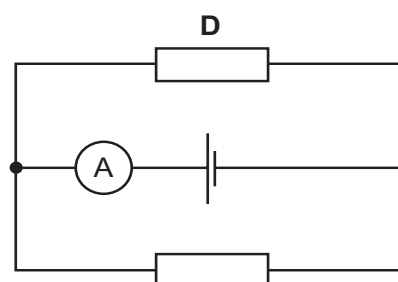
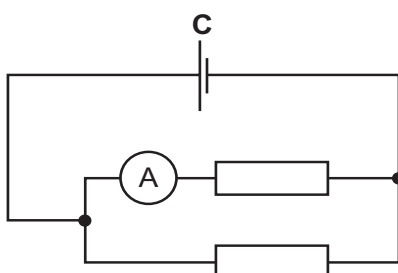
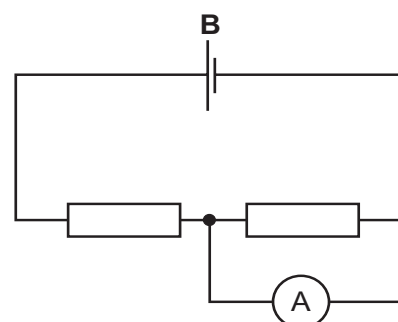
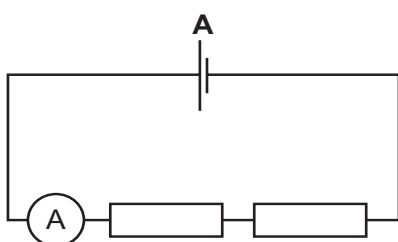


- A V decreases because there is a p.d. across R .
 B V decreases because there is a p.d. across r .
 C V remains the same because the decrease of p.d. across r is balanced by the increase of p.d. across R .
 D V remains the same because the sum of the p.d.s across r and R is still equal to E .

73 9702/11/O/N/11/Q37

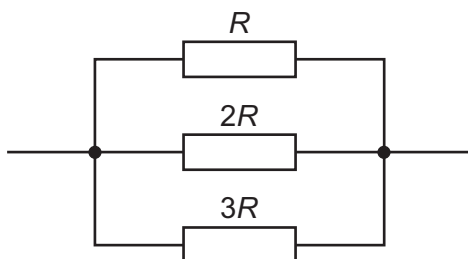
A cell, two resistors of equal resistance and an ammeter are used to construct four circuits. The resistors are the only parts of the circuits that have resistance.

In which circuit will the ammeter show the greatest reading?



74 9702/11/O/N/11/Q38

Three resistors of resistance R , $2R$ and $3R$ are connected in parallel.

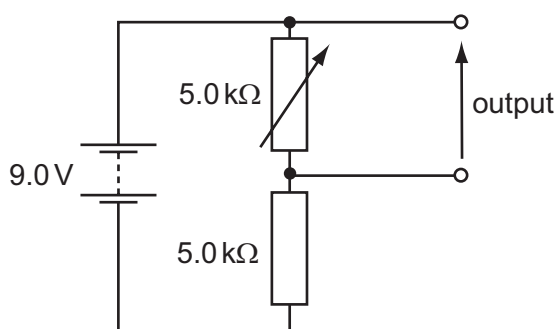


Using I to represent the current through the resistor of resistance R , which row represents the relationships between the currents through the resistors?

	resistor resistance		
	R	$2R$	$3R$
A	I	$\frac{1}{3} I$	$\frac{1}{2} I$
B	I	$\frac{1}{2} I$	$\frac{1}{3} I$
C	I	$\frac{2}{3} I$	$\frac{1}{3} I$
D	I	$2I$	$3I$

75 9702/11/O/N/11/Q39

The diagram shows a potential divider circuit designed to provide a variable output p.d.



Which row gives the available range of output p.d.?

	maximum output	minimum output
A	3.0 V	0
B	4.5 V	0
C	9.0 V	0
D	9.0 V	4.5 V

76 9702/12/O/N/11/Q36

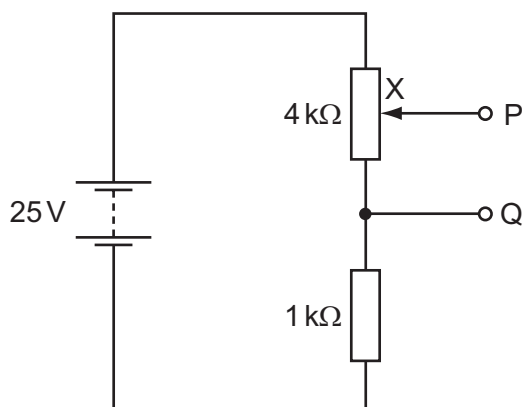
Each of Kirchhoff's laws is linked to the conservation of a physical quantity.

Which physical quantities are assumed to be conserved in the formulation of Kirchhoff's first law and of Kirchhoff's second law?

	Kirchhoff's first law	Kirchhoff's second law
A	energy	charge
B	energy	momentum
C	charge	energy
D	momentum	energy

77 9702/12/O/N/11/Q37

The diagram shows a potential divider circuit which, by adjustment of the contact X, can be used to provide a variable potential difference between the terminals P and Q.

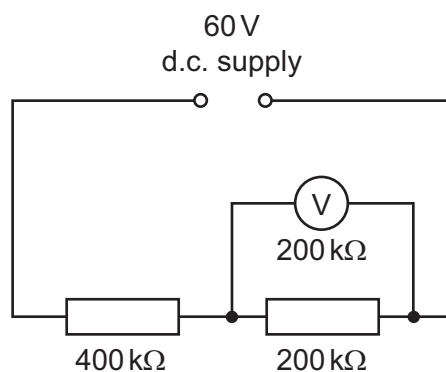


What are the limits of this potential difference?

- A** 0 and 5V **B** 0 and 20V **C** 0 and 25V **D** 5V and 25V

78 9702/12/O/N/11/Q38

A constant 60V d.c. supply is connected across two resistors of resistance 400 kΩ and 200 kΩ.

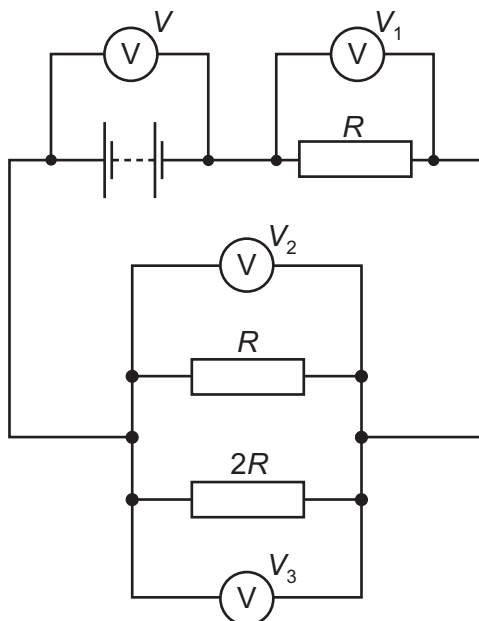


What is the reading on a voltmeter, also of resistance 200 kΩ, when connected across the 200 kΩ resistor as shown in the diagram?

- A** 12V **B** 15V **C** 20V **D** 30V

79 9702/01/M/J/05/Q36

The diagram shows a circuit with four voltmeter readings V , V_1 , V_2 and V_3 .

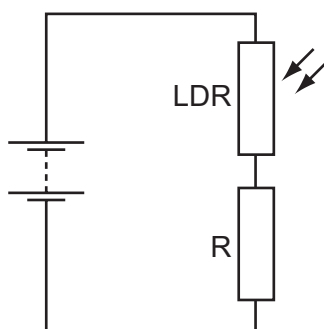


Which equation relating the voltmeter readings must be true?

- | | |
|--------------------------------|--------------------------|
| A $V = V_1 + V_2 + V_3$ | C $V_3 = 2(V_2)$ |
| B $V + V_1 = V_2 + V_3$ | D $V - V_1 = V_3$ |

80 9702/12/M/J/12/Q36

A light-dependent resistor (LDR) is connected in series with a resistor R and a battery.



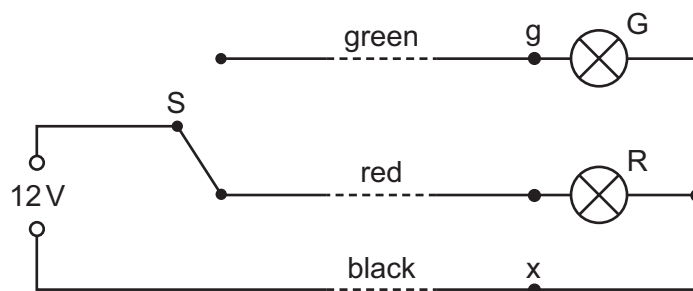
The resistance of the LDR is equal to the resistance of R when no light falls on the LDR.

When the light intensity falling on the LDR increases, which statement is correct?

- | | |
|--|---|
| A The current in R decreases. | C The p.d. across R decreases. |
| B The current in the LDR decreases. | D The p.d. across the LDR decreases. |

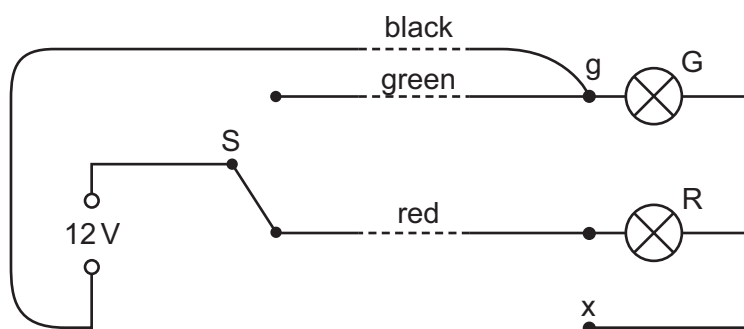
81 9702/12/M/J/12/Q37

The diagram shows the circuit for a signal to display a green or a red light. It is controlled by the switch S.



The signal is some way from S to which it is connected by a cable with green, red and black wires. At the signal, the green and red wires are connected to the corresponding lamp and the black wire is connected to a terminal x to provide a common return. The arrangement is shown correctly connected and with the switch set to illuminate the red lamp.

During maintenance, the wires at the signal are disconnected and, when reconnected, the black wire is connected in error to the green lamp (terminal g) instead of terminal x. The red wire is connected correctly to its lamp and connections at S remain as in the diagram.

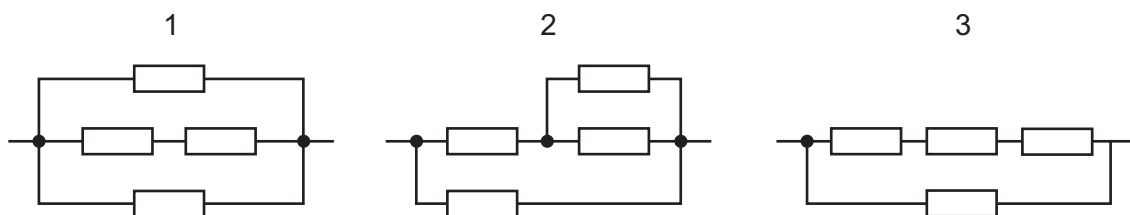


When the system is tested with the switch connection to the red wire, what does the signal show?

- A the green lamp illuminated normally
- B the red lamp illuminated normally
- C the red and green lamps both illuminated normally
- D the red and green lamps both illuminated dimly

82 9702/12/M/J/12/Q38

Four identical resistors are connected in the three networks below.

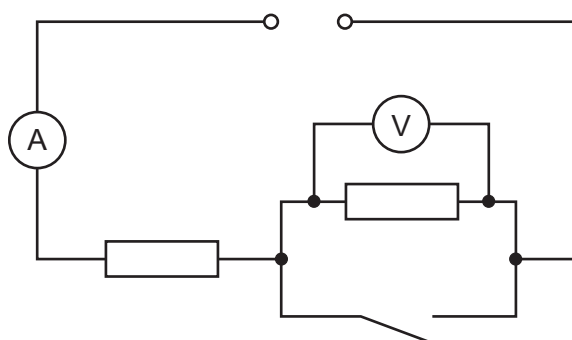


Which arrangement has the highest total resistance and which has the lowest?

	highest	lowest
A	1	2
B	1	3
C	3	1
D	3	2

83 9702/13/M/J/12/Q35

In the circuit below, the ammeter reading is I and the voltmeter reading is V .

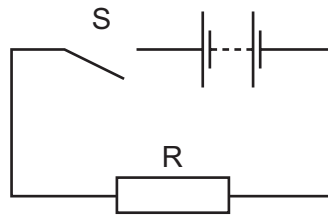


When the switch is closed, which row describes what happens to I and V ?

	I	V
A	decreases	decreases to zero
B	increases	decreases to zero
C	increases	stays the same
D	stays the same	increases

84 9702/13/M/J/12/Q36

The diagram shows a simple circuit.



Which statement is correct?

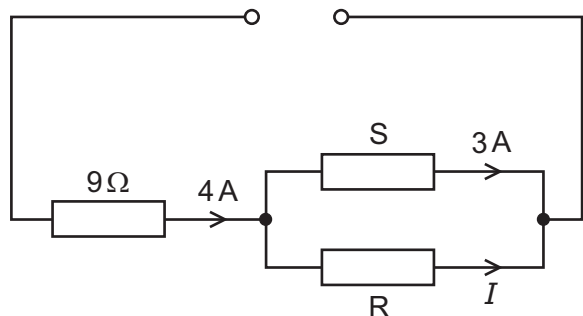
- A When switch S is closed, the electromotive force (e.m.f.) of the battery falls because work is done against the internal resistance of the battery.
- B When switch S is closed, the e.m.f. of the battery falls because work is done against the resistance R.
- C When switch S is closed, the potential difference across the battery falls because work is done against the internal resistance of the battery.
- D When switch S is closed, the potential difference across the battery falls because work is done against the resistance R.

85 9702/13/M/J/12/Q37

The circuit below has a current I in the resistor R.

What must be known in order to determine the value of I ?

- A e.m.f. of the power supply
- B resistance of resistor S
- C Kirchhoff's first law
- D Kirchhoff's second law

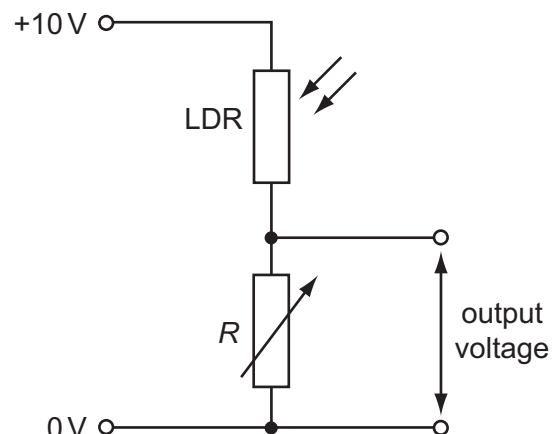


86 9702/13/M/J/12/Q38

A potential divider consists of a light-dependent resistor (LDR) in series with a variable resistor of resistance R . The resistance of the LDR decreases when the light level increases. The variable resistor can be set at either high resistance or low resistance.

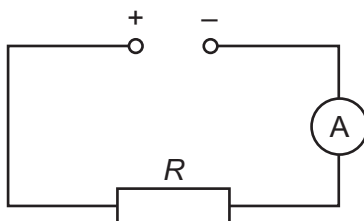
Which situation gives the largest output voltage?

	light level at LDR	R
A	high	high
B	high	low
C	low	high
D	low	low



87 9702/12/O/N/12/Q37

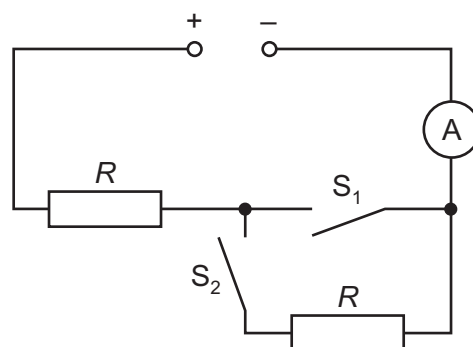
The ammeter reading in the circuit below is I .



Another circuit containing the same voltage supply, two switches, an ammeter and two resistors each of resistance R , is shown.

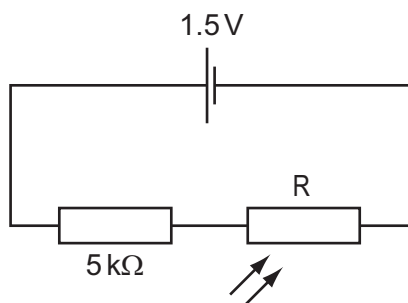
Which row is **not** correct?

	S_1	S_2	ammeter reading
A	closed	closed	I
B	closed	open	I
C	open	closed	I
D	open	open	0



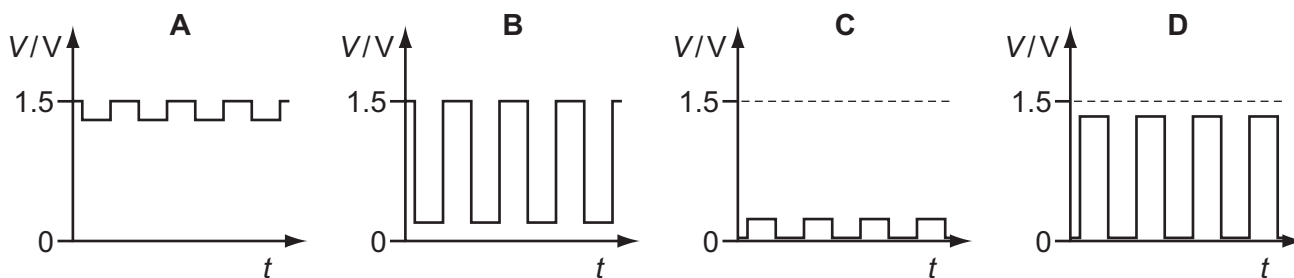
88 9702/12/O/N/12/Q38

A light-dependent resistor R has resistance of about $1\text{ M}\Omega$ in the dark and about $1\text{ k}\Omega$ when illuminated. It is connected in series with a $5\text{ k}\Omega$ resistor to a 1.5 V cell of negligible internal resistance.



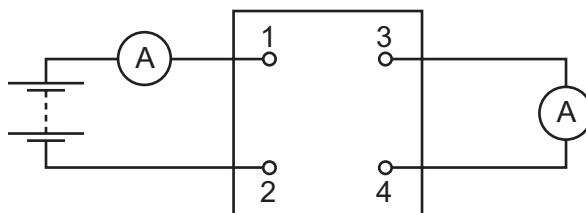
The light-dependent resistor is illuminated (in an otherwise dark room) by a flashing light.

Which graph best shows the variation with time t of potential difference V across R ?



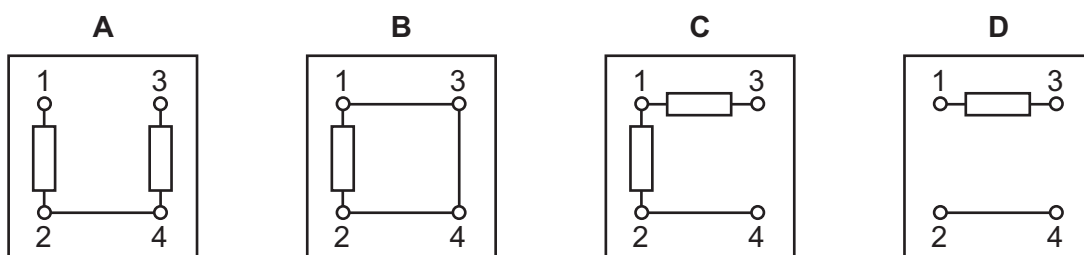
89 9702/13/O/N/12/Q35

The diagram shows a four-terminal box connected to a battery and two ammeters.



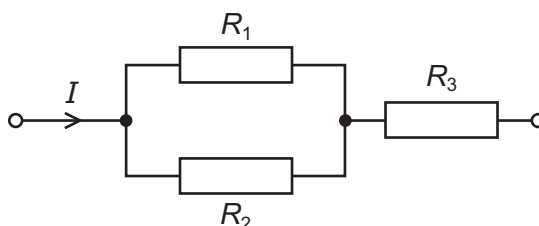
The currents in the two meters are identical.

Which circuit, within the box, will give this result?



90 9702/13/O/N/12/Q37

The diagram shows a resistor network. The potential difference across the network is V .



Is the equation shown below correct for the network? $V = I(1/R_1 + 1/R_2 + R_3)$

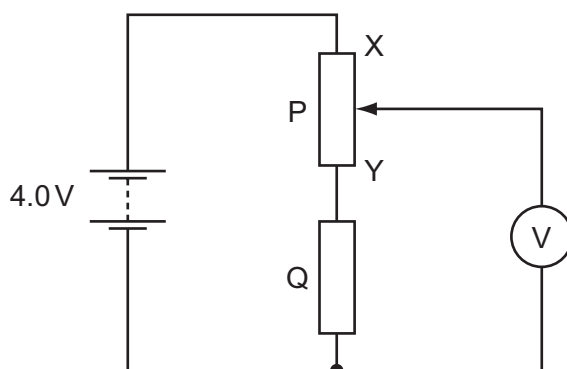
- A Yes, it correctly combines two series resistors with one parallel resistor, and correctly uses Ohm's Law.
- B Yes, it correctly combines two parallel resistors with one series resistor, and correctly uses Ohm's Law.
- C No, because it should read $V = I \div (1/R_1 + 1/R_2 + R_3)$.
- D No, because the terms $1/R_2$ and R_3 have different units and cannot be added.

91 9702/11/M/J/13/Q34

The principles of conservation of which two quantities are associated with Kirchhoff's first and second laws?

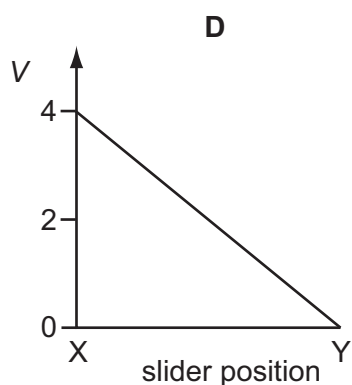
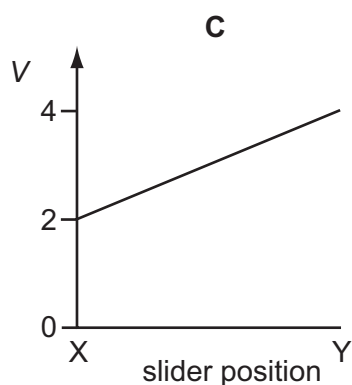
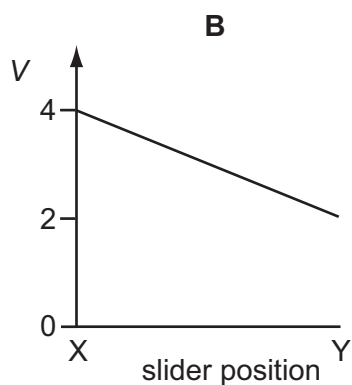
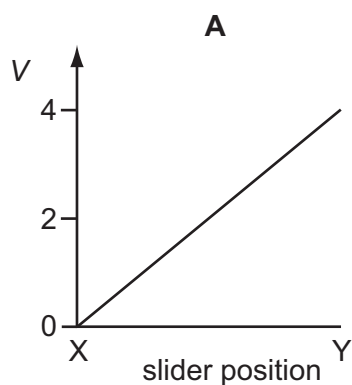
	first law	second law
A	charge	energy
B	charge	voltage
C	energy	charge
D	voltage	charge

In the circuit below, P is a potentiometer of total resistance 10Ω and Q is a fixed resistor of resistance 10Ω . The battery has an e.m.f. of 4.0V and negligible internal resistance. The voltmeter has a very high resistance.



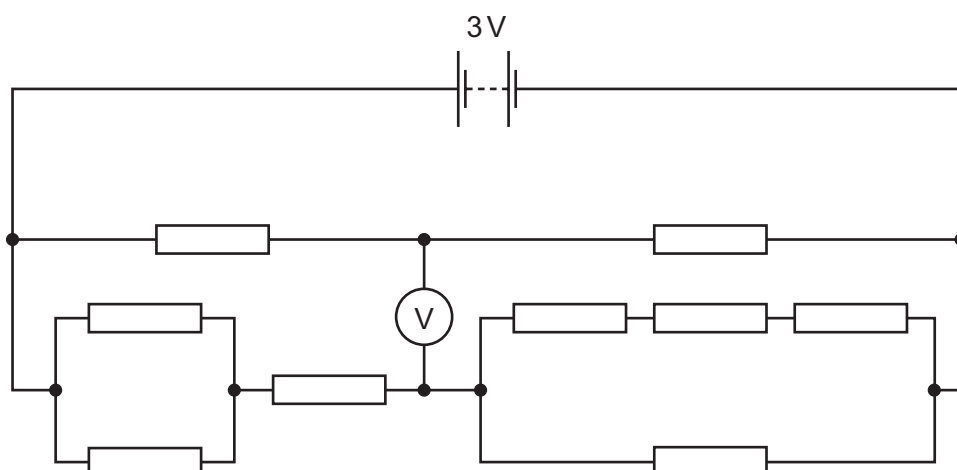
The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading V is plotted against slider position.

Which graph is obtained?



93 9702/11/M/J/13/Q35

A circuit is set up as shown, supplied by a 3 V battery. All resistances are $1\text{ k}\Omega$.

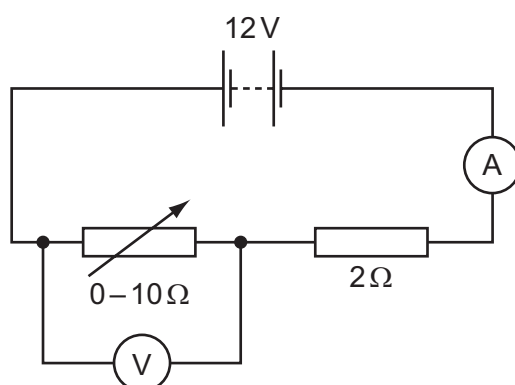


What will be the reading on the voltmeter?

- A 0 B 0.5 V C 1.0 V D 1.5 V

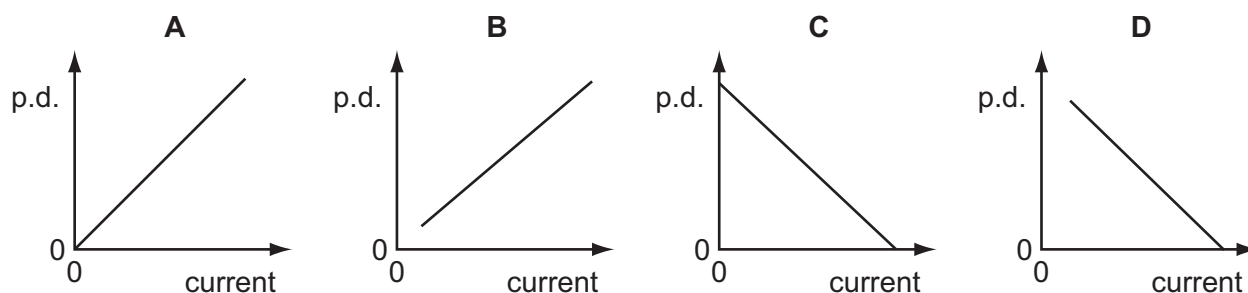
94 9702/11/M/J/13/Q37*

A 12 V battery is in series with an ammeter, a 2Ω fixed resistor and a $0-10\Omega$ variable resistor. A high-resistance voltmeter is connected across the variable resistor.



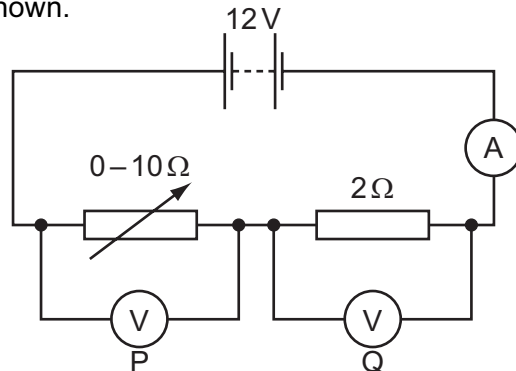
The resistance of the variable resistor is changed from zero to its maximum value.

Which graph shows how the potential difference (p.d.) measured by the voltmeter varies with the current measured by the ammeter?

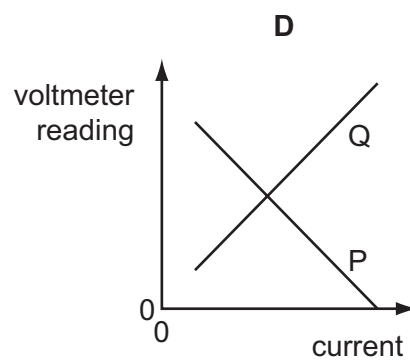
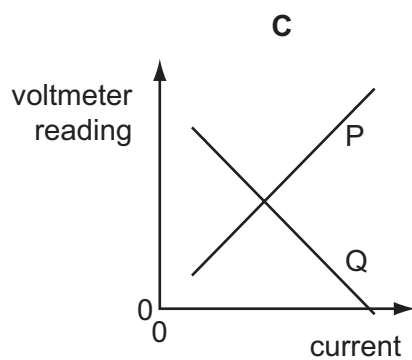
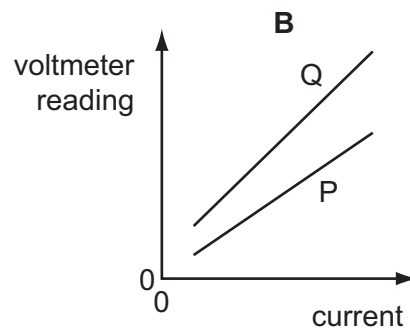
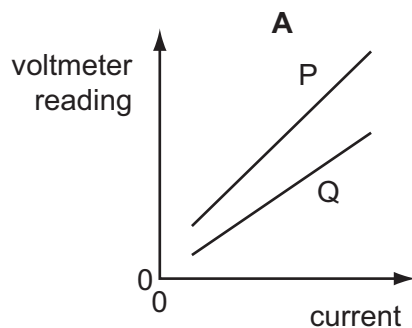


95 9702/12/M/J/13/Q36

A 12 V battery is in series with an ammeter, a 2Ω fixed resistor and a $0-10\Omega$ variable resistor. High-resistance voltmeters P and Q are connected across the variable resistor and the fixed resistor respectively, as shown.



The resistance of the variable resistor is changed from its maximum value to zero. Which graph shows the variation with current of the voltmeter readings?

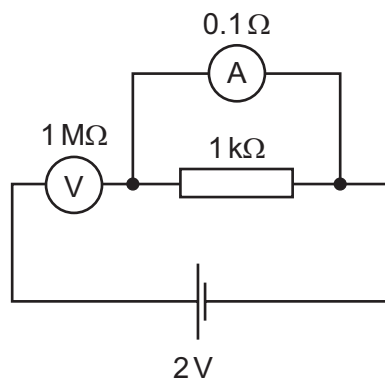


96 9702/11/M/J/13/Q36

The diagram shows an incorrectly connected circuit. The ammeter has a resistance of 0.1Ω and the voltmeter has a resistance of $1\text{ M}\Omega$.

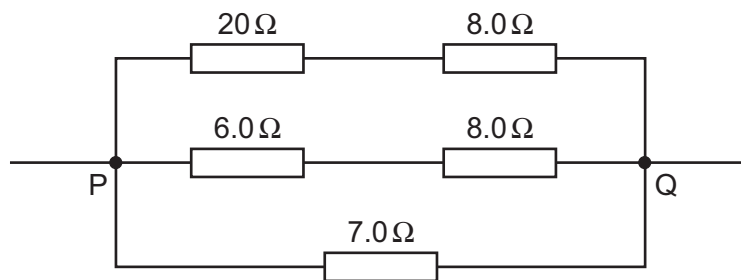
Which statement is correct?

- A** The ammeter reads 2 mA.
- B** The ammeter reads 20 A.
- C** The voltmeter reads zero.
- D** The voltmeter reads 2 V.



97 9702/12/M/J/13/Q37*

Five resistors are connected as shown.

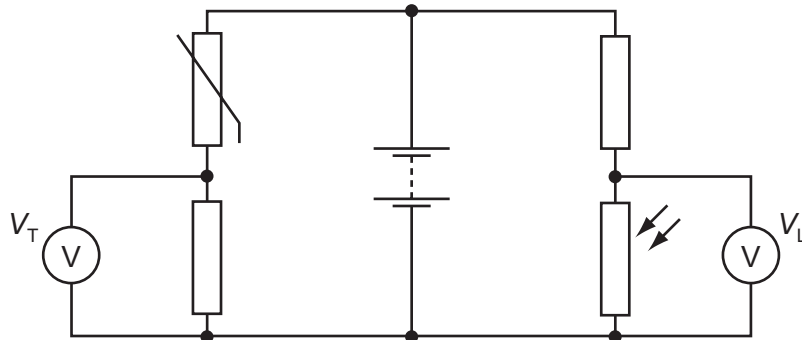


What is the total resistance between P and Q?

- A** $0.25\ \Omega$ **B** $0.61\ \Omega$ **C** $4.0\ \Omega$ **D** $16\ \Omega$

98 9702/12/M/J/13/Q38

In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



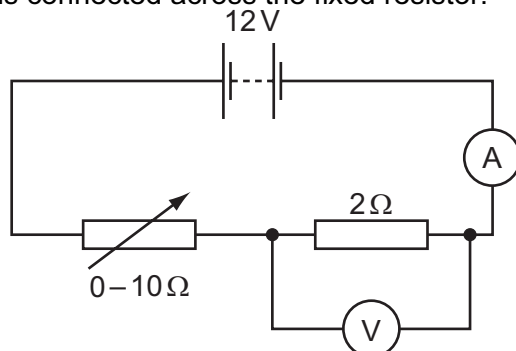
The readings V_T and V_L are both high.

What are the conditions of temperature and light level?

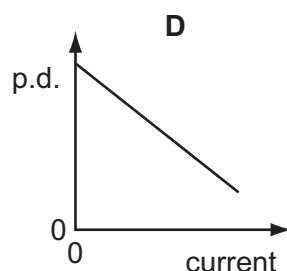
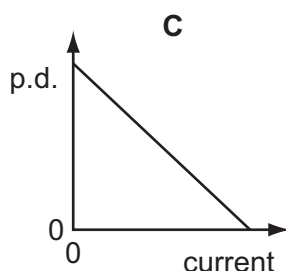
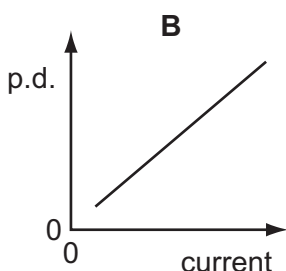
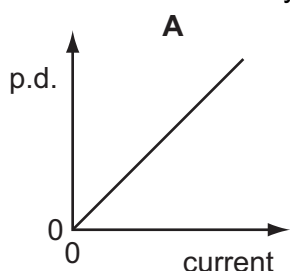
	temperature	light level
A	low	low
B	low	high
C	high	low
D	high	high

99 9702/13/M/J/13/Q36

A 12 V battery is in series with an ammeter, a 2Ω fixed resistor and a $0-10\Omega$ variable resistor. A high-resistance voltmeter is connected across the fixed resistor.

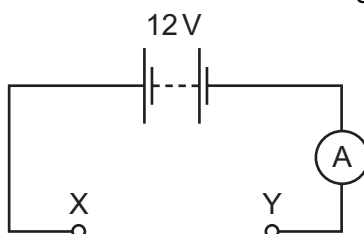


The resistance of the variable resistor is changed from zero to its maximum value. Which graph shows how the potential difference (p.d.) measured by the voltmeter varies with the current measured by the ammeter?



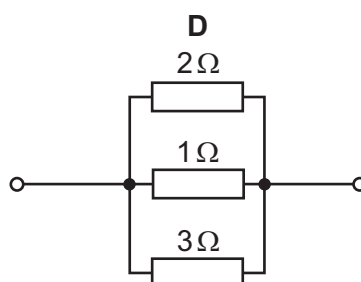
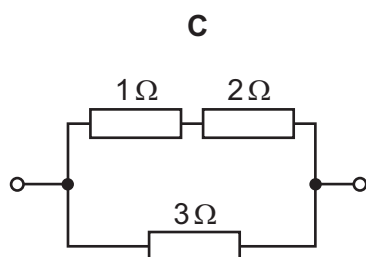
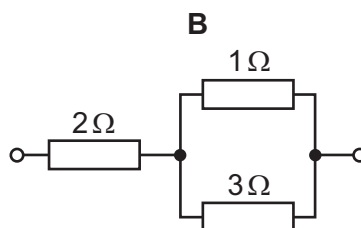
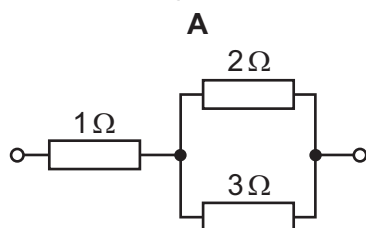
100 9702/13/M/J/13/Q37

In the circuit shown, the battery and ammeter each have negligible resistance.



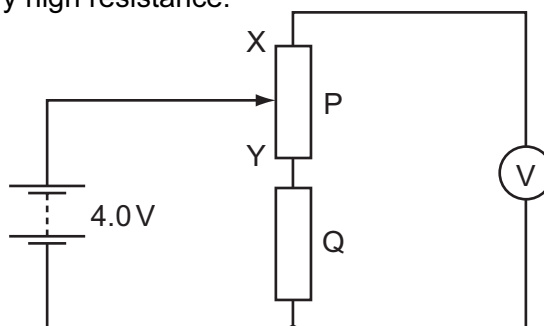
The following combinations of resistors are placed in turn between the terminals X and Y of the circuit.

Which combination would give an ammeter reading of 8 A?



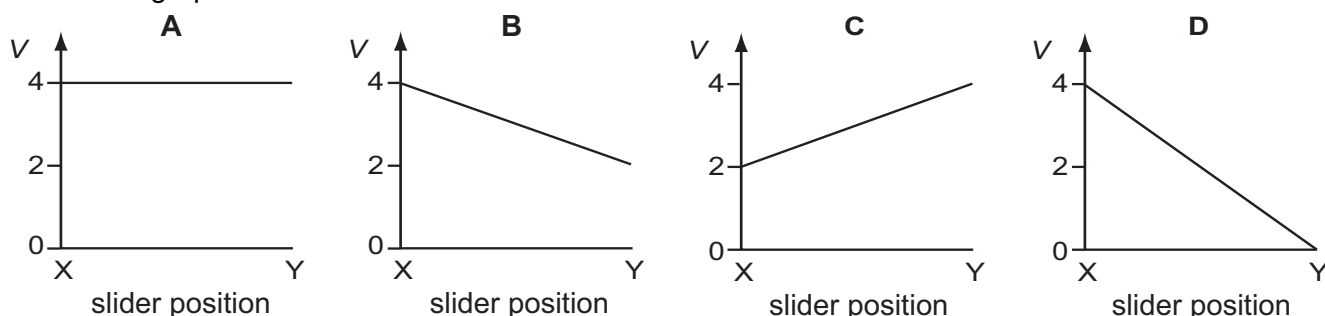
101 9702/12/O/N/13/Q36

In the circuit below, P is a potentiometer of total resistance 10Ω and Q is a fixed resistor of resistance 10Ω . The battery has an electromotive force (e.m.f.) of 4.0V and negligible internal resistance. The voltmeter has a very high resistance.



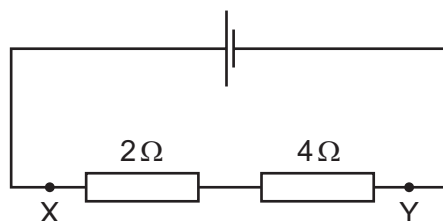
The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading V is plotted against slider position.

Which graph would be obtained?

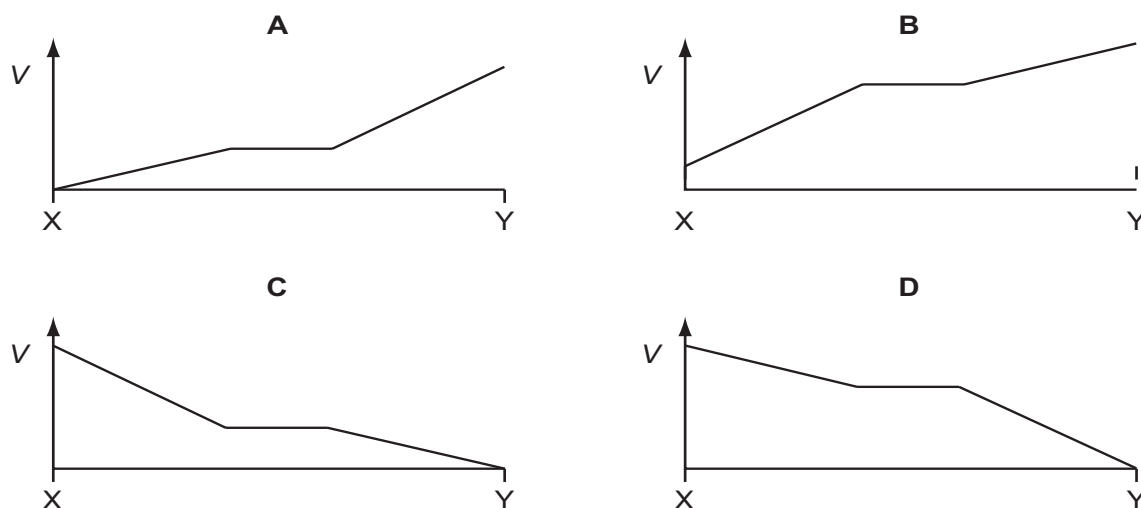


102 9702/12/O/N/13/Q37

A 2Ω resistor and a 4Ω resistor are connected to a cell.

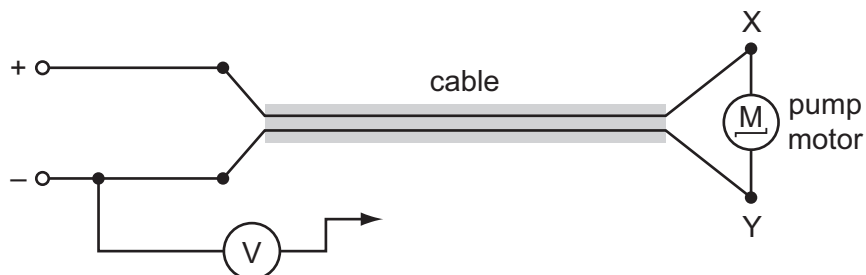


Which graph shows how the potential V varies with distance between X and Y?



103 9702/13/O/N/13/Q36

The diagram shows the electric motor for a garden pump connected to a 24 V power supply by an insulated two-core cable.



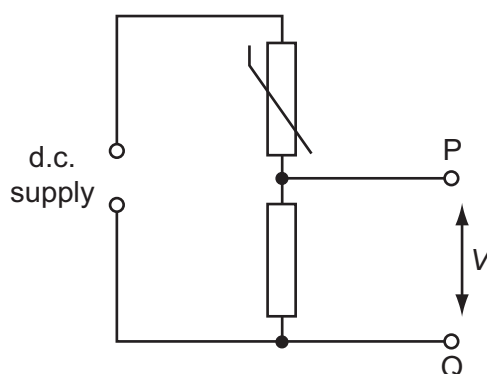
The motor does not work so, to find the fault, the negative terminal of a voltmeter is connected to the negative terminal of the power supply and its other end is connected in turn to terminals X and Y at the motor.

Which row represents two readings and a correct conclusion?

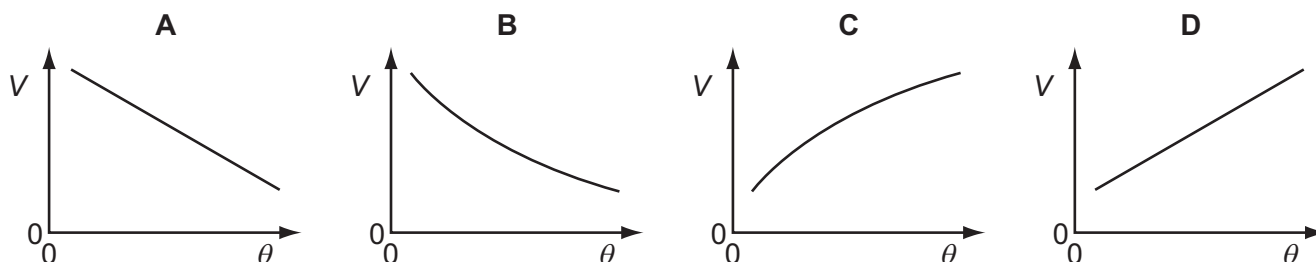
	voltmeter reading when connected to X/V	voltmeter reading when connected to Y/V	conclusion
A	24	0	break in positive wire of cable
B	24	12	break in negative wire of cable
C	24	24	break in connection within the motor
D	24	24	break in negative wire of cable

104 9702/13/O/N/13/Q37

In the circuit shown, the resistance of the thermistor decreases as temperature increases.

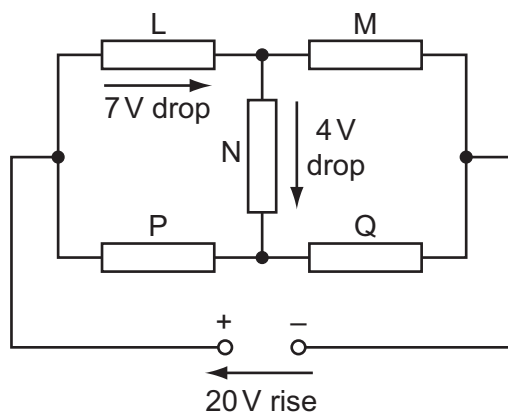


Which graph shows the variation with Celsius temperature θ of potential difference V between points P and Q?



105 9702/13/O/N/13/Q38

A 20 V d.c. supply is connected to a circuit consisting of five resistors L, M, N, P and Q.

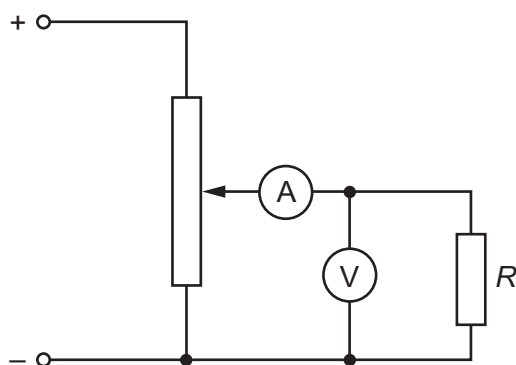


There is a potential drop of 7 V across L and a further 4 V potential drop across N.
What are the potential drops across M, P and Q?

	potential drop across M/V	potential drop across P/V	potential drop across Q/V
A	9	7	13
B	13	7	13
C	13	11	9
D	17	3	17

106 9702/11/M/J/14/Q33

In the circuit below, a voltmeter of resistance R_V and an ammeter of resistance R_A are used to measure the resistance R of the fixed resistor.

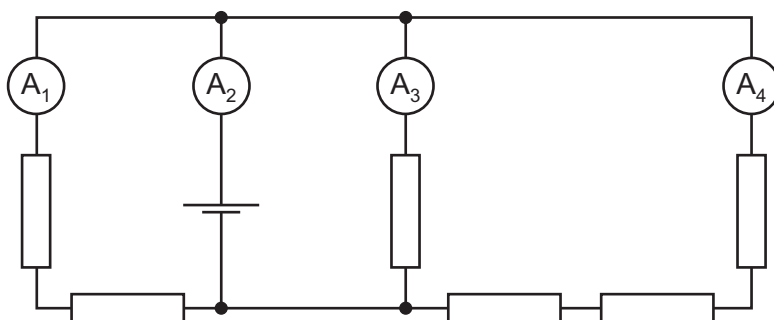


Which condition is necessary for an accurate value to be obtained for R ?

- | | |
|---|---|
| A R is much smaller than R_V . | C R is much greater than R_V . |
| B R is much smaller than R_A . | D R is much greater than R_A . |

107 9702/11/M/J/14/Q34

In the circuit shown, all the resistors are identical and all the ammeters have negligible resistance.



The reading on ammeter A_1 is 0.6 A .

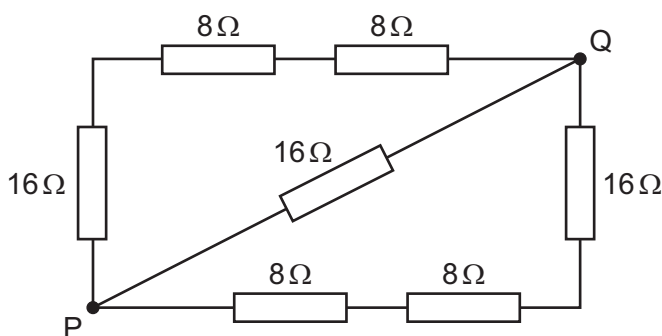
What are the readings on the other ammeters?

	reading on ammeter A_2/A	reading on ammeter A_3/A	reading on ammeter A_4/A
A	1.0	0.3	0.1
B	1.4	0.6	0.2
C	1.8	0.9	0.3
D	2.2	1.2	0.4

108 9702/11/M/J/14/Q36

What is the total resistance between points P and Q in this network of resistors?

- A** $8\ \Omega$
- B** $16\ \Omega$
- C** $24\ \Omega$
- D** $32\ \Omega$

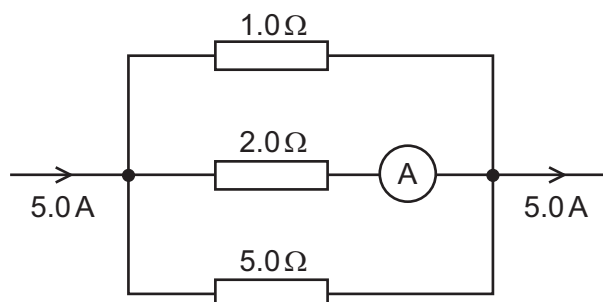


109 9702/13/M/J/15/Q37

The diagram shows part of a current-carrying circuit. The ammeter has negligible resistance.

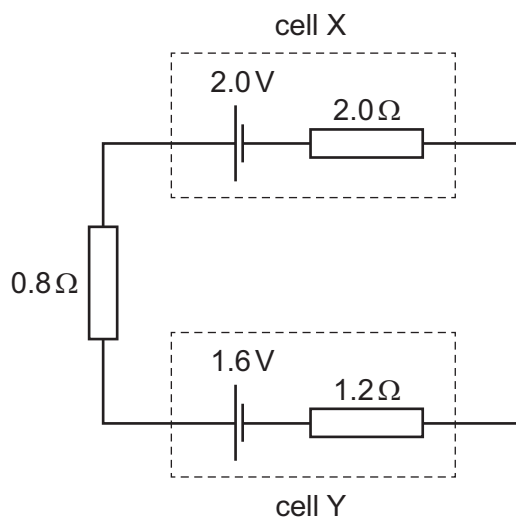
What is the reading on the ammeter?

- A** 0.7 A
- B** 1.3 A
- C** 1.5 A
- D** 1.7 A



110 9702/12/M/J/14/Q33

Cell X has an e.m.f. of 2.0 V and an internal resistance of $2.0\ \Omega$. Cell Y has an e.m.f. of 1.6 V and an internal resistance of $1.2\ \Omega$. These two cells are connected to a resistor of resistance $0.8\ \Omega$, as shown.

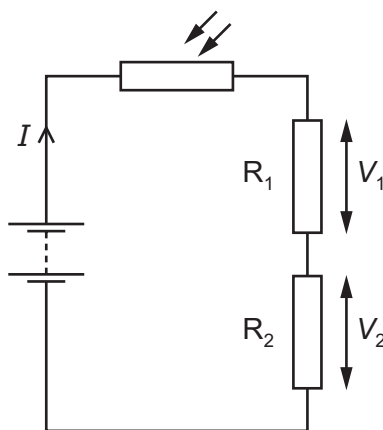


What is the current in cell X?

- A** 0.10 A **B** 0.50 A **C** 0.90 A **D** 1.0 A

111 9702/12/M/J/14/Q34

In the circuit shown, a light-dependent resistor (LDR) is connected to two resistors R_1 and R_2 . The potential difference (p.d.) across R_1 is V_1 and the p.d. across R_2 is V_2 . The current in the circuit is I .

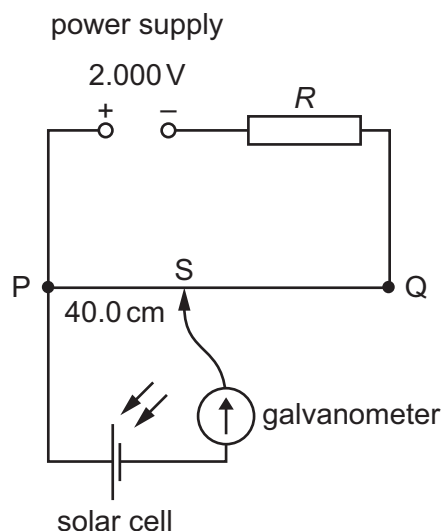


Which statement about this circuit is correct?

- A** The current I increases when the light intensity decreases.
B The LDR is an ohmic conductor.
C The p.d. V_2 increases when the light intensity decreases.
D The ratio $\frac{V_1}{V_2}$ is independent of light intensity.

112 9702/12/M/J/14/Q35

A power supply and a solar cell are compared using the potentiometer circuit shown.



The e.m.f. produced by the solar cell is measured on the potentiometer.

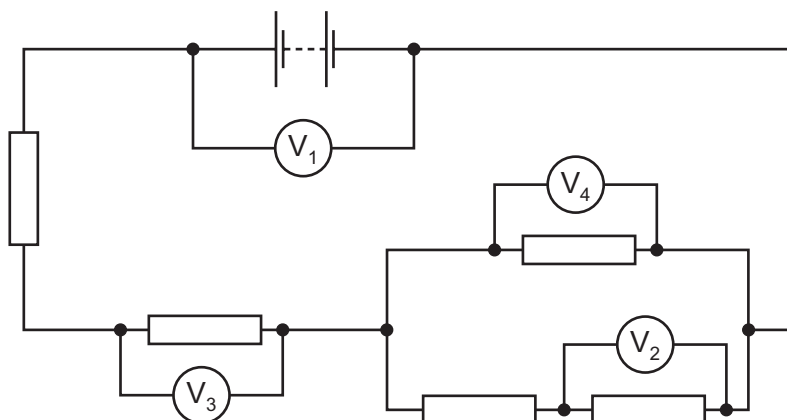
The potentiometer wire PQ is 100.0 cm long and has a resistance of $5.00\ \Omega$. The power supply has an e.m.f. of 2.000 V and the solar cell has an e.m.f. of 5.00 mV.

Which resistance R must be used so that the galvanometer reads zero when $PS = 40.0\text{ cm}$?

- A $395\ \Omega$ B $795\ \Omega$ C $995\ \Omega$ D $1055\ \Omega$

113 9702/12/M/J/14/Q36

In the circuit shown, all the resistors are identical.



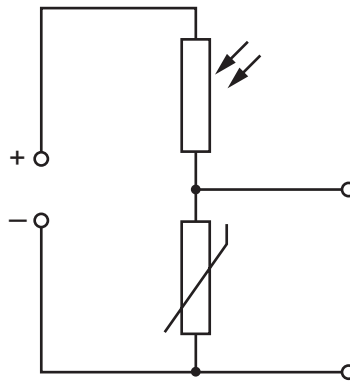
The reading on voltmeter V_1 is 8.0 V and the reading on voltmeter V_2 is 1.0 V.

What are the readings on the other voltmeters?

	reading on voltmeter V_3 / V	reading on voltmeter V_4 / V
A	1.5	1.0
B	3.0	2.0
C	4.5	3.0
D	6.0	4.0

114 9702/12/M/J/14/Q37

The diagram shows a light-dependent resistor (LDR) and a thermistor forming a potential divider.

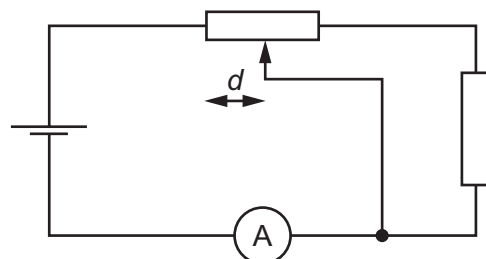
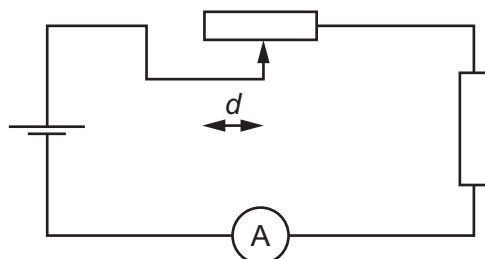
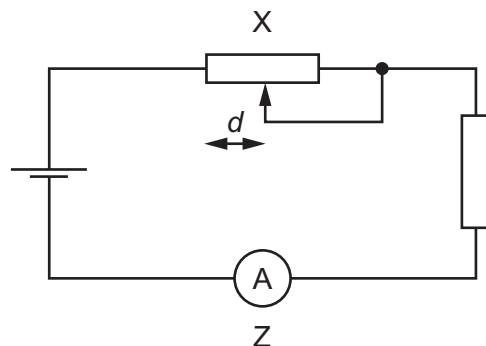
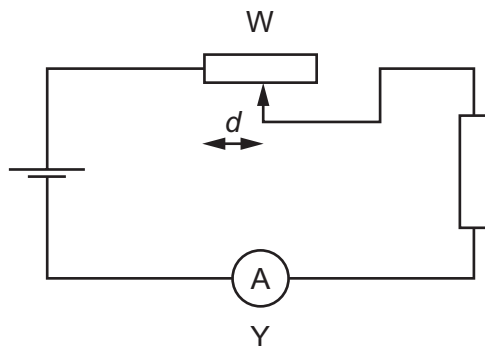


Under which set of conditions will the potential difference across the thermistor have the greatest value?

	illumination	temperature
A	low	low
B	high	low
C	low	high
D	high	high

115 9702/13/M/J/14/Q36

The diagrams show the same cell, ammeter, potentiometer and fixed resistor connected in different ways.



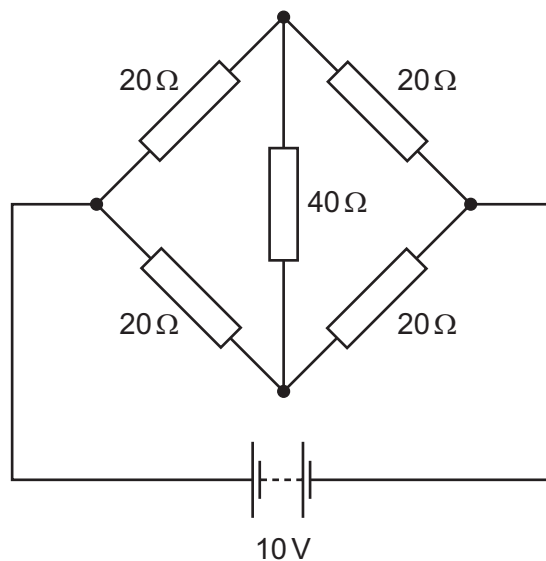
The distance d between the sliding contact and a particular end of the potentiometer is varied. The current measured is then plotted against the distance d .

For which two circuits will the graphs be identical?

- A** W and X **B** W and Y **C** X and Y **D** Y and Z

116 9702/13/M/J/14/Q37

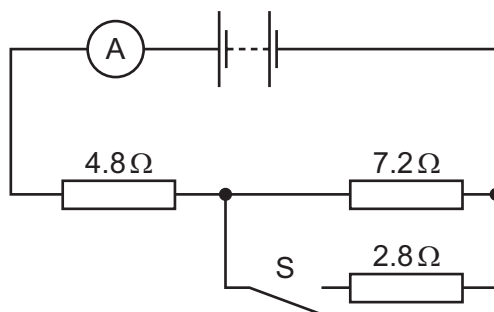
What is the current in the $40\ \Omega$ resistor of the circuit shown?



- A zero B 0.13 A C 0.25 A D 0.50 A

117 9702/13/M/J/14/Q38

A battery of negligible internal resistance is connected to a resistor network, an ammeter and a switch S, as shown.



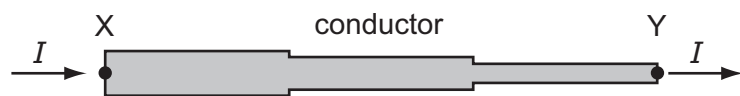
When S is open, the reading on the ammeter is 250 mA.

When S is closed, what is the **change** in the reading on the ammeter?

- A 1.07 A B 1.32 A C 190 mA D 440 mA

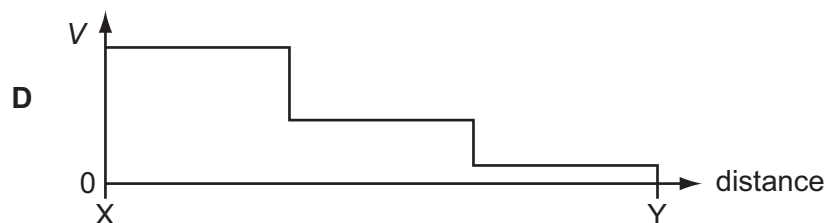
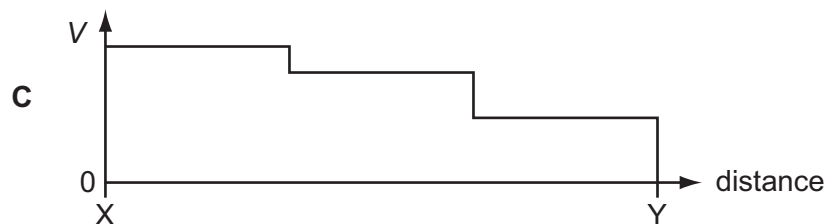
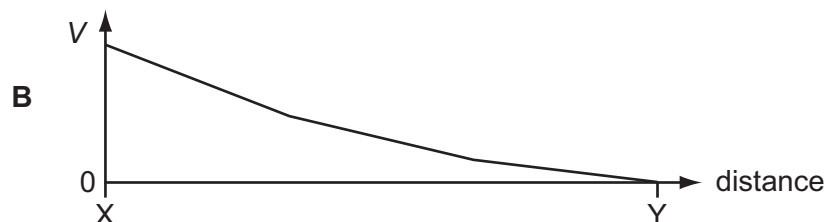
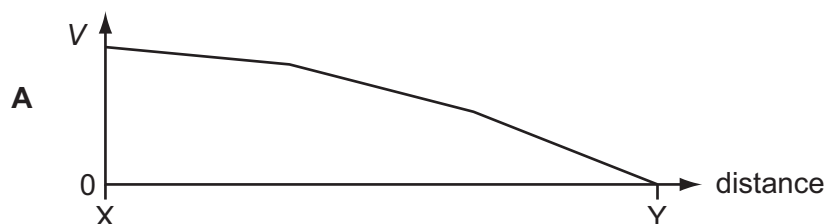
118 9702/11/O/N/14/Q33

A conductor consists of three wires connected in series. The wires are all made of the same metal but have different cross-sectional areas. There is a current I in the conductor.



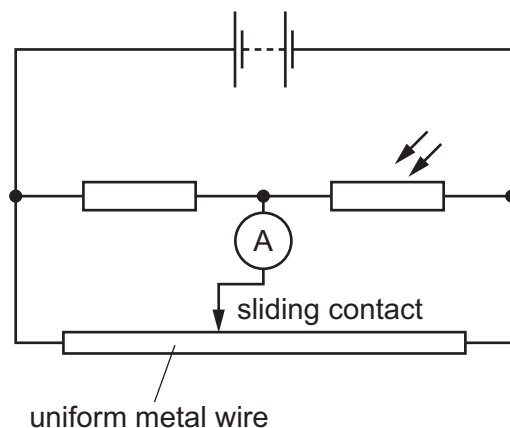
Point Y on the conductor is at zero potential.

Which graph best shows the variation of potential V with distance along the conductor?



119 9702/11/O/N/14/Q36

In the potentiometer circuit shown, the reading on the ammeter is zero.



The light-dependent resistor (LDR) is then covered up and the ammeter gives a non-zero reading.

Which change could return the ammeter reading to zero?

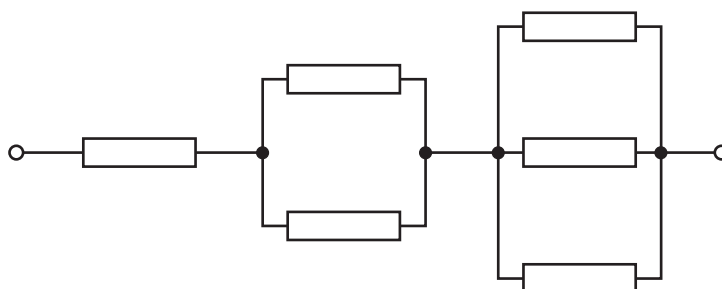
- | | |
|---------------------------------------|---|
| A Decrease the supply voltage. | C Move the sliding contact to the left. |
| B Increase the supply voltage. | D Move the sliding contact to the right. |

120 9702/11/O/N/14/Q37

Six resistors, each of resistance R , are connected as shown.

The combined resistance is $66 \text{ k}\Omega$.

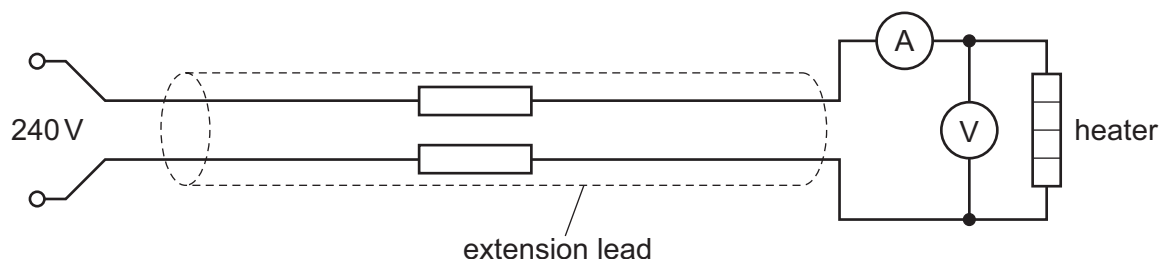
What is the value of R ?



- A** $11 \text{ k}\Omega$
B $18 \text{ k}\Omega$
C $22 \text{ k}\Omega$
D $36 \text{ k}\Omega$

121 9702/13/O/N/14/Q36

An extension lead is used to connect a 240 V electrical supply to a heater as shown.



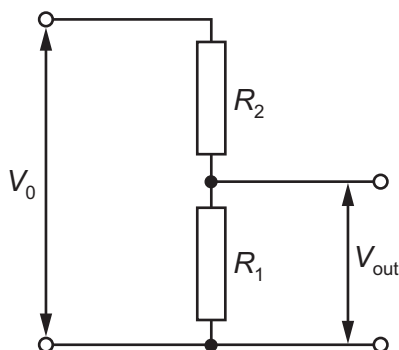
A voltmeter measures the potential difference (p.d.) across the heater as 216 V and an ammeter measures the current through the heater as 7.7 A .

What is the total resistance of the extension lead?

- | | | | |
|-----------------------|-----------------------|----------------------|----------------------|
| A 3.1Ω | B 6.2Ω | C 28Ω | D 31Ω |
|-----------------------|-----------------------|----------------------|----------------------|

122 9702/13/O/N/14/Q35

A potential divider consists of resistors of resistance R_1 and R_2 connected in series across a source of potential difference V_0 . The potential difference across R_1 is V_{out} .

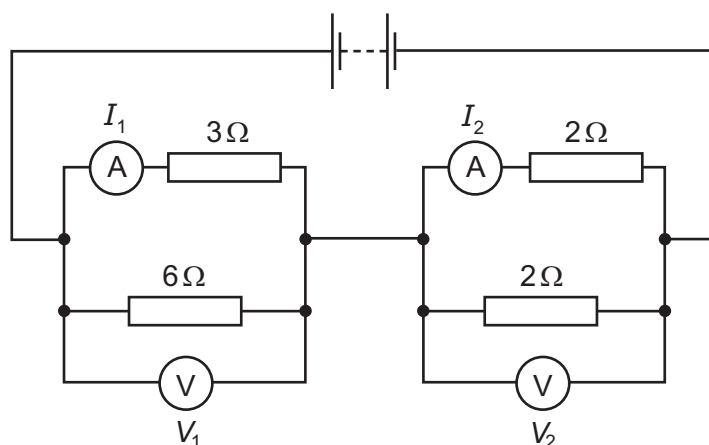


Which changes to R_1 and R_2 will increase the value of V_{out} ?

	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

123 9702/13/O/N/14/Q37

In the circuit shown, the ammeters have negligible resistance and the voltmeters have infinite resistance.



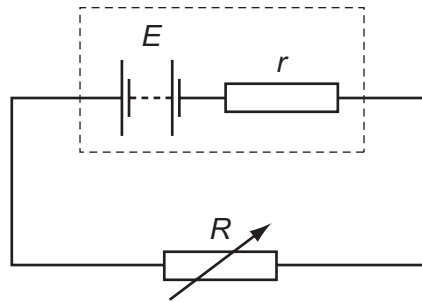
The readings on the meters are I_1 , I_2 , V_1 and V_2 , as labelled on the diagram.

Which statement is correct?

- A** $I_1 > I_2$ and $V_1 > V_2$ **B** $I_1 > I_2$ and $V_1 < V_2$
C $I_1 < I_2$ and $V_1 > V_2$ **D** $I_1 < I_2$ and $V_1 < V_2$

124 9702/11/M/J/15/Q36

A battery with e.m.f. E and internal resistance r is connected in series with a variable external resistor.



The value of the external resistance R is slowly increased from zero.

Which statement is correct? (Ignore any temperature effects.)

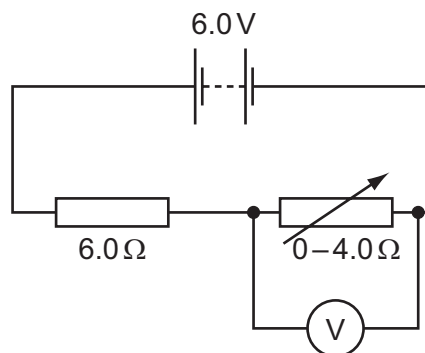
- A The potential difference across the external resistance decreases.
- B The potential difference across the internal resistance increases.
- C The power dissipated in r increases and then decreases.
- D The power dissipated in R increases and then decreases.

125 9702/11/M/J/15/Q37

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected in series with a resistor of resistance $6.0\ \Omega$ and a variable resistor of resistance from zero to $4.0\ \Omega$. A voltmeter is connected across the variable resistor. The resistance of the variable resistor is changed.

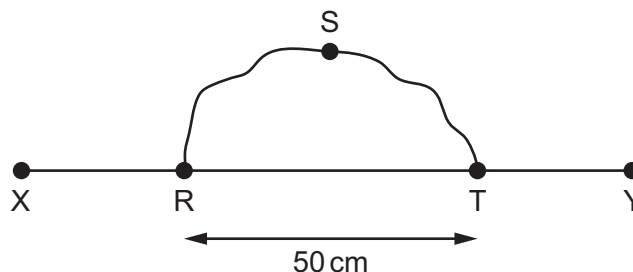
What is the range of the voltmeter reading?

- A $0\text{ V} - 2.4\text{ V}$
- B $0\text{ V} - 3.6\text{ V}$
- C $2.4\text{ V} - 6.0\text{ V}$
- D $3.6\text{ V} - 6.0\text{ V}$



126 9702/11/M/J/15/Q38

A wire RST is connected to another wire XY as shown.



Each wire is 100 cm long with a resistance per unit length of $10\ \Omega\text{ m}^{-1}$.

What is the total resistance between X and Y?

- A $3.3\ \Omega$
- B $5.0\ \Omega$
- C $8.3\ \Omega$
- D $13.3\ \Omega$

127 9702/12/M/J/15/Q35

Each of Kirchhoff's two laws presumes that some quantity is conserved.

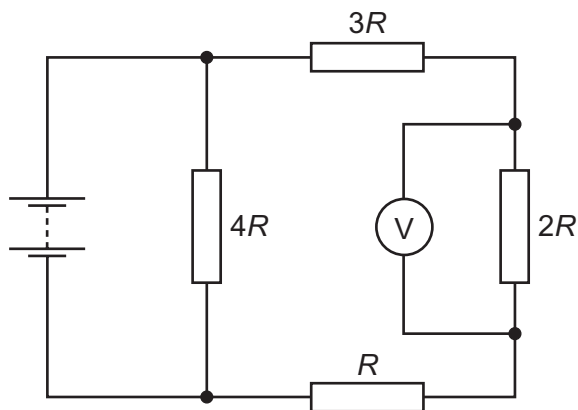
Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	the algebraic sum of currents into a junction is zero	charge
B	the algebraic sum of currents into a junction is zero	energy
C	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	charge
D	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	energy

128 9702/12/O/N/13/Q38

Four resistors of resistance R , $2R$, $3R$ and $4R$ are connected to form a network.

A battery of negligible internal resistance and a voltmeter are connected to the resistor network as shown.



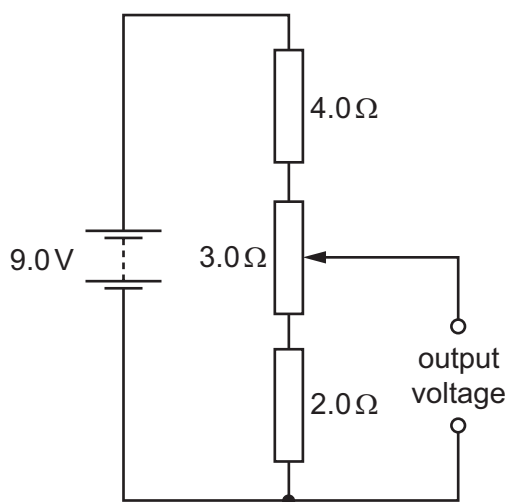
The voltmeter reading is 2 V.

What is the electromotive force (e.m.f.) of the battery?

- A** 2 V **B** 4 V **C** 6 V **D** 10 V

129 9702/12/M/J/15/Q36

A potential divider circuit consists of fixed resistors of resistance $2.0\ \Omega$ and $4.0\ \Omega$ connected in series with a $3.0\ \Omega$ resistor fitted with a sliding contact. These are connected across a battery of e.m.f. $9.0\ \text{V}$ and zero internal resistance, as shown.

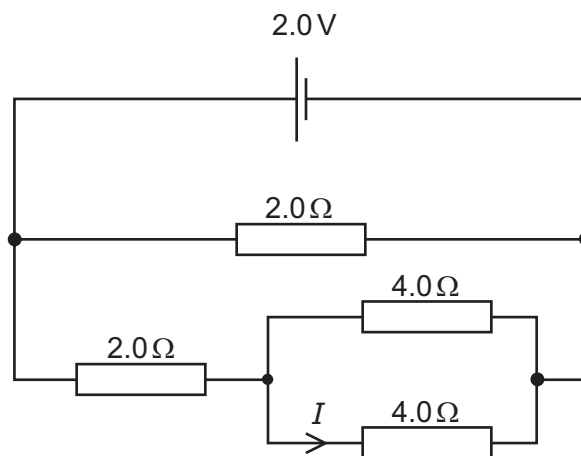


What are the maximum and the minimum output voltages of this potential divider circuit?

	maximum voltage / V	minimum voltage / V
A	4.0	2.0
B	5.0	2.0
C	9.0	0
D	9.0	2.0

130 9702/12/M/J/15/Q37

A cell of e.m.f. $2.0\ \text{V}$ and negligible internal resistance is connected to a network of resistors as shown.

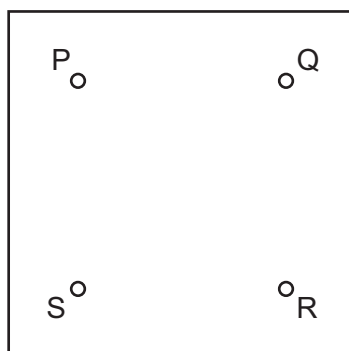


What is the current I ?

- A** 0.25 A **B** 0.33 A **C** 0.50 A **D** 1.5 A

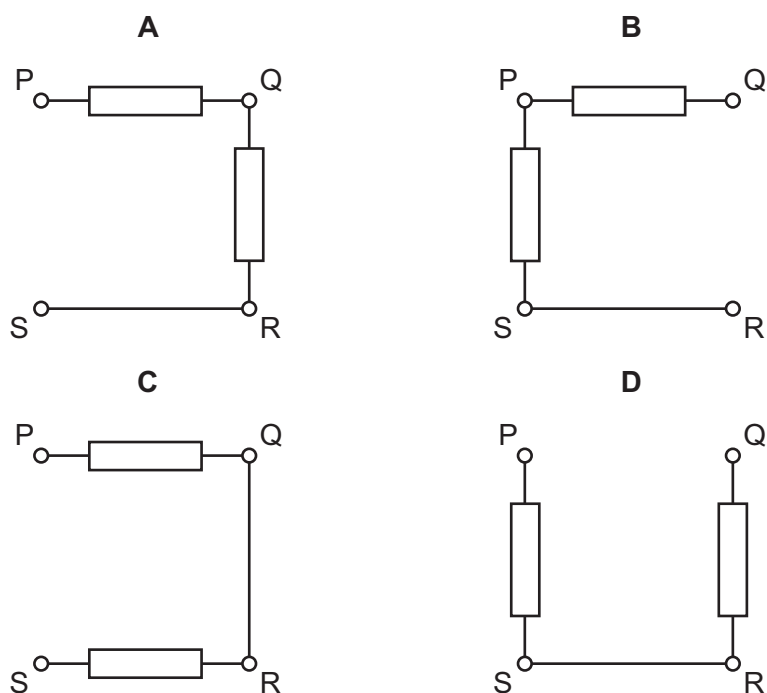
131 9702/13/M/J/15/Q36

A box with four terminals P, Q, R and S contains two identical resistors.



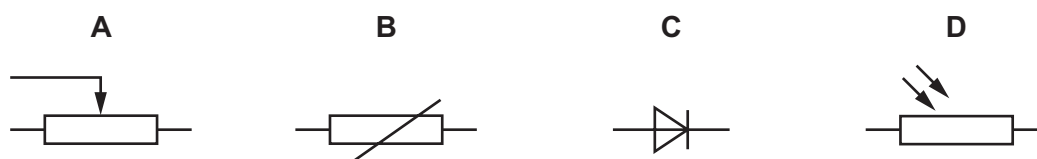
When a battery of electromotive force (e.m.f.) E and negligible internal resistance is connected across PS, a high-resistance voltmeter connected across QR reads $\frac{E}{2}$.

Which diagram shows the correct arrangement of the two resistors inside the box?



132 9702/11/M/J/14/Q32

Which symbol represents a component whose resistance is designed to change with temperature?



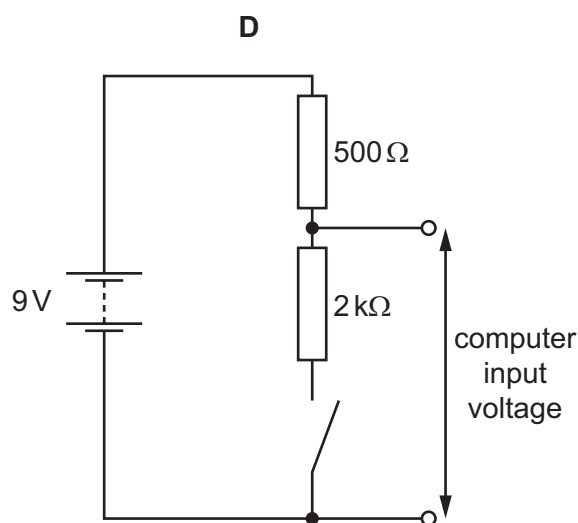
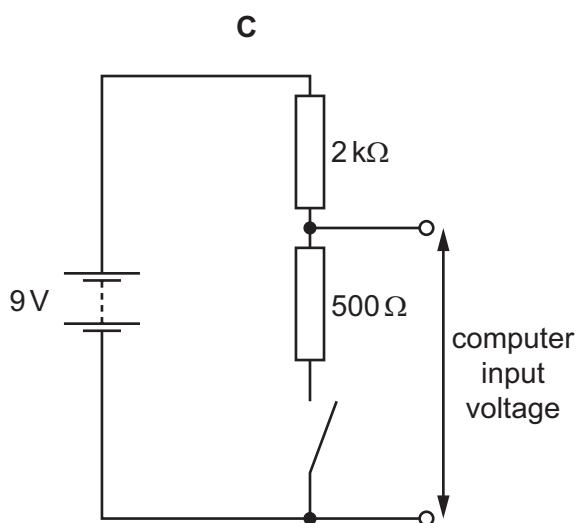
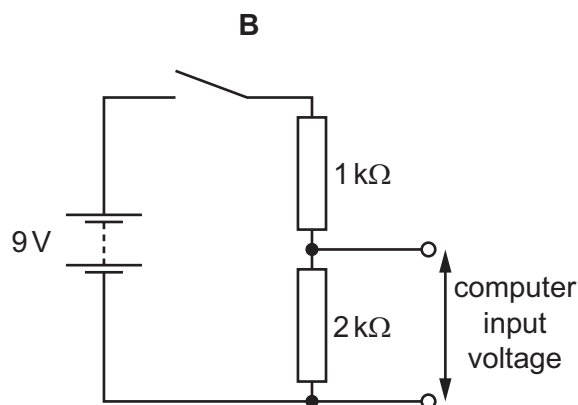
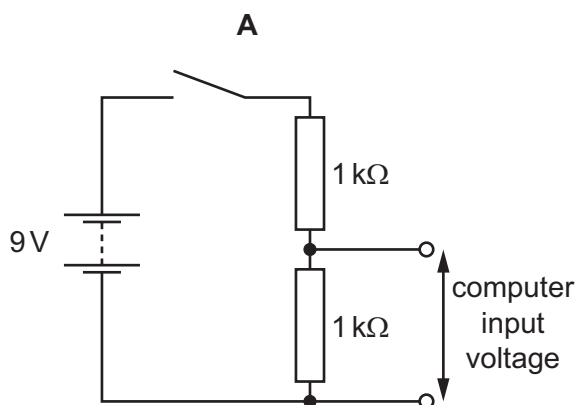
133 9702/11/M/J/17/Q36

A computer is used to detect the change of position of a switch.

To detect the change of position, the computer requires a potential difference (p.d.) of 0 V to its input at one switch position and a p.d. of between 5 V and 7 V at the other switch position.

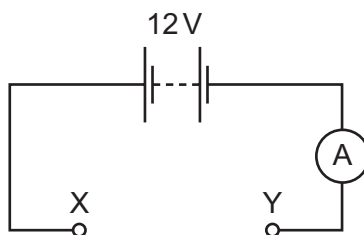
For each of the circuits, assume the battery has negligible internal resistance.

Which circuit provides an input voltage to the computer that enables it to detect the change of position of the switch?



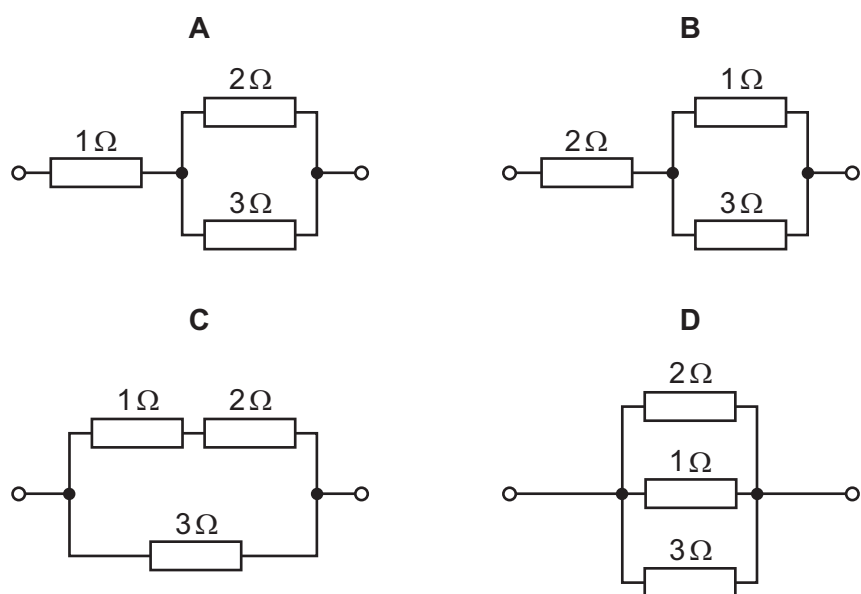
134 9702/11/M/J/17/Q37

In the circuit shown, the battery and ammeter have negligible resistance.



The following combinations of resistors are each separately placed between the terminals X and Y of the circuit.

Which combination would give an ammeter reading of 8 A?



135 9702/12/M/J/17/Q35

A cell has a constant electromotive force.

A variable resistor is connected between the terminals of the cell.

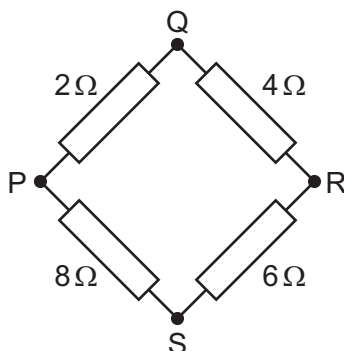
The resistance of the variable resistor is decreased.

Which statement about the change of the cell's terminal potential difference (p.d.) is correct?

- A** The terminal p.d. is decreased because more work is done moving unit charge through the internal resistance of the cell.
- B** The terminal p.d. is decreased because the current in the variable resistor is decreased.
- C** The terminal p.d. is increased because more work is done moving unit charge through the variable resistor.
- D** The terminal p.d. is increased because the current in the variable resistor is increased.

136 9702/12/M/J/17/Q36

Four resistors are connected in a square as shown.



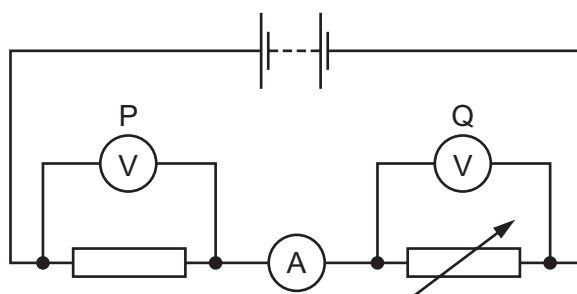
The resistance may be measured between any two junctions.

Between which two junctions is the measured resistance greatest?

- A** P and Q **B** Q and S **C** R and S **D** S and P

137 9702/12/M/J/17/Q37

A circuit is set up as shown.



The variable resistor is adjusted so that the ammeter reading decreases.

How do the readings of the voltmeters change?

	reading on voltmeter P	reading on voltmeter Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

138 9702/11/O/N/17/Q34

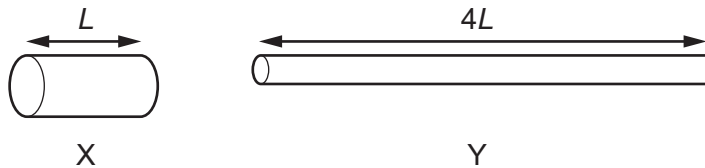
A filament lamp has a resistance of $180\ \Omega$ when the current in it is $500\ \text{mA}$.

What is the power dissipated in the lamp?

- A** 45 W **B** 90 W **C** 290 W **D** 360 W

139 9702/11/O/N/17/Q35

Two copper wires X and Y have the same volume. Wire Y is four times as long as wire X.



What is the ratio $\frac{\text{resistance of wire Y}}{\text{resistance of wire X}}$?

- A** 1 **B** 4 **C** 8 **D** 16

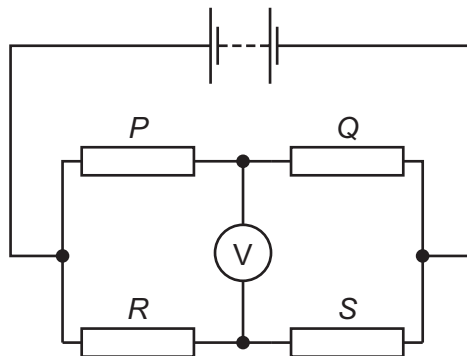
140 9702/11/O/N/17/Q38

The circuit diagram shows four resistors of different resistances P , Q , R and S connected to a battery.

The voltmeter reading is zero.

Which equation is correct?

- A** $P - Q = R - S$
B $P - S = Q - R$
C $PQ = RS$
D $PS = QR$



141 9702/12/O/N/17/Q19

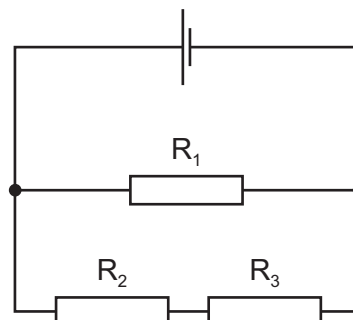
During refuelling, a petrol car receives 50 litres of fuel in 90 seconds. The petrol has 34 MJ of energy per litre.

For an electric car to receive the same amount of energy in the same time from a 230 V supply, what is the minimum current required?

- A** 2700 A **B** $8.2 \times 10^4\ \text{A}$ **C** $7.4 \times 10^6\ \text{A}$ **D** $6.6 \times 10^8\ \text{A}$

142 9702/12/O/N/17/Q34

A cell of negligible internal resistance is connected to resistors R_1 , R_2 and R_3 , as shown. The cell provides power to the circuit and power is dissipated in the resistors.

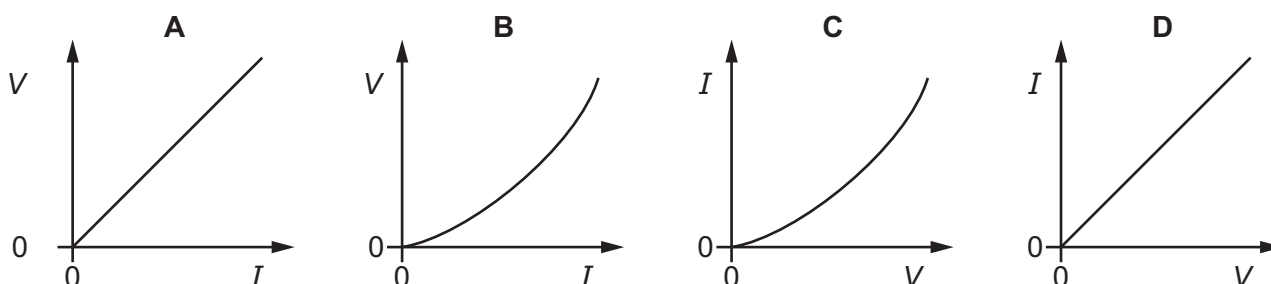


Which word equation **must** be correct?

- A power loss in R_1 = power loss in R_2 + power loss in R_3
- B power loss in R_2 = power loss in R_3
- C power output of cell = power loss in R_1 + power loss in R_2 + power loss in R_3
- D power output of cell = power loss in R_1

143 9702/12/O/N/17/Q35

Which graph represents a metallic conductor, where the resistance of the conductor is given by the gradient of the graph?



144 9702/12/O/N/17/Q36

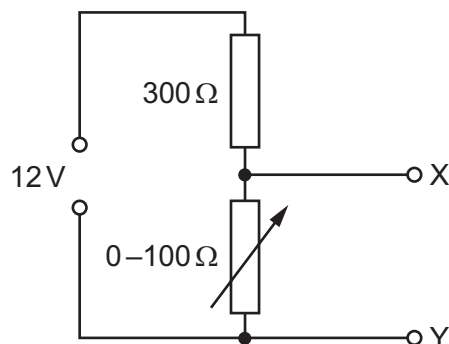
A typical mobile phone battery has an e.m.f. of 5.0 V and an internal resistance of $200\text{ m}\Omega$.

What is the terminal p.d. of the battery when it supplies a current of 500 mA?

- A 4.8 V
- B 4.9 V
- C 5.0 V
- D 5.1 V

145 9702/12/O/N/17/Q38

The diagram shows a potential divider connected to a 12V supply of negligible internal resistance.



Which range of voltages can be obtained between X and Y?

- A 0 to 3V B 0 to 4V C 0 to 8V D 0 to 9V

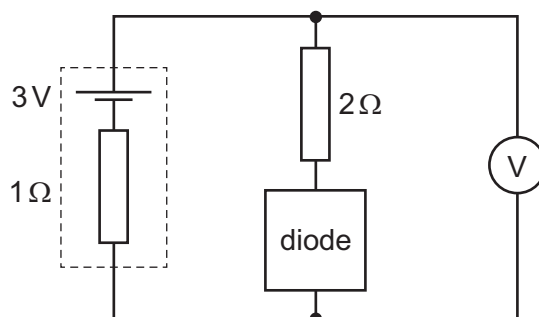
146 9702/13/O/N/17/Q35

An ideal diode has zero resistance when forward biased and infinite resistance when reverse biased. The diode is connected in series with a 2Ω resistor across the terminals of a source having electromotive force (e.m.f.) 3V and internal resistance 1Ω , as shown.

A high-resistance voltmeter is connected across the diode and resistor.

Which row gives the readings of the voltmeter for the two ways of connecting the diode?

	forward biased	reverse biased
A	1V	3V
B	2V	0V
C	2V	3V
D	3V	0V

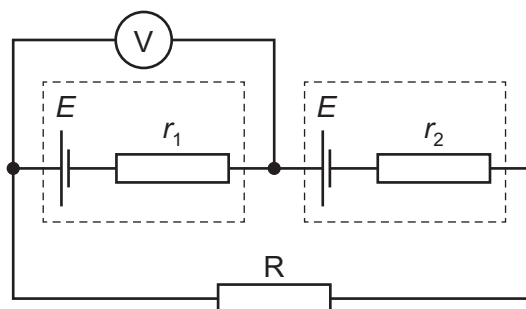


147 9702/13/O/N/17/Q36

Two cells, each with electromotive force (e.m.f.) E , but different internal resistances r_1 and r_2 , are connected in series to a resistor R . The reading on the voltmeter is 0V.

What is the resistance of R ?

- A 0 B $r_1 - r_2$
C $r_1 + r_2$ D $\frac{r_1 r_2}{r_1 + r_2}$

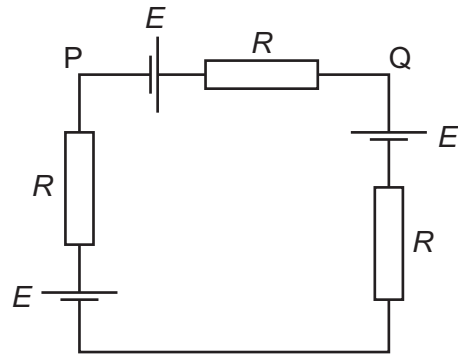


148 9702/13/O/N/17/Q37

Three identical cells each have electromotive force (e.m.f.) E and negligible internal resistance. The cells are connected to three identical resistors, each of resistance R , as shown.

What is the potential difference between P and Q?

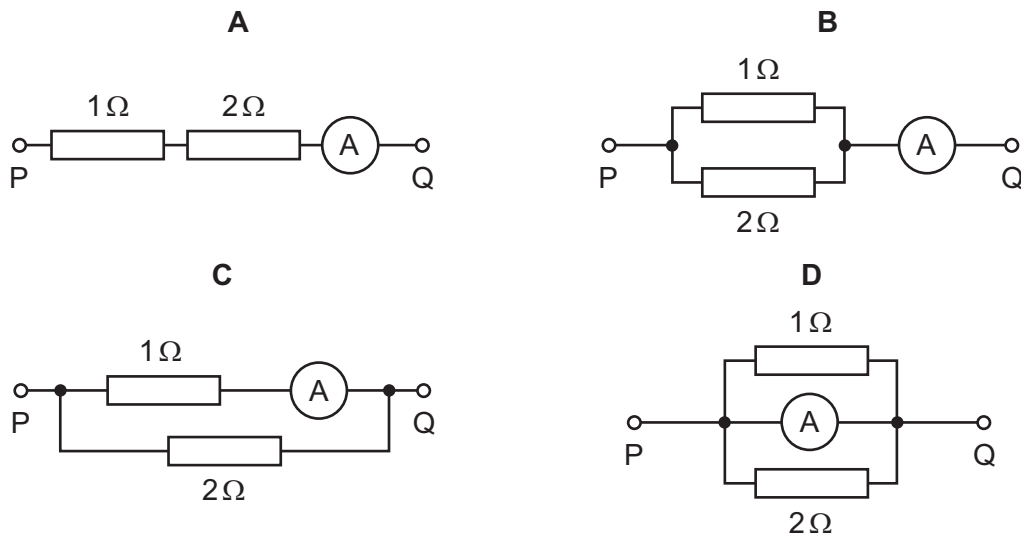
- A 0 B $\frac{E}{3}$
C $\frac{2E}{3}$ D E



149 9702/13/O/N/17/Q38

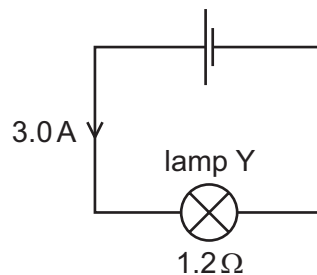
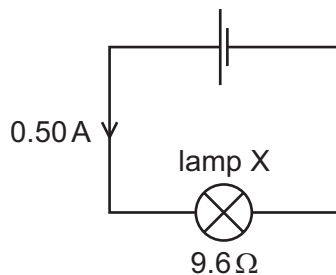
In each arrangement of resistors, the ammeter has a resistance of 2Ω .

Which arrangement gives the largest reading on the ammeter when the same potential difference is applied between points P and Q?



150 9702/11/M/J/18/Q31

The circuit diagrams show two lamps X and Y each connected to a cell. The current in lamp X is 0.50 A and its resistance is 9.6Ω . The current in lamp Y is 3.0 A and its resistance is 1.2Ω .



What is the ratio $\frac{\text{power in lamp X}}{\text{power in lamp Y}}$?

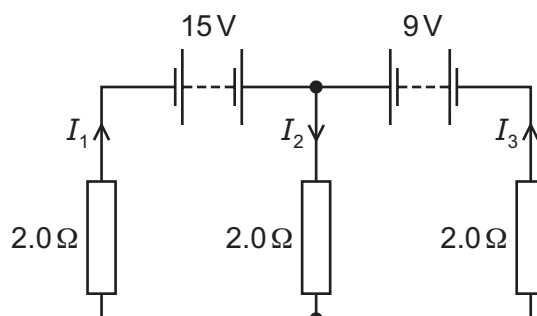
- A 0.22 B 0.75 C 1.3 D 4.5

151 9702/11/M/J/18/Q34

In the circuit shown, the batteries have negligible internal resistance.

What are the values of the currents I_1 , I_2 and I_3 ?

	I_1/A	I_2/A	I_3/A
A	-5.5	1.0	6.5
B	0.5	4.0	3.5
C	3.5	4.0	0.5
D	6.5	1.0	-5.5



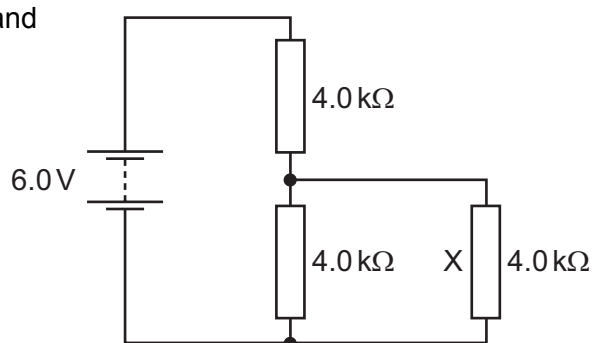
152 9702/11/M/J/18/Q35

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected to three resistors as shown.

Each resistor has a resistance of 4.0 kΩ.

What is the current in resistor X?

- A** 0.25 mA **B** 0.50 mA
C 0.75 mA **D** 1.0 mA



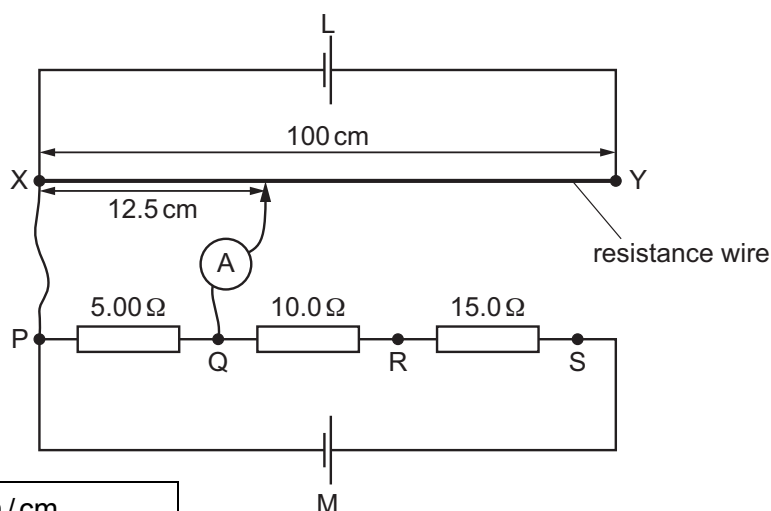
153 9702/11/M/J/18/Q36

A uniform resistance wire XY of length 100 cm is connected in series with a cell L. Another cell M is connected in series with resistors of resistances 5.00 Ω, 10.0 Ω and 15.0 Ω.

The potential difference (p.d.) between P and Q is balanced against 12.5 cm of the resistance wire, so that the ammeter reads zero.

The p.d. across the other resistors is then balanced against other lengths of the resistance wire.

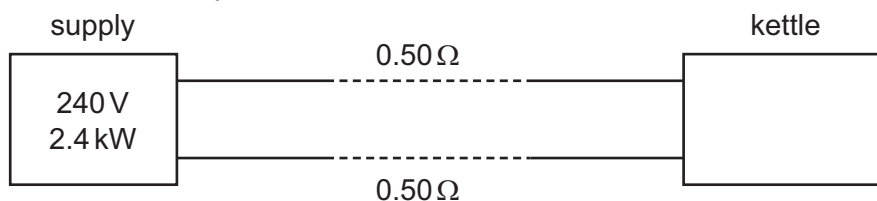
Which balanced lengths of resistance wire correspond to the connection points given in the table?



connection points	balanced length / cm			
	A	B	C	D
Q and R	12.5	25.0	25.0	25.0
Q and S	62.5	62.5	75.0	62.5
P and R	37.5	37.5	37.5	12.5

154 9702/12/M/J/18/Q32

The power output of an electrical supply is 2.4 kW at a potential difference (p.d.) of 240 V. The two wires between the supply and a kettle each have a resistance of $0.50\ \Omega$, as shown.

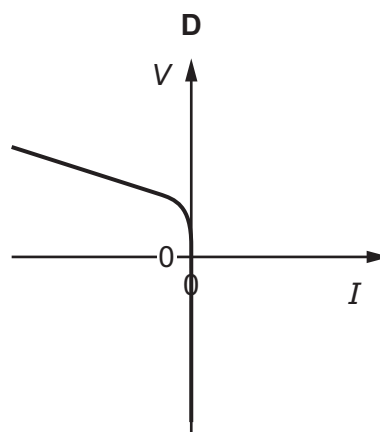
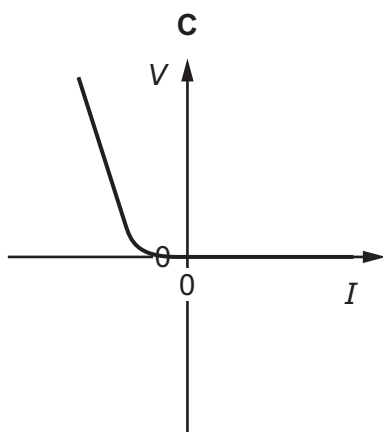
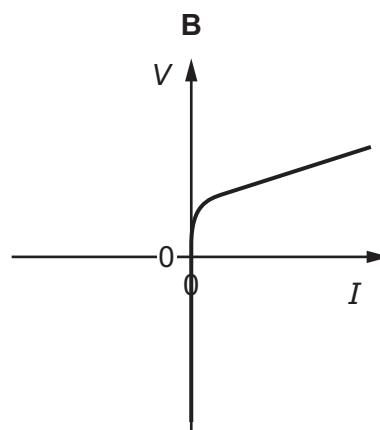
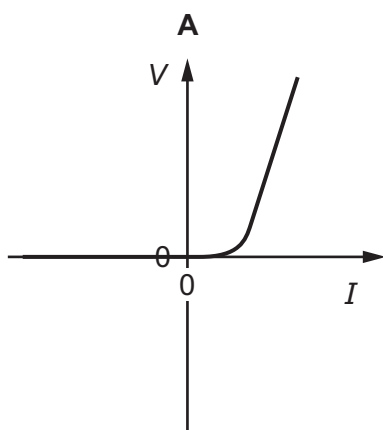


What is the power supplied to the kettle and what is the p.d. across the kettle?

	power / kW	p.d. / V
A	2.3	230
B	2.3	235
C	2.4	230
D	2.4	235

155 9702/12/M/J/18/Q33

Which graph shows the variation of voltage V with current I for a semiconductor diode?

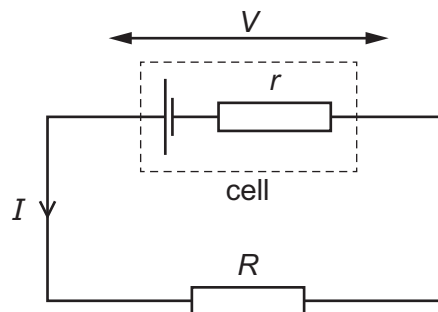


156 9702/12/M/J/18/Q35

A cell of constant electromotive force drives a current I through an external resistor of resistance R . The terminal potential difference (p.d.) across the cell is V .

When the internal resistance r of the cell increases, what is the effect on V and on I ?

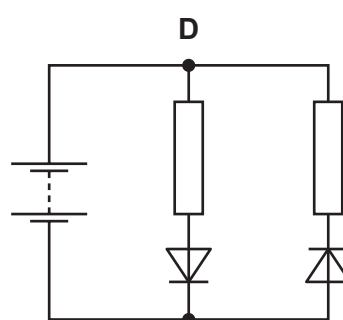
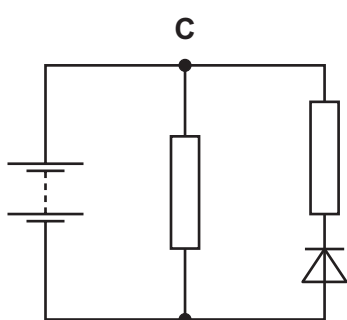
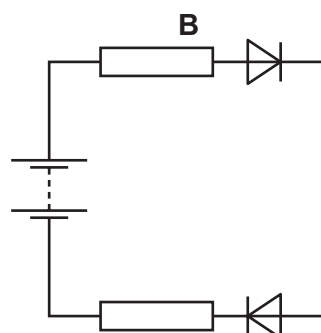
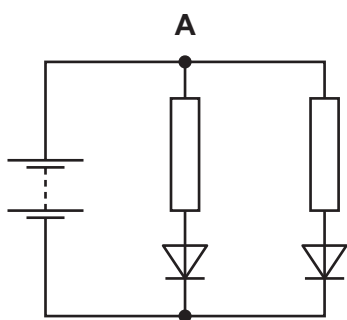
	V	I
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases



157 9702/12/M/J/18/Q36

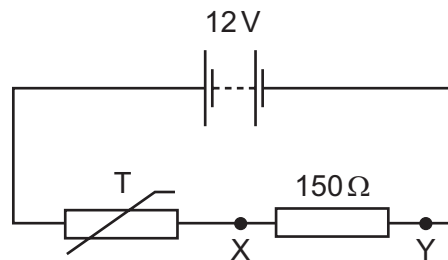
In the circuits shown, the batteries are identical and all have negligible internal resistance. All of the resistors have the same resistance. The diodes have zero resistance when conducting and infinite resistance when not conducting.

In which circuit is the current in the battery greatest?



158 9702/12/M/J/18/Q37

A thermistor is an electrical component with a resistance that varies with temperature. A thermistor T is used in a fire alarm system. The alarm is triggered when the potential difference between X and Y is 4.5 V .

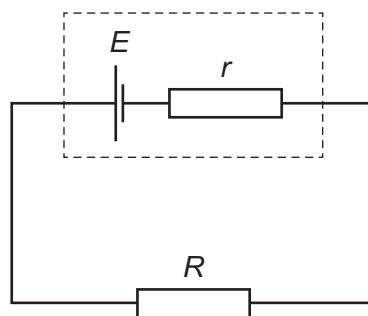


What is the resistance of T when the alarm is triggered?

- A $90\ \Omega$ B $150\ \Omega$ C $250\ \Omega$ D $400\ \Omega$

159 9702/13/M/J/18/Q31

A cell of electromotive force (e.m.f.) E and internal resistance r is connected to an external resistor of resistance R , as shown.



What is the power dissipated in the external resistor?

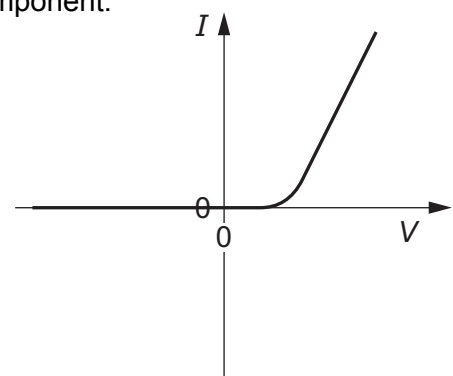
- A $\frac{E^2(R+r)}{R^2}$ B $\frac{E^2R}{(R+r)^2}$ C $\frac{E^2(R+r)}{r^2}$ D $\frac{E^2r}{(R+r)^2}$

160 9702/13/M/J/18/Q32

The graph shows the I - V characteristic of an electrical component.

What is the component?

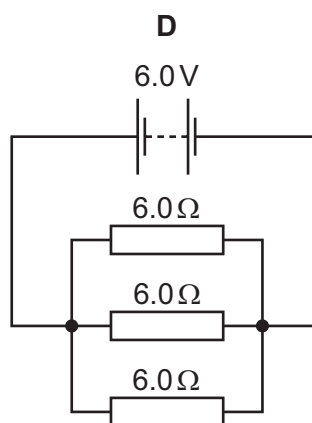
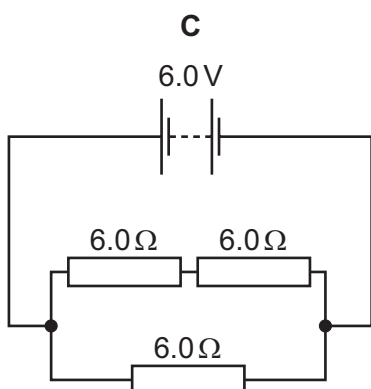
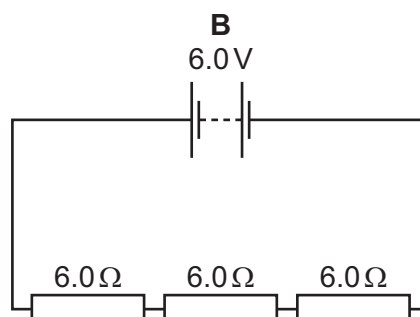
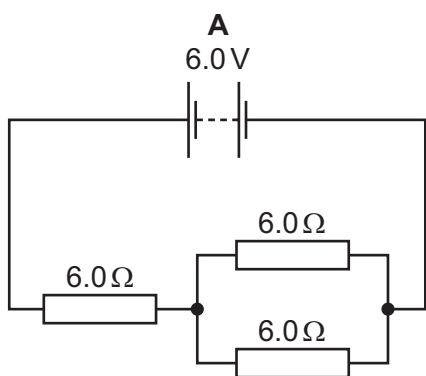
- A a filament lamp
B a metallic conductor at constant temperature
C a resistor
D a semiconductor diode



161 9702/13/M/J/18/Q34

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected to three resistors each of resistance $6.0\ \Omega$.

Which circuit will produce a current through the battery of 0.67 A ?



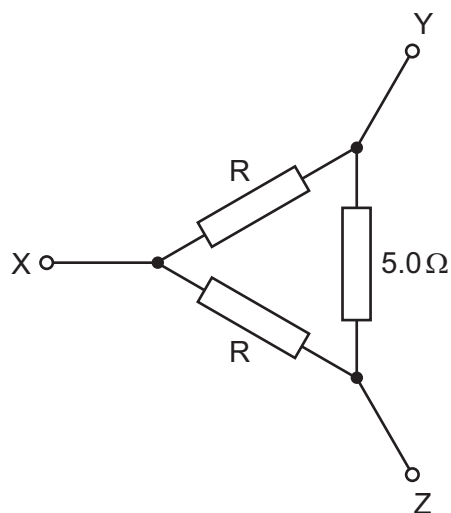
162 9702/13/M/J/18/Q35

The diagram shows a network of three resistors. Two of these, marked R , are identical. The other resistor has a resistance of $5.0\ \Omega$.

The resistance between Y and Z is found to be $2.5\ \Omega$.

What is the resistance between X and Y ?

- | | | | |
|----------|----------------|----------|----------------|
| A | $0.30\ \Omega$ | B | $0.53\ \Omega$ |
| C | $1.9\ \Omega$ | D | $3.3\ \Omega$ |



163 9702/13/M/J/18/Q36

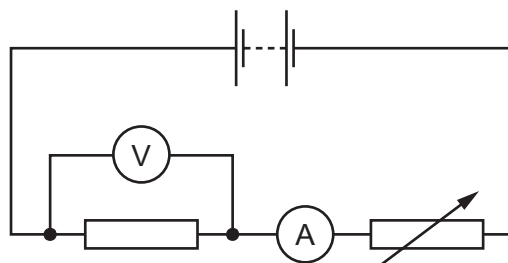
The diagram shows a battery, a fixed resistor, an ammeter and a variable resistor connected in series.

A voltmeter is connected across the fixed resistor.

The resistance of the variable resistor is reduced.

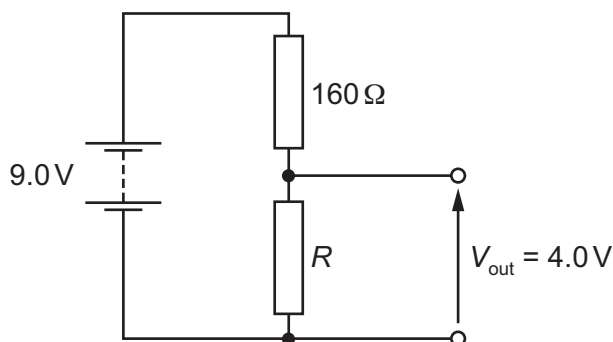
Which row describes the changes in the readings of the ammeter and of the voltmeter?

	ammeter	voltmeter
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase



164 9702/13/M/J/18/Q37

The circuit diagram shows a battery of electromotive force (e.m.f.) 9.0 V and negligible internal resistance. It is connected to two resistors of resistances $160\ \Omega$ and R . The output potential difference V_{out} is 4.0 V .

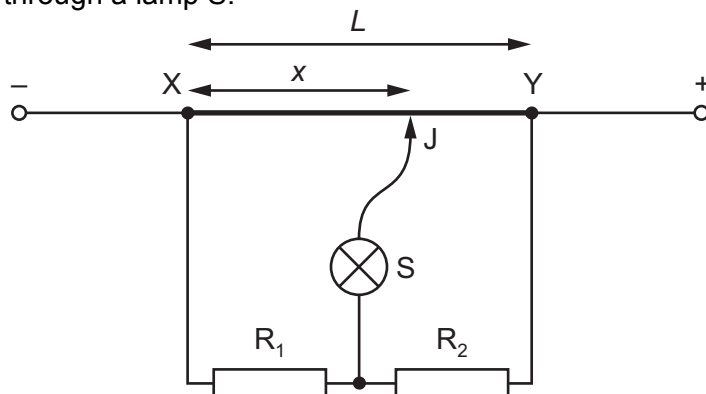


What is the resistance R ?

- A** $32\ \Omega$ **B** $49\ \Omega$ **C** $71\ \Omega$ **D** $128\ \Omega$

165 9702/13/M/J/18/Q38

In the circuit shown, XY is a length L of uniform resistance wire. A potential difference is applied across XY. R_1 and R_2 are unknown resistors. J is a sliding contact that joins the junction of R_1 and R_2 to points on XY through a lamp S.



J is moved along XY to a point at which the lamp is off. This point is at a distance x from X. The potential difference across R_1 is V_1 and the potential difference across R_2 is V_2 .

What is the value of the ratio $\frac{V_1}{V_2}$?

- A $\frac{L}{x}$ B $\frac{x}{L}$ C $\frac{L-x}{x}$ D $\frac{x}{L-x}$

166 9702/12/F/M/18/Q33

A resistor has resistance R . When the potential difference across the resistor is V , the current in the resistor is I . The power dissipated in the resistor is P . Work W is done when charge Q flows through the resistor.

What is **not** a valid relationship between these variables?

- A $I = \frac{P}{V}$ B $Q = \frac{W}{V}$ C $R = \frac{P}{I^2}$ D $R = \frac{V}{P}$

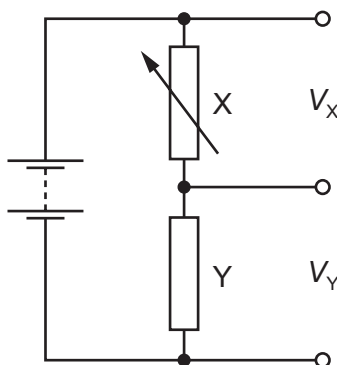
167 9702/12/F/M/18/Q36

A potential divider circuit is constructed with one variable resistor X and one fixed resistor Y, as shown.

The potential difference across resistor X is V_X and the potential difference of resistor Y is V_Y .

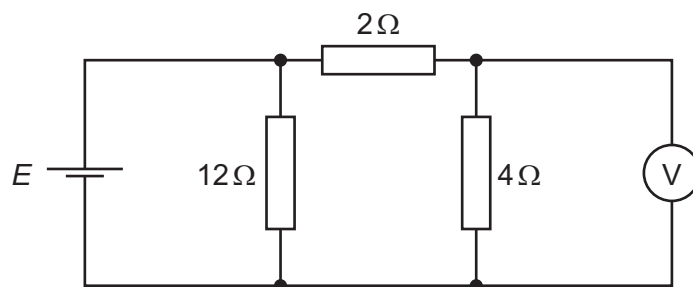
As the resistance of X is increased, what happens to V_X and to V_Y ?

	V_X	V_Y
A	falls	rises
B	falls	stays the same
C	rises	falls
D	rises	stays the same



168 9702/12/F/M/18/Q37

A cell of electromotive force (e.m.f.) E and negligible internal resistance is connected into a circuit, as shown.



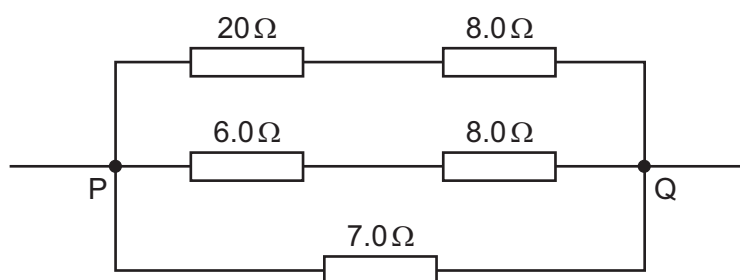
The voltmeter has a very high resistance and reads a potential difference V_{out} .

What is the ratio $\frac{V_{\text{out}}}{E}$?

- A** $\frac{1}{6}$ **B** $\frac{1}{3}$ **C** $\frac{1}{2}$ **D** $\frac{2}{3}$

169 9702/12/F/M/18/Q38

Five resistors are connected as shown.

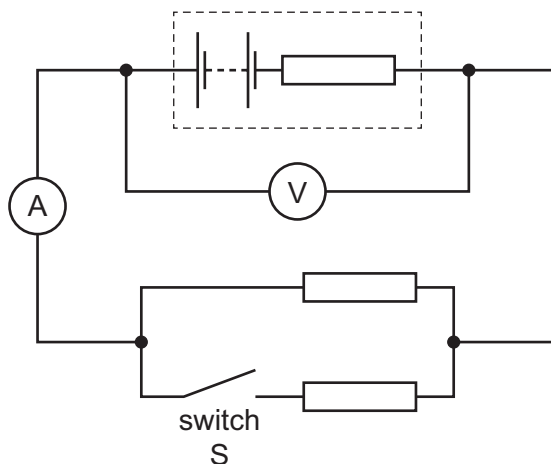


What is the total resistance between points P and Q?

- A** $0.25\ \Omega$ **B** $0.61\ \Omega$ **C** $4.0\ \Omega$ **D** $16\ \Omega$

170 9702/11/O/N/18/Q36

A battery, with internal resistance, is connected to a parallel arrangement of two resistors and a switch S, as shown.



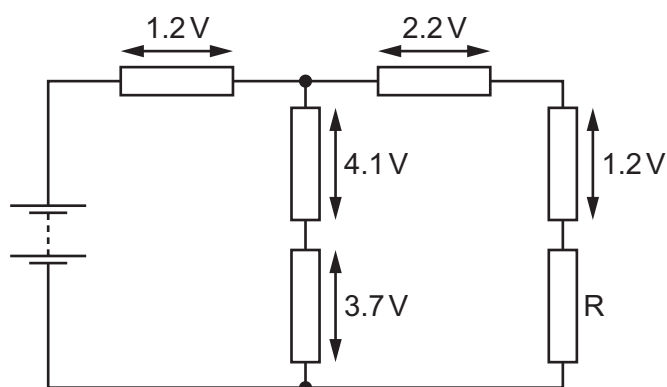
Initially switch S is open.

What happens to the voltmeter and ammeter readings when switch S is closed?

	voltmeter reading	ammeter reading
A	decreases	increases
B	decreases	decreases
C	increases	increases
D	increases	decreases

171 9702/11/O/N/18/Q37

A battery is connected to a network of six resistors, as shown.



The potential differences across five of the resistors are labelled on the diagram.

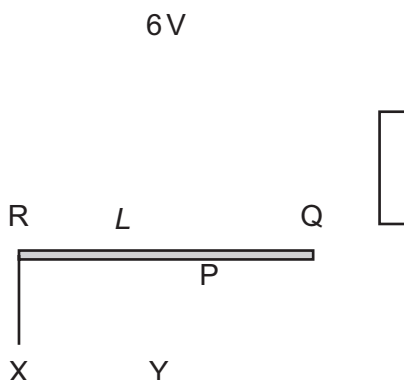
What is the potential difference across resistor R?

- A** 4.4 V **B** 4.6 V **C** 6.6 V **D** 11.2 V

172 9702/11/O/N/18/Q38

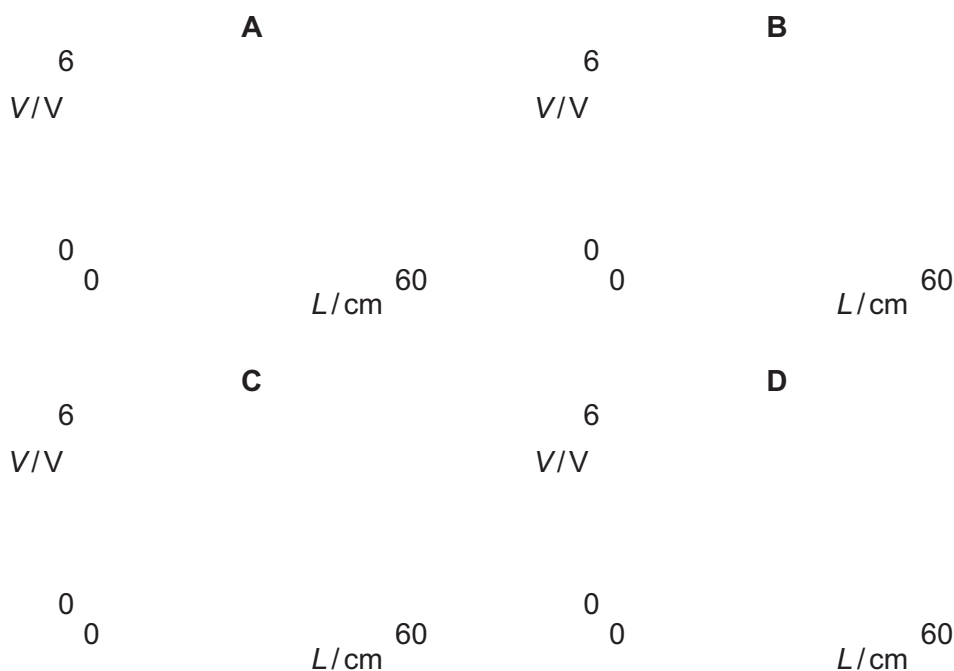
The diagram shows a battery of electromotive force (e.m.f.) 6 V, connected in series with a resistor and a uniform resistance wire RQ of length 60 cm.

The resistance of RQ is equal to the resistance of the resistor.



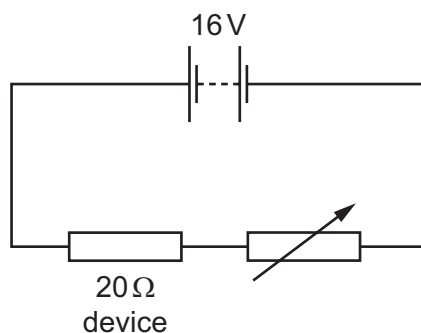
Terminal X is connected to fixed point R. Terminal Y is connected to point P, a connection that may be made at any position along the wire. L is the distance between R and P.

Which graph shows the variation with L of the potential difference (p.d.) V across XY?



173 9702/12/O/N/18/Q34

An electrical device of fixed resistance $20\ \Omega$ is connected in series with a variable resistor and a battery of electromotive force (e.m.f.) 16 V and negligible internal resistance.

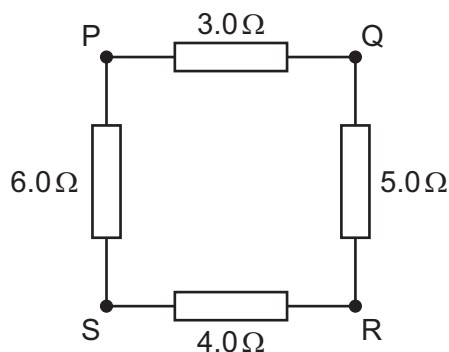


What is the resistance of the variable resistor when the power dissipated in the electrical device is 4.0 W ?

- A** $16\ \Omega$ **B** $36\ \Omega$ **C** $44\ \Omega$ **D** $60\ \Omega$

174 9702/12/O/N/18/Q36

A battery of negligible internal resistance may be connected between any two points P, Q, R and S of the network of resistors shown.



Which connections will give the largest current and the smallest current in the battery?

	largest current	smallest current
A	PQ	PR
B	PQ	QS
C	RS	PR
D	RS	QS

175 9702/12/O/N/18/Q37

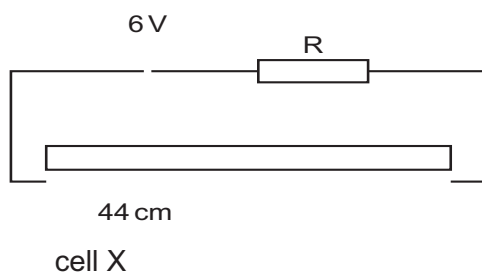
Kirchhoff's second law is a consequence of a basic principle.

What is this principle?

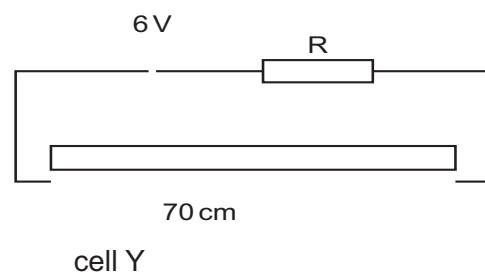
- A The charge flowing in an electric circuit is conserved.
- B The energy in an electric circuit is conserved.
- C The sum of the electric currents entering a point in an electrical circuit is equal to the sum of the electric currents leaving that point.
- D The sum of the potential differences in a circuit is equal to the sum of the products of the current and resistance.

176 9702/12/O/N/18/Q38

Two cells are investigated using a potentiometer. At the balance point, cell X gives a reading of 44 cm and cell Y gives a reading of 70 cm.



galvanometer



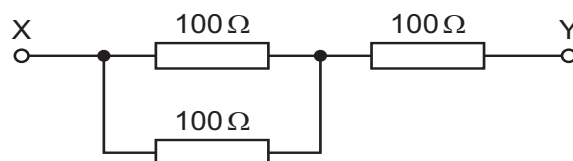
galvanometer

Which statement is **not** correct?

- A A potentiometer balance point results in zero current through the galvanometer.
- B At the balance point, the current through resistor R in both circuits is the same.
- C The electromotive force (e.m.f.) of cell X is larger than that of cell Y.
- D The value of the e.m.f. of each of the cells X and Y is less than 6 V.

177 9702/13/O/N/18/Q33

Three resistors are to be connected into a circuit with the arrangement shown.



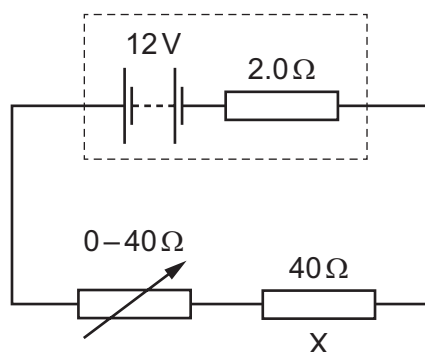
The power in any resistor must not be greater than 4.0 W.

What is the maximum voltage across XY?

- A 24 V
- B 30 V
- C 40 V
- D 60 V

178 9702/13/O/N/18/Q35

A resistor X of resistance $40\ \Omega$ and a variable resistor are connected to a battery of electromotive force (e.m.f.) 12 V and internal resistance $2.0\ \Omega$, as shown.



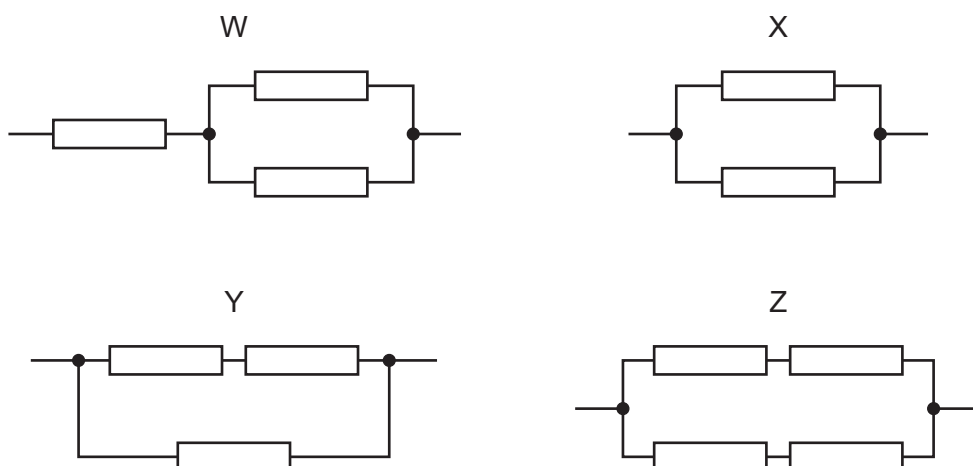
The resistance of the variable resistor is changed from 0 to $40\ \Omega$.

What is the change in power dissipated in resistor X?

- A** 2.4 W **B** 2.7 W **C** 3.6 W **D** 5.6 W

179 9702/13/O/N/18/Q36

All the resistors shown in the resistor networks W, X, Y and Z have the same resistance.

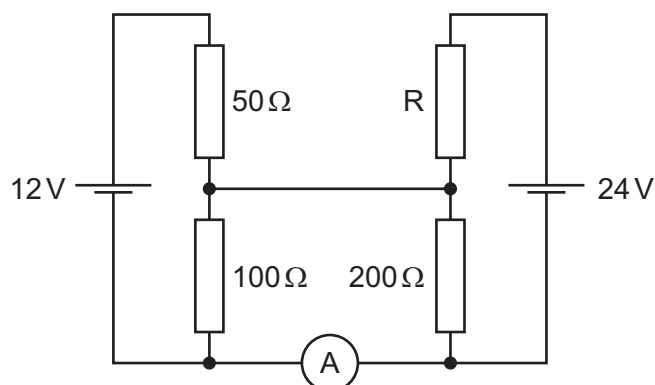


Which list gives the networks in order of increasing total resistance?

- A** $W \rightarrow Z \rightarrow Y \rightarrow X$ **C** $X \rightarrow Y \rightarrow W \rightarrow Z$
B $X \rightarrow W \rightarrow Y \rightarrow Z$ **D** $X \rightarrow Y \rightarrow Z \rightarrow W$

180 9702/13/O/N/18/Q37

In the circuit shown, the ammeter reading is zero.



What is the resistance of resistor R?

- A 100 Ω B 200 Ω C 400 Ω D 600 Ω

181 9702/11/M/J/19/Q35

When a battery is connected to a resistor, the battery gradually becomes warm. This causes the internal resistance of the battery to increase whilst its electromotive force (e.m.f.) stays unchanged.

As the internal resistance of the battery increases, how do the terminal potential difference and the output power change, if at all?

	terminal potential difference	output power
A	decreases	decreases
B	decreases	unchanged
C	unchanged	decreases
D	unchanged	unchanged

182 9702/11/M/J/19/Q36

A cell is connected to a resistor of resistance 3.00 Ω . The current in the resistor is 1.00 A.

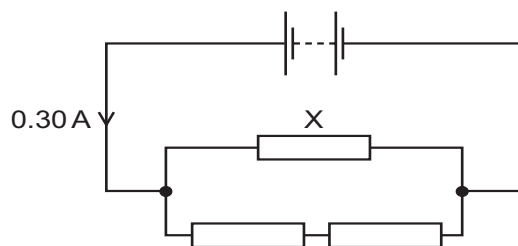
A second identical resistor is added in parallel. The current becomes 1.93 A.

What are the e.m.f. E and internal resistance r of the cell?

	E/V	r/Ω
A	0.113	3.11
B	3.04	0.0358
C	3.11	0.113
D	9.34	6.34

183 9702/11/M/J/19/Q37

A battery with negligible internal resistance is connected to three resistors, as shown.



All three resistors have the same resistance.

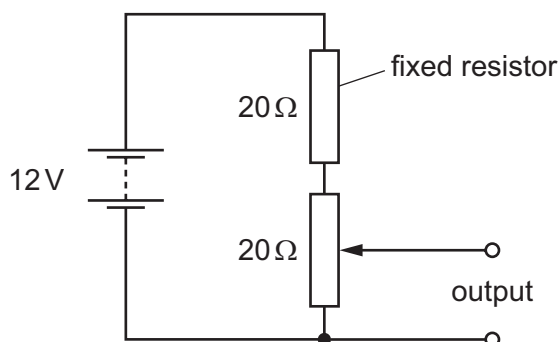
The current in the battery is 0.30 A.

What is the current in resistor X?

- A** 0.10 A **B** 0.15 A **C** 0.20 A **D** 0.30 A

184 9702/11/M/J/19/Q38

The diagram shows a potentiometer and a fixed resistor connected across a 12 V battery of negligible internal resistance.



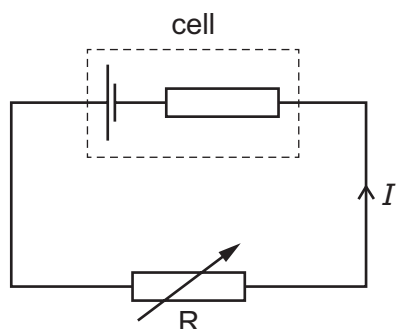
The fixed resistor and the potentiometer each have resistance $20\ \Omega$. The circuit is designed to provide a variable output voltage.

What is the range of output voltages?

- A** 0–6 V **B** 0–12 V **C** 6–12 V **D** 12–20 V

185 9702/12/M/J/19/Q35

A cell with internal resistance is connected to a variable resistor R as shown.



The resistance of R is gradually decreased.

How do the current I and the terminal potential difference across the cell change?

	current I	terminal potential difference across cell
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

186 9702/12/M/J/19/Q36

Kirchhoff's first law states that the sum of the currents entering a junction in a circuit is equal to the sum of the currents leaving it.

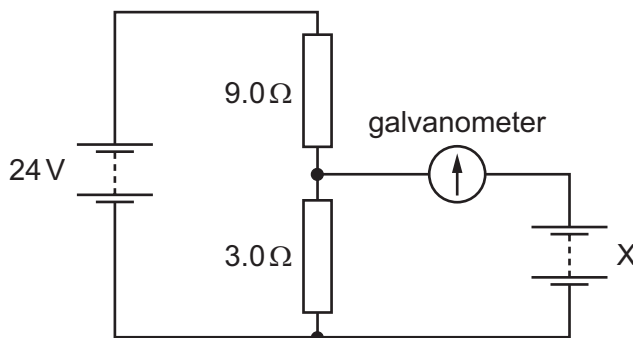
The law is based on the conservation of a physical quantity.

What is this physical quantity?

- | | |
|-----------------|-------------------|
| A charge | C mass |
| B energy | D momentum |

187 9702/12/M/J/19/Q37

A circuit contains two batteries, each of negligible internal resistance, and two resistors as shown.



The galvanometer has a current reading of zero.

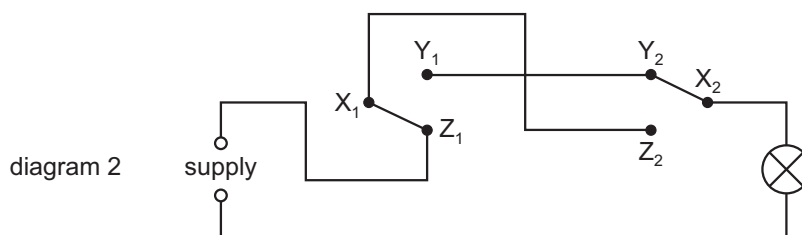
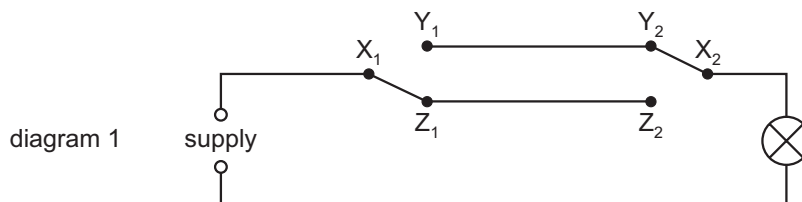
What is the electromotive force (e.m.f.) of battery X?

- A** 6.0 V **B** 8.0 V **C** 16.0 V **D** 18.0 V

188 9702/13/M/J/19/Q34

Diagram 1 shows a lamp connected to a supply through two switches.

During repairs, an electrician mistakenly reverses the connections X_1 and Z_1 , so that Z_1 is connected to the supply and X_1 to the other switch at Z_2 , as shown in diagram 2.



Which switch positions will now light the lamp?

A	X_1 to Y_1	X_2 to Y_2
B	X_1 to Y_1	X_2 to Z_2
C	X_1 to Z_1	X_2 to Y_2
D	X_1 to Z_1	X_2 to Z_2

189 9702/13/M/J/19/Q35

A wire supplying a shower heater with a current of 35 A has a resistance of 25 m Ω .

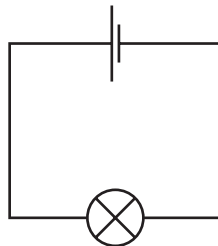
What is the power dissipated in the wire?

- A** 31 W **B** 49 W **C** 31 kW **D** 49 kW

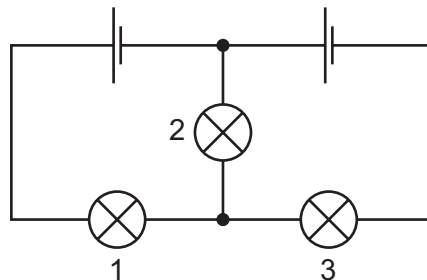
190 9702/13/M/J/19/Q36

A student has a set of identical cells and identical lamps. The cells have negligible internal resistance.

A lamp connected to a cell lights with normal brightness.



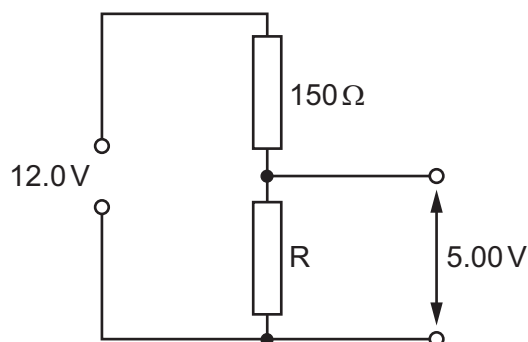
What happens when the student connects the lamps and the cells as shown?



- A** All three lamps light with normal brightness.
B Only lamp 2 lights with normal brightness.
C Only lamps 1 and 3 light with normal brightness.
D None of the lamps light with normal brightness.

191 9702/13/M/J/19/Q37

A potential divider circuit is shown.

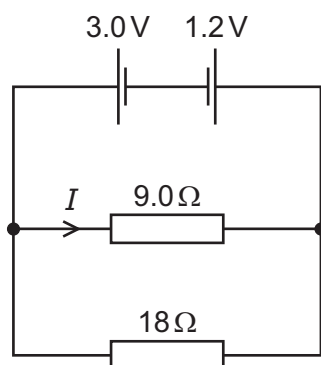


What is the resistance of resistor R in the potential divider circuit?

- A** 62.5 Ω **B** 107 Ω **C** 210 Ω **D** 360 Ω

192 9702/13/M/J/19/Q38

Two cells of electromotive force (e.m.f.) 3.0 V and 1.2 V and negligible internal resistance are connected to resistors of resistance 9.0 Ω and 18 Ω as shown.



What is the current I in the 9.0 Ω resistor?

- A** 0.10 A **B** 0.20 A **C** 0.30 A **D** 0.47 A

193 9702/11/O/N/19/Q34

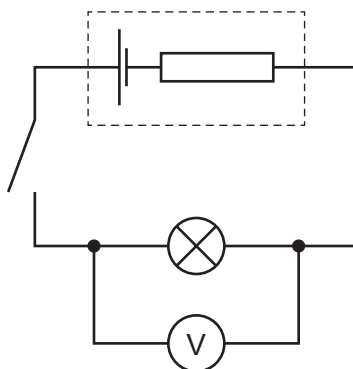
Kirchhoff's two laws for electric circuits can be derived by using conservation laws.

On which conservation laws do Kirchhoff's laws depend?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	current
B	charge	energy
C	current	mass
D	energy	current

194 9702/11/O/N/19/Q35

The diagram shows a circuit.

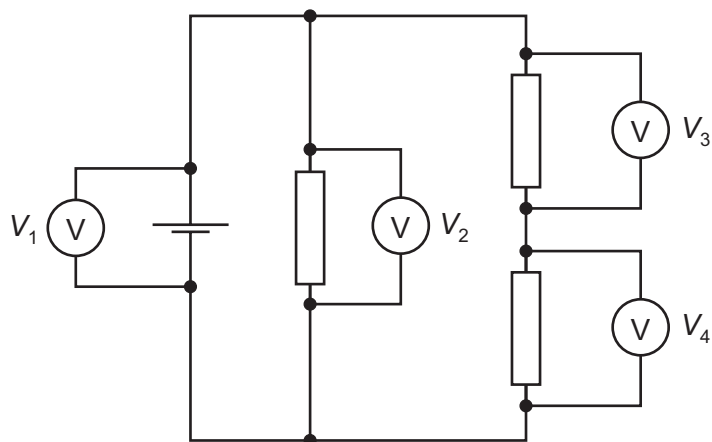


Which statement about the circuit is **not** correct?

- A** Electromotive force is the energy transferred per unit charge.
- B** Energy is transferred from chemical potential energy in the cell to other forms when the switch is closed.
- C** The electromotive force of the cell is greater than the terminal potential difference when the switch is closed.
- D** When the switch is open, the voltmeter measures the electromotive force of the cell.

195 9702/11/O/N/19/Q36

The diagram shows a circuit containing four voltmeters. The readings on the voltmeters are V_1 , V_2 , V_3 and V_4 . All the readings are positive.

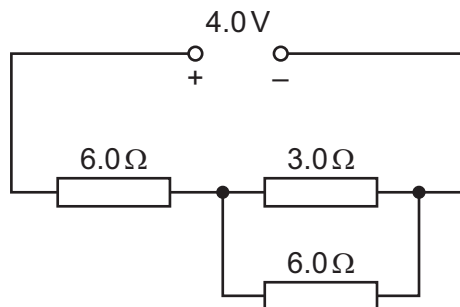


Which equation relating the voltmeter readings is correct?

- | | |
|----------------------------------|--------------------------------|
| A $V_1 = V_2 + V_4$ | C $V_2 + V_3 = V_4$ |
| B $V_1 = V_2 + V_3 + V_4$ | D $V_3 + V_4 - V_2 = 0$ |

196 9702/11/O/N/19/Q37

A network consists of a $3.0\ \Omega$ resistor and two $6.0\ \Omega$ resistors, as shown.



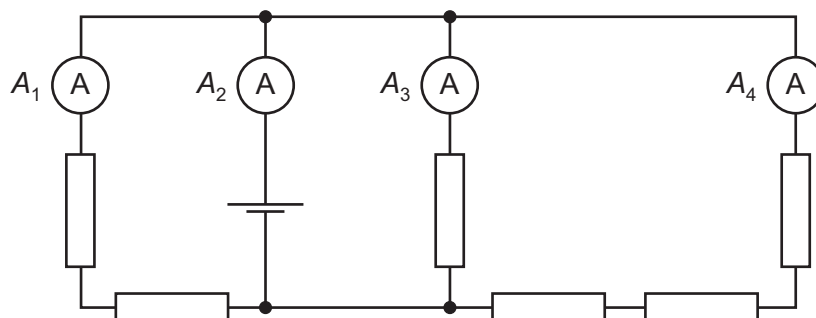
The potential difference (p.d.) across the network is 4.0 V .

What is the current through the $3.0\ \Omega$ resistor?

- | | | | |
|--------------------------|--------------------------|--------------------------|-------------------------|
| A 0.17 A | B 0.25 A | C 0.33 A | D 1.3 A |
|--------------------------|--------------------------|--------------------------|-------------------------|

197 9702/11/O/N/19/Q38

In the circuit shown, all the resistors are identical and all the ammeters have negligible resistance.



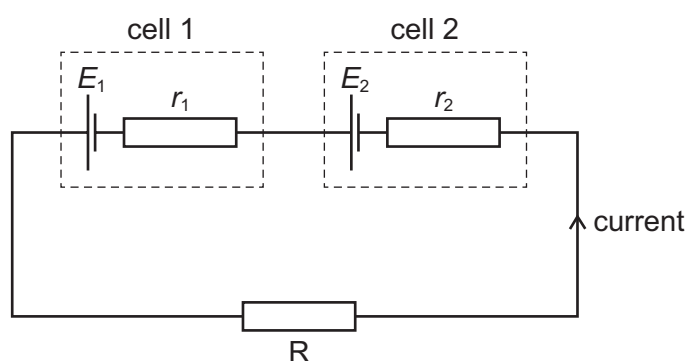
The reading A_1 is 0.6 A.

What are the readings on the other ammeters?

	A_2 / A	A_3 / A	A_4 / A
A	1.0	0.3	0.1
B	1.4	0.6	0.2
C	1.8	0.9	0.3
D	2.2	1.2	0.4

198 9702/12/O/N/19/Q34

Two cells with electromotive forces E_1 and E_2 and internal resistances r_1 and r_2 are connected to a resistor R as shown.



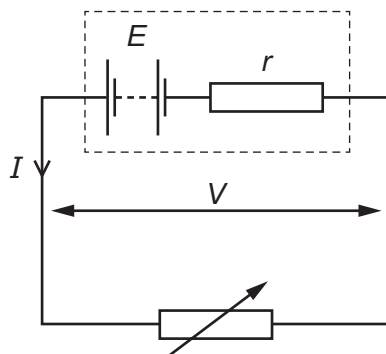
The terminal potential difference across cell 1 is zero.

Which expression gives the resistance of resistor R ?

- A** $\frac{E_2 r_1 - E_1 r_2}{E_1}$ **B** $\frac{E_2 r_1 - E_1 r_2}{E_2}$ **C** $\frac{E_1 r_2 - E_2 r_1}{E_1}$ **D** $\frac{E_1 r_2 - E_2 r_1}{E_2}$

199 9702/12/O/N/19/Q35

A battery has an electromotive force (e.m.f.) E and internal resistance r . The battery delivers a current I to a variable resistor and the potential difference (p.d.) across its terminals is V .



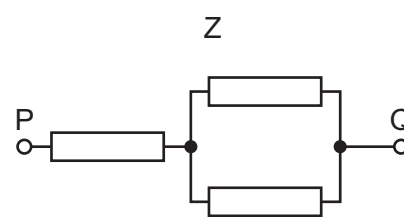
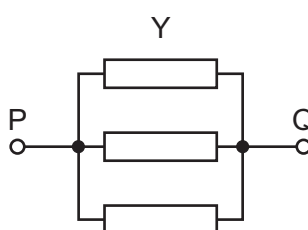
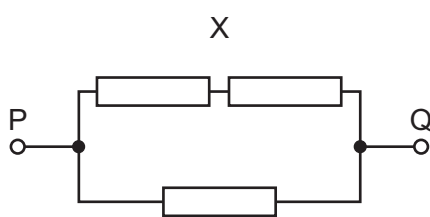
The variable resistor is adjusted so that I increases.

Why does V decrease?

- A The e.m.f. E decreases.
- B The internal resistance r increases.
- C The p.d. across r increases.
- D The resistance of the variable resistor increases.

200 9702/12/O/N/19/Q36

Three identical resistors are connected between terminals P and Q in different networks X, Y and Z as shown.

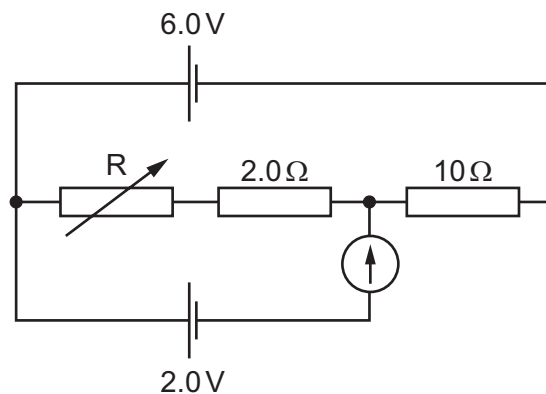


What is the order of increasing combined resistance between P and Q (lowest first)?

- A $X \rightarrow Y \rightarrow Z$
- B $X \rightarrow Z \rightarrow Y$
- C $Y \rightarrow X \rightarrow Z$
- D $Y \rightarrow Z \rightarrow X$

201 9702/12/O/N/19/Q37

The diagram shows a variable resistor R and two fixed resistors connected in series in a circuit to act as a potential divider.



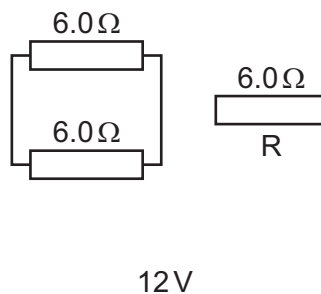
The cell of electromotive force (e.m.f.) 6.0V has negligible internal resistance. A cell of e.m.f. 2.0V and a galvanometer are connected into the potential divider. The resistance of R is varied until the galvanometer reads zero.

What is the resistance of resistor R ?

- A** 3.0Ω **B** 5.0Ω **C** 8.0Ω **D** 18Ω

202 9702/13/O/N/19/Q34

A battery of electromotive force (e.m.f.) 12V and negligible internal resistance is connected to three resistors, each of resistance 6.0Ω , as shown.

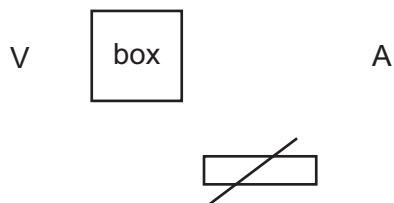


What is the power dissipated in resistor R ?

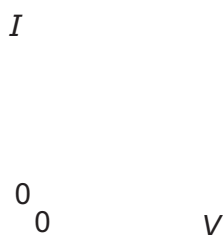
- A** 2.7W **B** 6.0W **C** 11W **D** 24W

203 9702/13/O/N/19/Q35

A box containing two electrical components is connected into a circuit.



The variable resistor is adjusted and measurements are taken to determine the I – V characteristic for the box, as shown.



Which arrangement of two electrical components in the box would create the best fit to the measured I – V characteristic?

- A a filament lamp and a fixed resistor in parallel
- B a filament lamp and a fixed resistor in series
- C a semiconductor diode and a filament lamp in parallel
- D a semiconductor diode and a filament lamp in series

204 9702/13/O/N/19/Q36

A cell of internal resistance $0.5\ \Omega$ is connected to a fixed resistor of resistance $10\ \Omega$.

The resistance of the resistor is changed to $20\ \Omega$.

Which statement is **not** correct?

- A The current in the circuit will halve.
- B The e.m.f. of the cell will remain constant.
- C The power dissipated by the fixed resistor will decrease.
- D The terminal p.d. of the cell will increase.

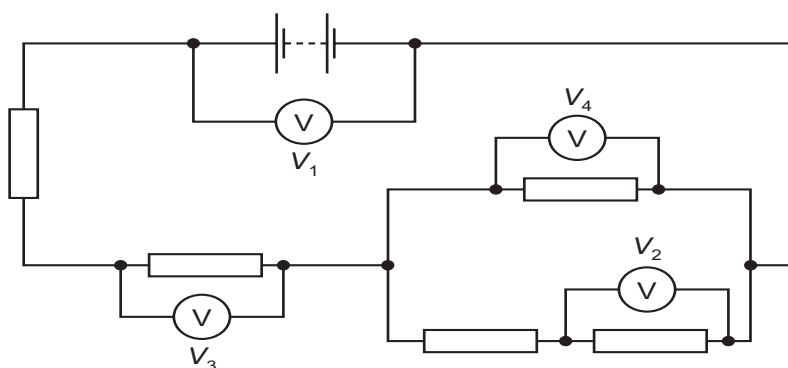
205 9702/13/O/N/19/Q37

Which row correctly describes Kirchhoff's laws?

	Kirchhoff's first law	physics principle applied for first law	Kirchhoff's second law	physics principle applied for second law
A	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy
B	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge
C	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge
D	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy

206 9702/13/O/N/19/Q38

In the circuit shown, all the resistors are identical.



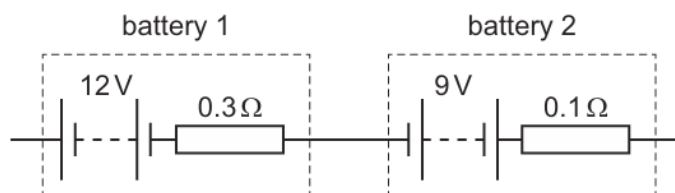
The reading V_1 is 8.0 V and the reading V_2 is 1.0 V.

What are the readings on the other voltmeters?

	V_3 / V	V_4 / V
A	1.5	1.0
B	3.0	2.0
C	4.5	3.0
D	6.0	4.0

207. 9702/12/F/M/20/Q35

Two batteries are connected together, as shown.



Battery 1 has electromotive force (e.m.f.) 12 V and internal resistance $0.3\ \Omega$.

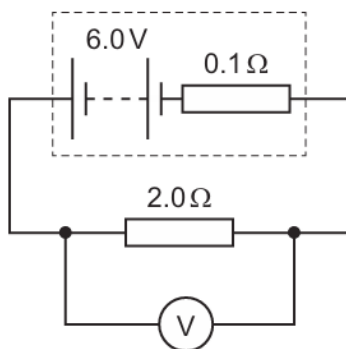
Battery 2 has e.m.f. 9 V and internal resistance $0.1\ \Omega$.

What are the e.m.f. and the internal resistance of a single battery that has the same effect as the combination?

	e.m.f./V	internal resistance/ Ω
A	3	0.2
B	3	0.4
C	21	0.2
D	21	0.4

208. 9702/12/F/M/20/Q36

The diagram shows a circuit.



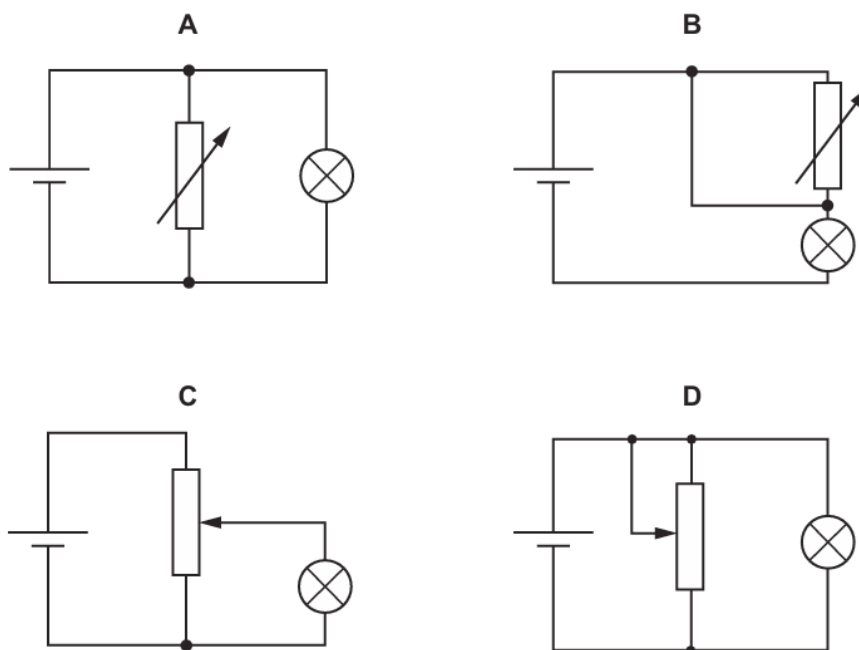
What is the reading on the voltmeter?

- A** 0.3 V **B** 5.7 V **C** 6.0 V **D** 6.3 V

209. 9702/12/F/M/20/Q37

In the circuits shown, the cell has negligible internal resistance.

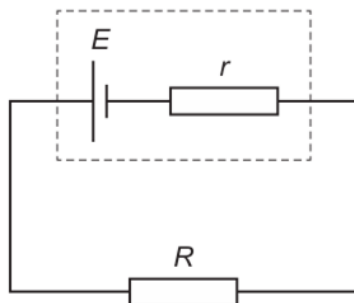
Which diagram shows a potential divider circuit that can vary the potential difference (p.d.) across the lamp?



210. 9702/11/M/J/20/Q36

A cell of electromotive force (e.m.f.) E and internal resistance r is connected to a resistor of resistance R .

A maximum power P can be dissipated by the resistor without overheating.



What is the maximum value of E if the resistor does not overheat?

- A** $R\sqrt{\frac{P}{(R-r)}}$
 B $R\sqrt{\frac{P}{(R+r)}}$
 C $(R-r)\sqrt{\frac{P}{R}}$
 D $(R+r)\sqrt{\frac{P}{R}}$

211. 9702/11/M/J/20/Q37

Three identical resistors can be connected together in four different ways.

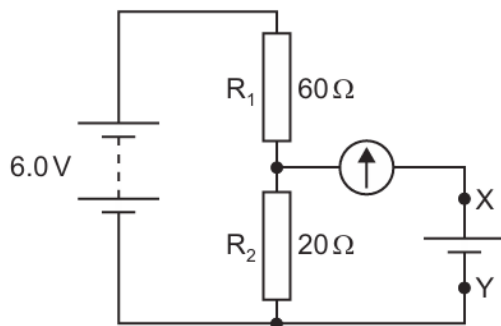
The resistances of two of these combinations are $4.0\ \Omega$ and $9.0\ \Omega$.

What is the resistance of each individual resistor?

- A** $3.0\ \Omega$
 B $6.0\ \Omega$
 C $12\ \Omega$
 D $18\ \Omega$

212. 9702/11/M/J/20/Q38

In the circuit shown, a battery of negligible internal resistance is connected in series with a pair of fixed resistors R_1 and R_2 .



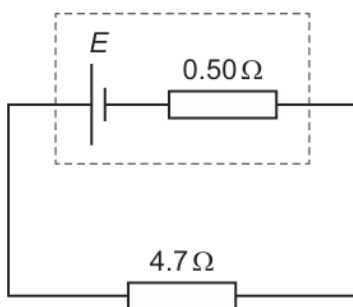
The circuit is to be used to test whether the electromotive force (e.m.f.) of a particular cell is 1.5 V. The cell is connected between terminals X and Y in parallel with R_2 and in series with a galvanometer.

Which statement about the test is correct?

- A** Any non-zero reading on the galvanometer means the cell has an e.m.f. of 1.5 V.
- B** The battery does not need to have an e.m.f. of 6.0 V.
- C** The cell may be connected either way round between X and Y.
- D** The galvanometer does not need a scale calibrated in amperes.

213. 9702/12/M/J/20/Q36

A cell of electromotive force (e.m.f.) E and internal resistance $0.50\ \Omega$ is connected to a resistor of resistance $4.7\ \Omega$.



The maximum power that can be dissipated by the resistor without overheating is 0.50 W.

What is the maximum value of E for the resistor **not** to overheat?

- A** 1.4 V
- B** 1.5 V
- C** 1.7 V
- D** 2.9 V

214. 9702/12/M/J/20/Q37

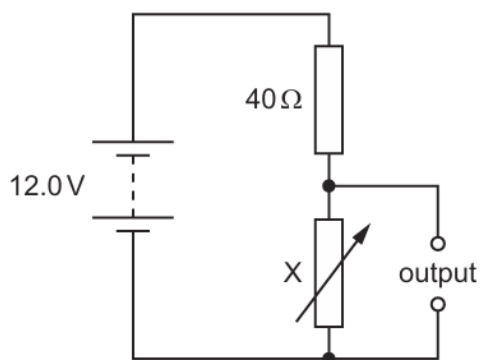
Kirchhoff's first and second laws link to the conservation of physical quantities.

Which quantities do they link to?

	first law	second law
A	charge	energy
B	charge	momentum
C	energy	charge
D	energy	momentum

215. 9702/12/M/J/20/Q38

In the circuit shown, X is a variable resistor whose resistance can be changed from 5.0Ω to 500Ω . The electromotive force (e.m.f.) of the battery is 12.0V . It has negligible internal resistance.

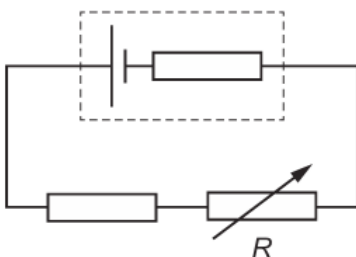


What is the maximum range of values of potential difference across the output?

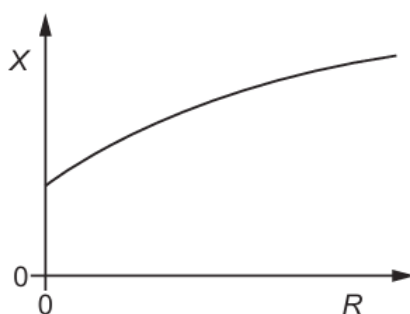
- A** 1.3V to 11.1V
- B** 1.3V to 12.0V
- C** 1.5V to 11.1V
- D** 1.5V to 12.0V

216. 9702/13/M/J/20/Q37

A fixed resistor and a variable resistor are connected in series with a cell that has an internal resistance, as shown.



The graph shows the variation of a quantity X with the resistance R of the variable resistor as R is increased from zero to its maximum value.

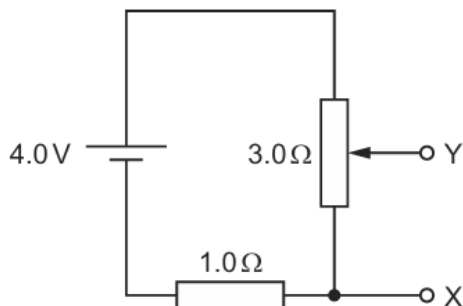


What could X represent?

- A** the current in the circuit
- B** the electromotive force of the cell
- C** the potential difference across the internal resistance
- D** the terminal potential difference across the cell

217. 9702/13/M/J/20/Q38

A cell of electromotive force (e.m.f.) 4.0 V and negligible internal resistance is connected to a fixed resistor of resistance $1.0\ \Omega$ and a potentiometer of maximum resistance $3.0\ \Omega$, as shown.



Which range of potential differences can be obtained between the terminals X and Y ?

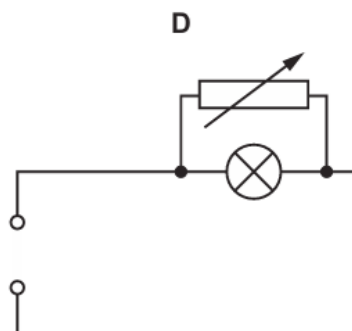
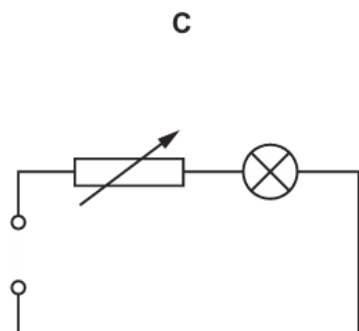
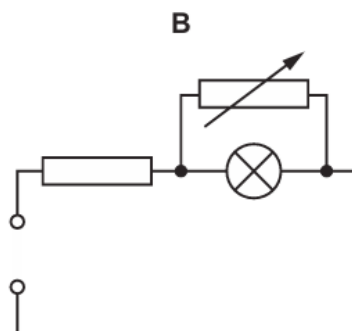
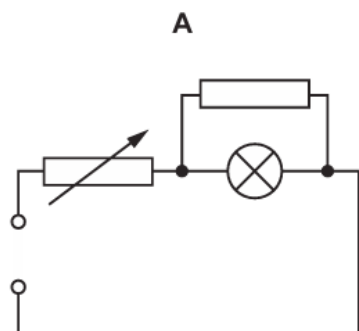
- A** 0 V to 3.0 V
- B** 0 V to 4.0 V
- C** 1.0 V to 3.0 V
- D** 1.0 V to 4.0 V

218. 9702/11/O/N/20/Q35

In the circuits shown, the power supply has an electromotive force (e.m.f.) greater than the normal operating voltage of the lamp. The internal resistance of the power supply is negligible.

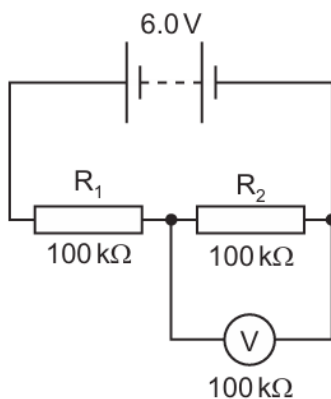
The resistance of the variable resistor is adjusted from zero to its maximum value.

In which circuit could the voltage across the lamp change from zero to its normal operating voltage and **not** exceed its normal operating voltage?



219. 9702/11/O/N/20/Q37

In the circuit shown, the 6.0 V battery has negligible internal resistance. Resistors R_1 and R_2 and the voltmeter each have a resistance of $100\text{ k}\Omega$.

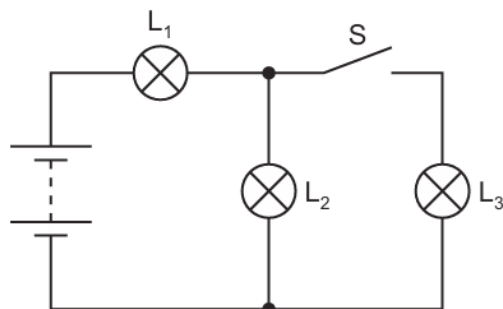


What is the current in the resistor R_2 ?

- A** $20\text{ }\mu\text{A}$ **B** $30\text{ }\mu\text{A}$ **C** $40\text{ }\mu\text{A}$ **D** $60\text{ }\mu\text{A}$

220. 9702/11/O/N/20/Q36

Three identical lamps L_1 , L_2 and L_3 are connected to a battery with negligible internal resistance, as shown.

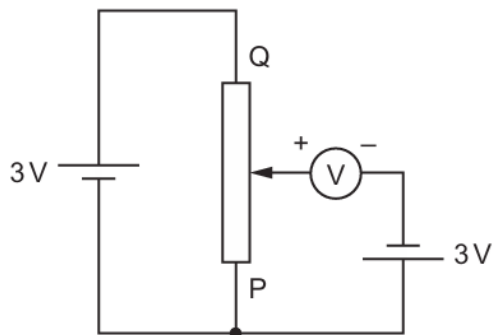


What happens to the brightness of lamps L_1 and L_2 when the switch S is closed?

	lamp L_1	lamp L_2
A	brighter	brighter
B	brighter	dimmer
C	dimmer	brighter
D	dimmer	dimmer

221. 9702/12/O/N/20/Q38

A voltmeter is connected into a circuit with the polarity shown in the diagram.



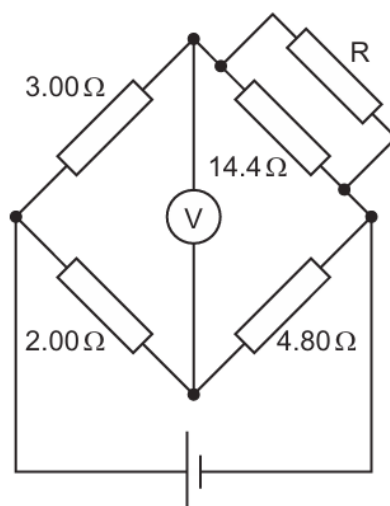
The sliding contact is moved to end P of the potentiometer and then to end Q .

What are the two readings of the voltmeter?

	sliding contact at end P	sliding contact at end Q
A	0V	3V
B	0V	6V
C	3V	3V
D	3V	6V

222. 9702/12/O/N/20/Q37

A cell of negligible internal resistance is connected to a network of resistors and a voltmeter, as shown.



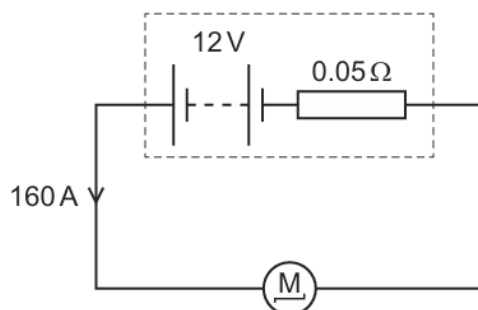
The reading on the voltmeter is zero.

What is the resistance of resistor R?

- A** $1.20\ \Omega$ **B** $1.80\ \Omega$ **C** $7.20\ \Omega$ **D** $14.4\ \Omega$

223. 9702/12/O/N/20/Q36

A car battery has an electromotive force (e.m.f.) of 12 V and an internal resistance of $0.05\ \Omega$. The battery is connected to the starter motor of a car. The current in the motor is 160 A.



What is the terminal p.d. across the battery?

- A** 0 V **B** 4 V **C** 8 V **D** 12 V

224. 9702/13/O/N/20/Q36

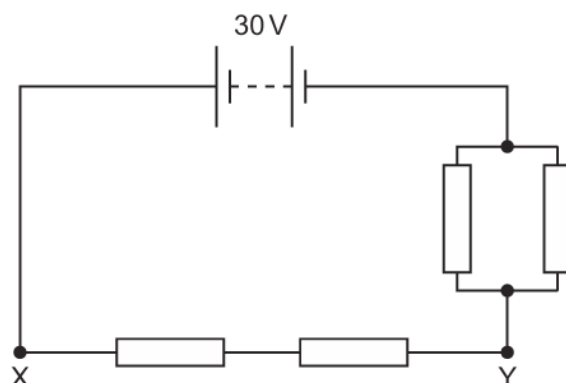
A cell is connected to a fixed resistor. Over a long period of time, the internal resistance of the cell increases.

What is the effect of the increase in internal resistance on the electromotive force (e.m.f.) of the cell and on the power dissipated by the fixed resistor?

	e.m.f.	power dissipated
A	decreases	decreases
B	decreases	no change
C	no change	decreases
D	no change	no change

225. 9702/13/O/N/20/Q37

Four identical resistors are connected in a circuit, as shown.



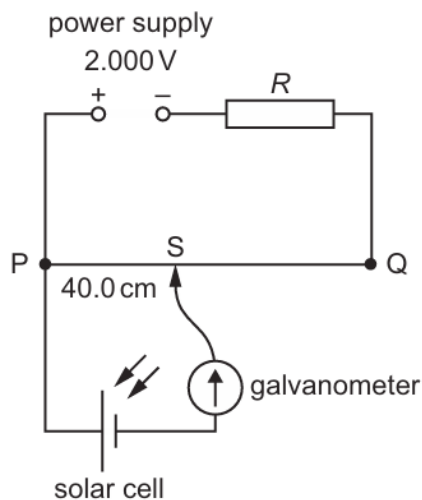
The battery has negligible internal resistance and an e.m.f. of 30 V.

What is the potential difference between the two points X and Y?

- A** 6.0 V **B** 15 V **C** 20 V **D** 24 V

226. 9702/13/O/N/20/Q38

A power supply and a solar cell are compared using the potentiometer circuit shown.



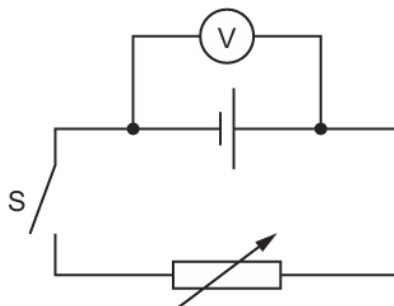
The potentiometer wire PQ is 100.0 cm long and has a resistance of $5.00\ \Omega$. The power supply has an e.m.f. of 2.000 V and the solar cell has an e.m.f. of 5.00 mV.

Which resistance R must be used so that the galvanometer reads zero when $PS = 40.0\text{ cm}$?

- A** $395\ \Omega$ **B** $405\ \Omega$ **C** $795\ \Omega$ **D** $805\ \Omega$

227. 9702/12/F/M/21/Q36

A cell that has internal resistance is connected to a switch S and a variable resistor. A voltmeter is connected between the terminals of the cell, as shown.



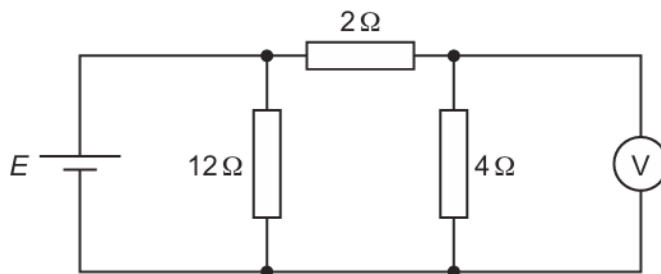
When switch S is open, the voltmeter reads 1.5 V. The switch is then closed and the variable resistor is adjusted to have a resistance of $4.0\ \Omega$. The voltmeter now reads 0.75 V.

What is the internal resistance of the cell?

- A** $1.0\ \Omega$ **B** $2.0\ \Omega$ **C** $4.0\ \Omega$ **D** $8.0\ \Omega$

228. 9702/12/F/M/21/Q37

A cell of electromotive force (e.m.f.) E and negligible internal resistance is connected into a circuit, as shown.



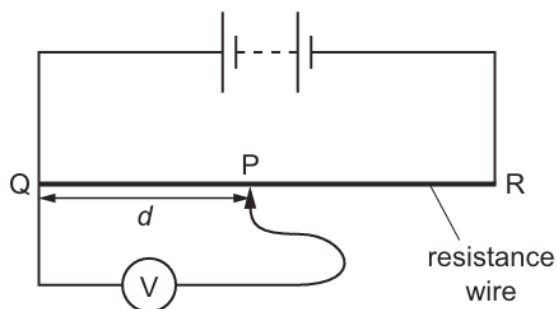
The voltmeter has a very high resistance and reads a potential difference V_{out} .

What is the ratio $\frac{V_{\text{out}}}{E}$?

- A** $\frac{1}{6}$ **B** $\frac{1}{3}$ **C** $\frac{1}{2}$ **D** $\frac{2}{3}$

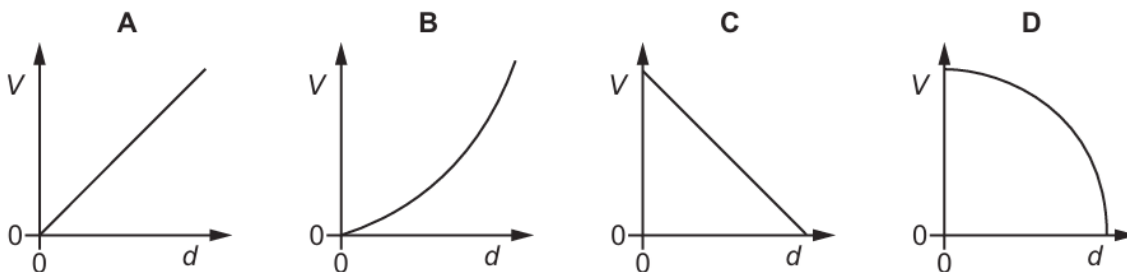
229. 9702/12/F/M/21/Q38

A battery is connected to a potentiometer. The potentiometer consists of a uniform resistance wire and a sliding contact P.



The potential difference (p.d.) V between the sliding contact P and end Q of the wire is measured using a voltmeter. The sliding contact P is moved from end Q to end R of the wire. Sliding contact P is distance d from Q.

Which graph shows the variation with distance d of the p.d. V ?



230. 9702/11/M/J/21/Q35

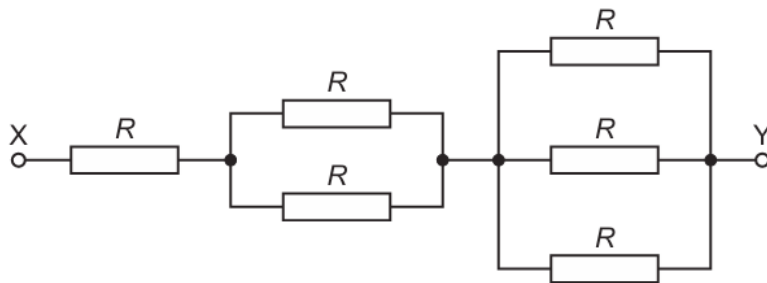
A cell is described as having an electromotive force (e.m.f.) of 6 V.

What does this mean?

- A** 1 coulomb of charge always dissipates 6 J of energy in the internal resistance of the cell.
- B** 1 electron gains 6 J of energy when passing through the cell.
- C** There is a potential difference of 6 V applied across any external circuit connected to the cell.
- D** When 1 coulomb of charge passes through the cell, 6 J of chemical energy is transformed.

231. 9702/11/M/J/21/Q36

The diagram shows a network of resistors. Each resistor has resistance R .

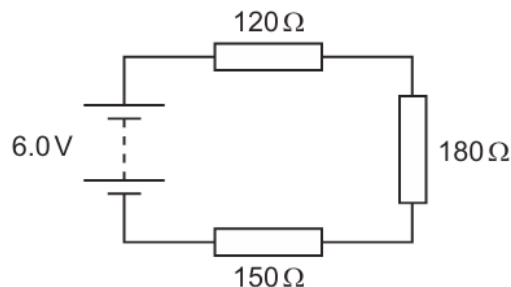


What is the total resistance of the network between points X and Y?

- A** $\frac{R}{6}$
- B** $\frac{6R}{11}$
- C** $\frac{11R}{6}$
- D** $6R$

232. 9702/11/M/J/21/Q37

Three resistors are connected in series with a battery, as shown. The battery has negligible internal resistance.



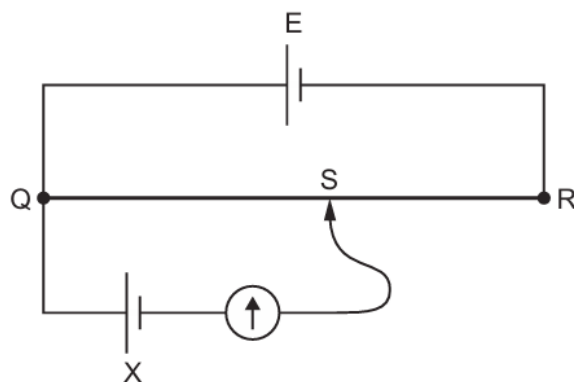
What is the potential difference across the $180\ \Omega$ resistor?

- A** 1.6 V
- B** 2.4 V
- C** 3.6 V
- D** 4.0 V

233. 9702/11/M/J/21/Q38

A potentiometer circuit is used to determine the unknown electromotive force (e.m.f.) of a cell X.

In the circuit shown, E is a cell with an e.m.f. that is known accurately. QR is the potentiometer wire, which has a movable contact S. Contact S is connected to a galvanometer and to cell X.

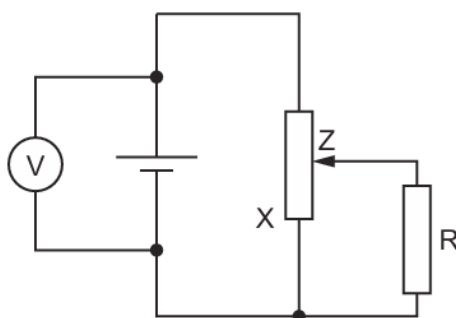


What is **not** a necessary requirement to determine the e.m.f. of X from the circuit?

- A The e.m.f. of cell X must be lower than the e.m.f. of cell E.
- B The internal resistance of cell X must be known.
- C The lengths QS and QR must be determined accurately.
- D The resistance of the wire QR must be proportional to its length.

234. 9702/12/M/J/21/Q36

A cell of constant electromotive force (e.m.f.) but with internal resistance is connected to a fixed resistor R using a potentiometer. A voltmeter measures the potential difference (p.d.) between the terminals of the cell.



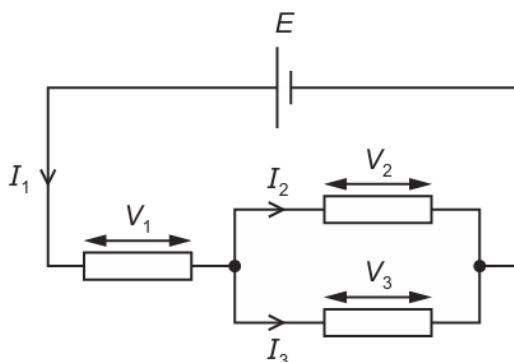
Which statement explains the change to the reading of the voltmeter as contact Z is moved towards end X of the potentiometer?

- A The voltmeter reading decreases because the current through the cell decreases.
- B The voltmeter reading decreases because the current through the cell increases.
- C The voltmeter reading increases because the current through the cell decreases.
- D The voltmeter reading increases because the current through the cell increases.

235. 9702/12/M/J/21/Q37

A cell of electromotive force (e.m.f.) E and negligible internal resistance is connected to a circuit.

The circuit has currents I_1 , I_2 and I_3 , and potential differences V_1 , V_2 and V_3 , as shown.

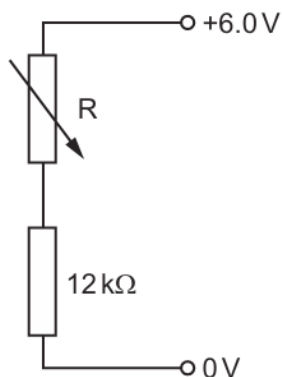


Which equation represents a statement of Kirchhoff's first law?

- A** $I_1 = I_2 + I_3$ **B** $I_1 = I_2 = I_3$ **C** $E = V_1 + V_2$ **D** $V_1 = V_2 = V_3$

236. 9702/12/M/J/21/Q38

Two resistors are connected in series with a 6.0 V power supply, as shown.

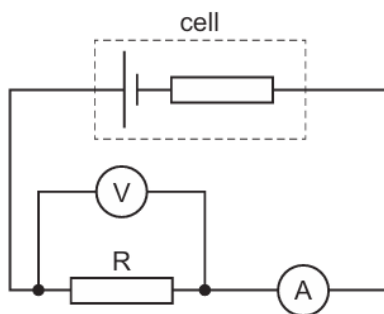


What is the resistance of the variable resistor R to give a potential difference of 1.0 V across the 12 kΩ resistor?

- A** 2.0 kΩ **B** 10 kΩ **C** 60 kΩ **D** 72 kΩ

237. 9702/13/M/J/21/Q36

The circuit shown includes a cell of constant internal resistance and an external resistor R.



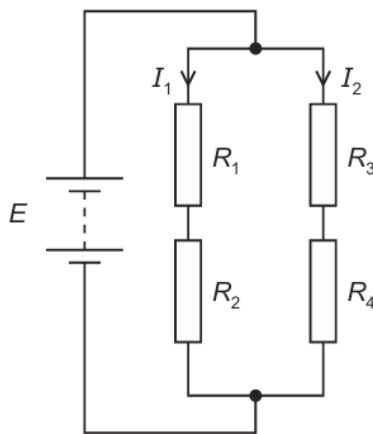
A student records the ammeter and voltmeter readings. She then connects a second identical external resistor in parallel with the first external resistor.

What happens to the ammeter reading and to the voltmeter reading?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	stays the same
C	increases	decreases
D	increases	stays the same

238. 9702/13/M/J/21/Q37

A battery of electromotive force (e.m.f.) E and negligible internal resistance is connected to four resistors of resistances R_1 , R_2 , R_3 and R_4 .



The currents I_1 and I_2 in the resistors are as shown.

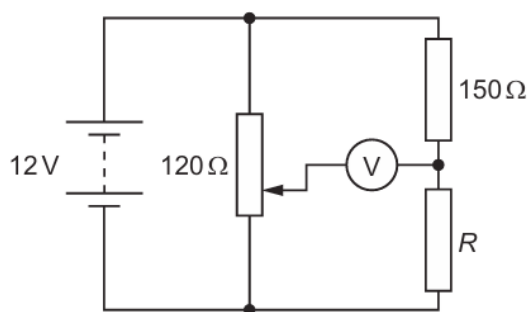
Which equation is correct?

- A** $0 = I_1(R_1 + R_2) + I_2(R_3 + R_4)$
- B** $0 = I_1(R_1 + R_2) - I_2(R_3 + R_4)$
- C** $E = I_1(R_1 + R_2) + I_2(R_3 + R_4)$
- D** $E = I_1(R_1 + R_2) - I_2(R_3 + R_4)$

239. 9702/13/M/J/21/Q38

In the circuit shown, a potentiometer of total resistance $120\ \Omega$ is connected in parallel with a resistor of resistance $150\ \Omega$ and a resistor of resistance R .

The battery has electromotive force (e.m.f.) 12 V and negligible internal resistance.



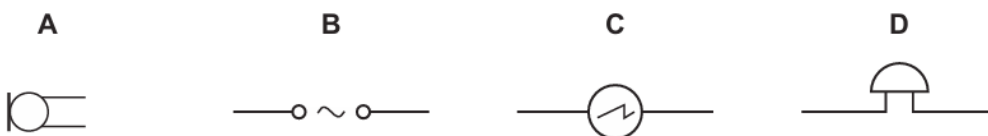
The voltmeter reads 0 V when the slider of the potentiometer is $\frac{1}{4}$ of the way from its lower end, as shown.

What is resistance R ?

- A** $30\ \Omega$ **B** $38\ \Omega$ **C** $50\ \Omega$ **D** $450\ \Omega$

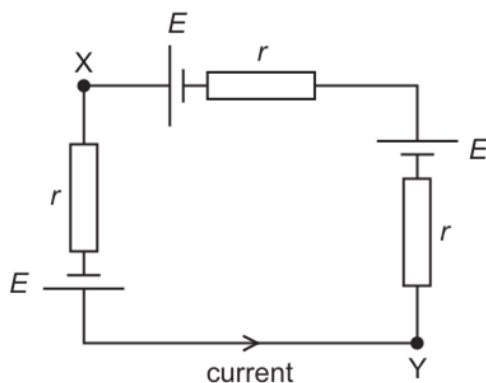
240. 9702/11/O/N/21/Q36

What is the circuit symbol for an oscilloscope?



241. 9702/11/O/N/21/Q37

Three identical cells, each of electromotive force (e.m.f.) E and internal resistance r , are connected as shown.

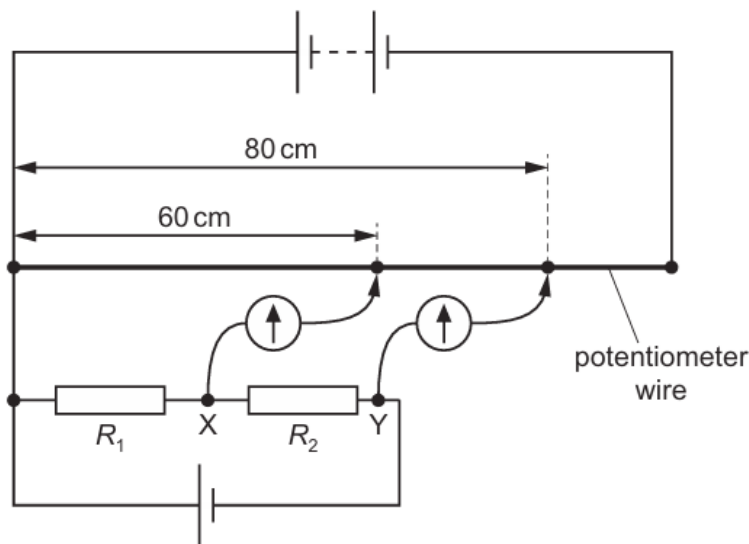


What is the potential difference between points X and Y?

- A** 0 **B** E **C** $2E$ **D** $3E$

242. 9702/11/O/N/21/Q38

Potential differences across two resistors of resistances R_1 and R_2 are compared using a potentiometer wire (uniform resistance wire) in the electrical circuit shown.



One terminal of a galvanometer is connected to point X. The galvanometer reads zero when its other terminal is connected to a point that is a distance of 60 cm from one end of the potentiometer wire.

One terminal of a second galvanometer is connected to point Y. This galvanometer reads zero when its other terminal is connected to a point that is a distance of 80 cm from the same end of the potentiometer wire.

What is the ratio $\frac{R_2}{R_1}$?

- A** $\frac{1}{3}$ **B** $\frac{3}{4}$ **C** $\frac{3}{1}$ **D** $\frac{4}{3}$

243. 9702/12/O/N/21/Q36

The electromotive force (e.m.f.) of a cell is 6.0 V. It has negligible internal resistance and is connected across a resistor. The potential difference (p.d.) across the resistor is also 6.0 V.

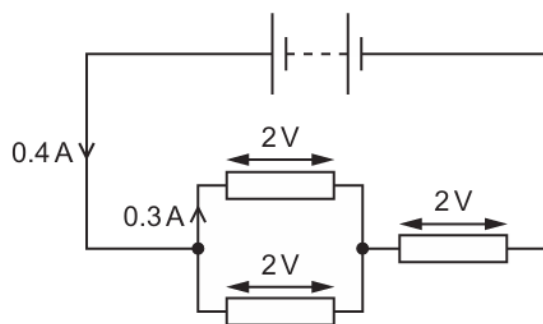
The e.m.f. and the p.d. have the same numerical value but represent different processes.

Which statement about the different processes is correct?

- A** The e.m.f. is the energy transferred from chemical energy to electrical energy in the cell and the p.d. is the energy transferred from electrical energy to thermal energy in the resistor.
- B** The p.d. is the energy transferred from chemical energy to electrical energy in the cell and the e.m.f. is the energy transferred from electrical energy to thermal energy in the resistor.
- C** The e.m.f. is the energy transferred per unit charge from chemical energy to electrical energy in the cell and the p.d. is the energy transferred per unit charge from electrical energy to thermal energy in the resistor.
- D** The p.d. is the energy transferred per unit charge from chemical energy to electrical energy in the cell and the e.m.f. is the energy transferred per unit charge from electrical energy to thermal energy in the resistor.

244. 9702/12/O/N/21/Q37

A battery of negligible internal resistance is connected to three resistors, as shown.



The potential difference across each resistor is 2 V.

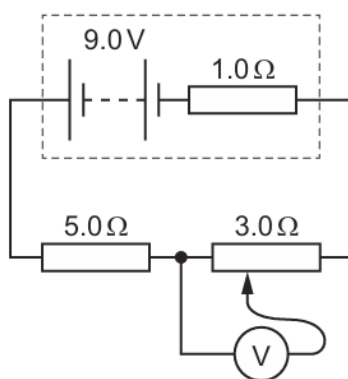
The current from the battery is 0.4 A and the current through one of the resistors connected in parallel is 0.3 A.

What is the current through the other resistor connected in parallel and what is the electromotive force (e.m.f.) of the battery?

	current / A	e.m.f. / V
A	0.1	4
B	0.3	4
C	0.1	6
D	0.3	6

245. 9702/12/O/N/21/Q38

A battery of electromotive force (e.m.f.) 9.0 V and internal resistance $1.0\ \Omega$ is connected to a fixed resistor of resistance $5.0\ \Omega$ and a potentiometer of maximum resistance $3.0\ \Omega$, as shown.



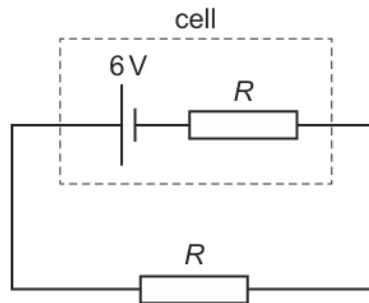
The sliding contact of the potentiometer is moved over its full range of movement.

What is the maximum value of the potential difference that is measured by the voltmeter?

- A** 3.0 V **B** 3.4 V **C** 4.5 V **D** 5.4 V

246. 9702/13/O/N/21/Q34

A cell has an electromotive force (e.m.f.) of 6 V and internal resistance R . An external resistor, also of resistance R , is connected across this cell, as shown.



Power P is dissipated by the external resistor.

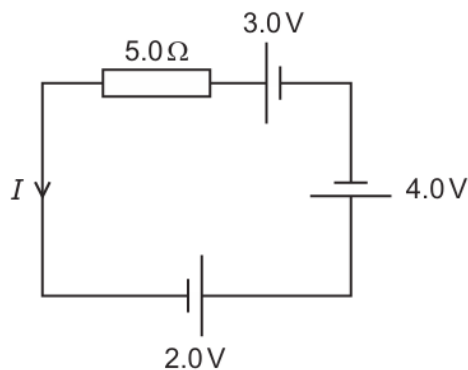
The cell is replaced by a different cell that has an e.m.f. of 6 V and negligible internal resistance.

What is the new power that is dissipated in the external resistor?

- A** $0.5P$ **B** P **C** $2P$ **D** $4P$

247. 9702/13/O/N/21/Q37

The circuit shown contains three cells of electromotive forces 3.0 V , 2.0 V and 4.0 V , in series with a resistor of resistance $5.0\ \Omega$. The cells have negligible internal resistance.



What is the current I in the circuit?

- A** 0.20 A **B** 0.60 A **C** 1.0 A **D** 1.8 A

248. 9702/12/F/M/22/Q33

Ten cells, each of electromotive force (e.m.f.) 1.5 V , are connected together, as shown.

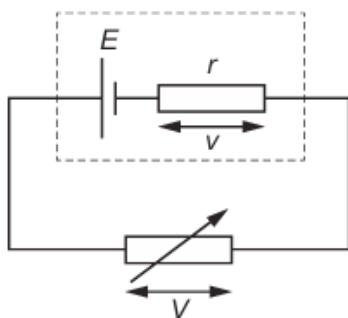


What is the combined e.m.f. between terminals X and Y?

- A** 8 V **B** 9 V **C** 12 V **D** 15 V

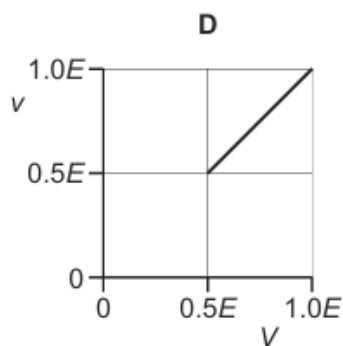
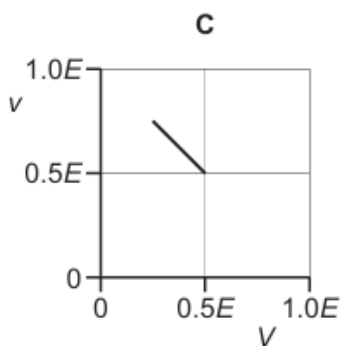
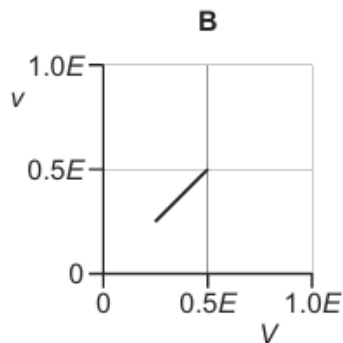
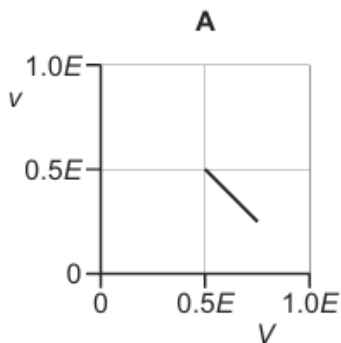
249. 9702/12/F/M/22/Q34

A cell of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor, as shown.



The resistance of the variable resistor is gradually increased from r to $3r$.

Which graph shows the variation of the potential difference (p.d.) v across the internal resistance with the p.d. V across the variable resistor?



250. 9702/12/F/M/22/Q35

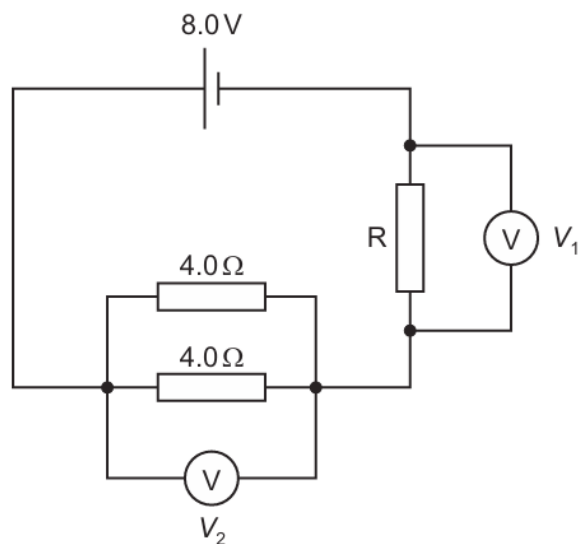
Each of Kirchhoff's two laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	the algebraic sum of currents into a junction is zero	charge
B	the algebraic sum of currents into a junction is zero	energy
C	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	charge
D	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	energy

251. 9702/12/F/M/22/Q36

A cell has an electromotive force (e.m.f.) of 8.0 V and negligible internal resistance. The cell forms part of a circuit, as shown.



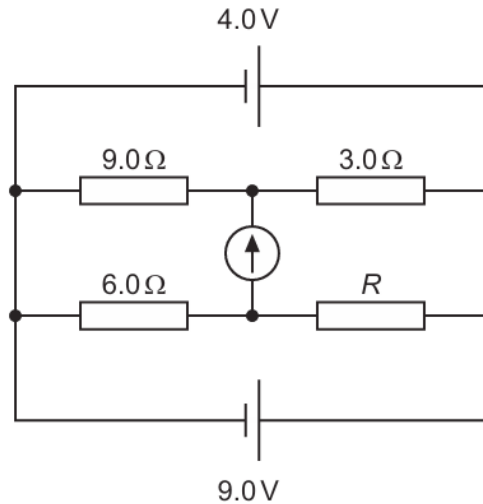
The reading V_1 is 4.0 V and the reading V_2 is also 4.0 V.

What is the resistance of resistor R ?

- A** $0.50\ \Omega$ **B** $2.0\ \Omega$ **C** $4.0\ \Omega$ **D** $8.0\ \Omega$

252. 9702/12/F/M/22/Q37

In the circuit shown, the cells have negligible internal resistance and the reading on the galvanometer is zero.

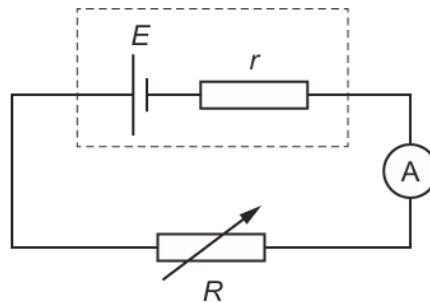


What is the value of resistor R ?

- A** 2.0Ω **B** 6.0Ω **C** 12Ω **D** 18Ω

253. 9702/11/M/J/22/Q35

A cell has internal resistance r and electromotive force (e.m.f.) E . The cell is connected in series with an ammeter and a variable resistor of resistance R .



When R is 10Ω the ammeter reads 0.3 A .

When R is 5Ω the ammeter reads 0.4 A .

What is the value of E ?

- A** 0.5 V **B** 2 V **C** 3 V **D** 6 V

254. 9702/11/M/J/22/Q36

The sum of the currents entering a junction in an electrical circuit is always equal to the sum of the currents leaving the junction.

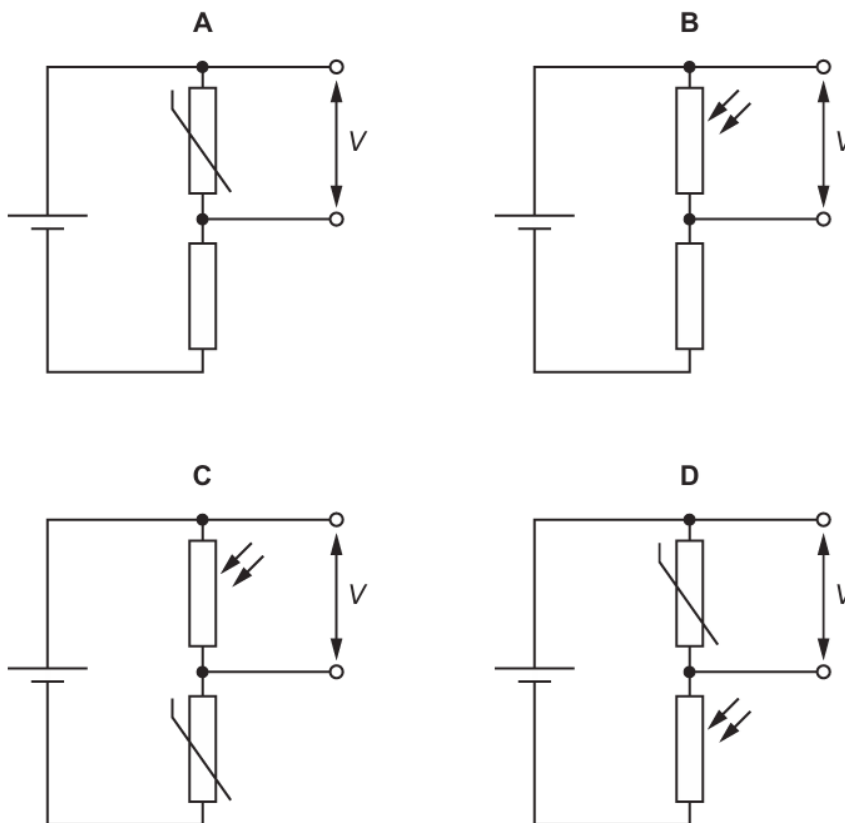
Why is this?

- A** It is a consequence of the conservation of charge.
B It is a consequence of the conservation of electromotive force.
C It is a consequence of the conservation of energy.
D It is a consequence of the conservation of potential difference.

255. 9702/11/M/J/22/Q37

In the circuits shown, the temperature remains constant.

In which circuit does the potential difference (p.d.) V increase with increasing light intensity?



256. 9702/12/M/J/22/Q35

A cell with constant electromotive force (e.m.f.) is connected across a fixed resistor. Over time, the internal resistance of the cell increases.

Which change occurs as the internal resistance of the cell increases?

- A** a decrease in the charge of each charge carrier
- B** a decrease in the potential difference measured across the cell
- C** an increase in the energy dissipated per unit time in the fixed resistor
- D** an increase in the number of charge carriers leaving the cell per unit time

257. 9702/12/M/J/22/Q36

Kirchhoff's first and second laws are consequences of the conservation of different quantities.

What are those quantities?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	energy
B	energy	current
C	current	charge
D	energy	charge

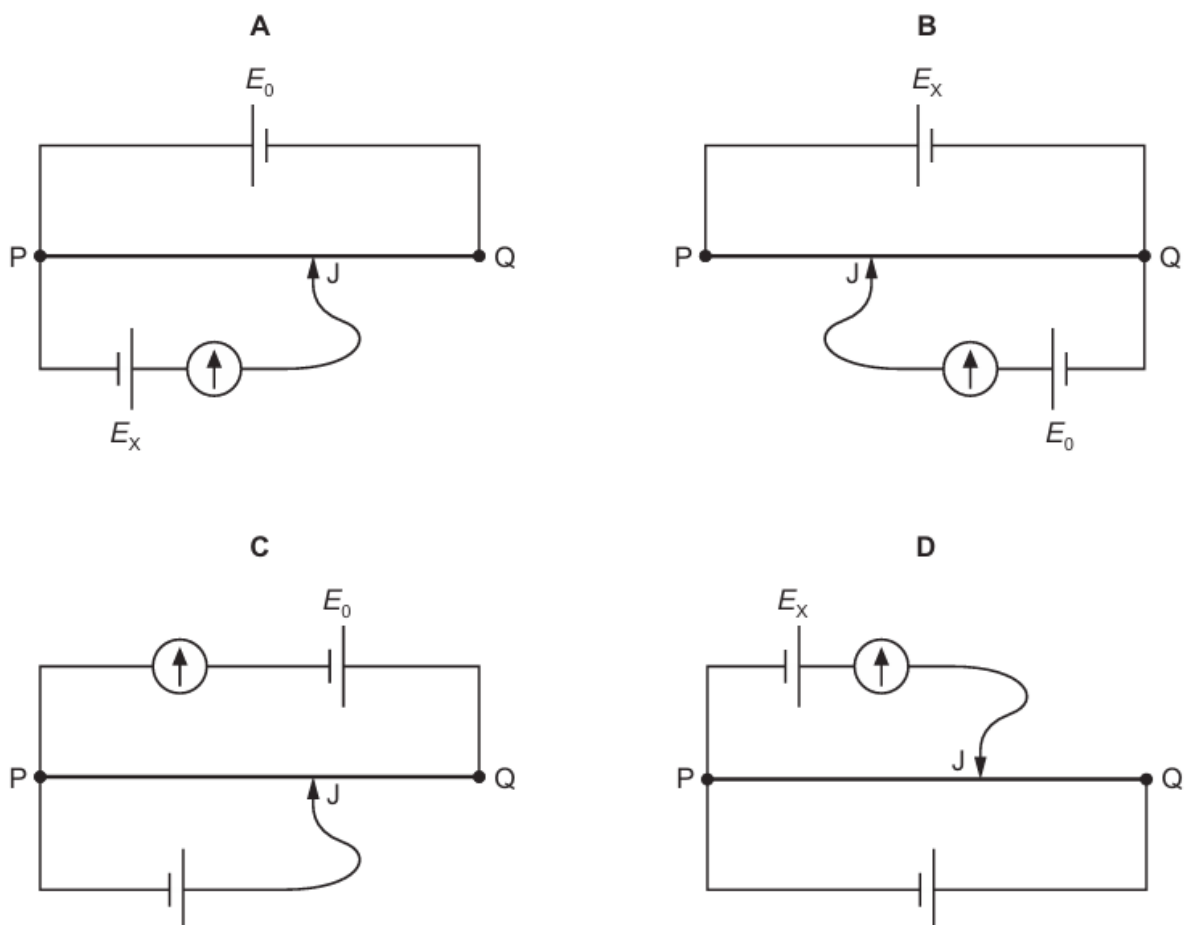
258. 9702/12/M/J/22/Q37

A potentiometer circuit is used to determine the electromotive force (e.m.f.) E_x of a cell. The circuit includes a second cell of known e.m.f. E_0 and negligible internal resistance, and a uniform resistance wire PQ of known length.

E_x is less than E_0 .

The movable connection J can be positioned anywhere along the length of the resistance wire.

Which circuit is suitable for determining E_x ?



259. 9702/13/M/J/22/Q36

The diagram shows the symbol for a component that may be used in an electrical circuit.



Which component is represented by this circuit symbol?

- A** buzzer
- B** electric bell
- C** loudspeaker
- D** microphone

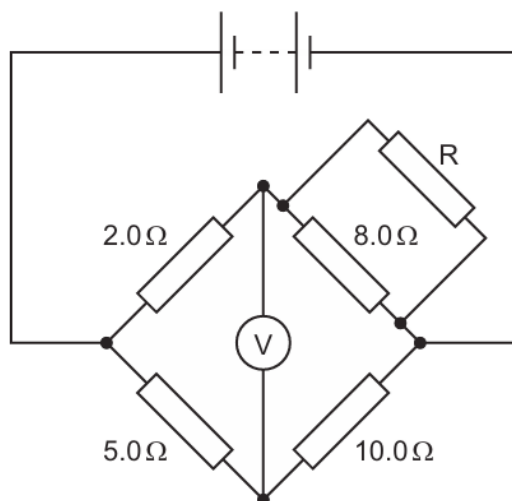
260. 9702/13/M/J/22/Q37

Which row correctly describes Kirchhoff's laws?

	Kirchhoff's first law	physics principle applied for first law	Kirchhoff's second law	physics principle applied for second law
A	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy
B	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge
C	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge
D	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy

261. 9702/11/O/N/22/Q36

A circuit consists of a battery, a voltmeter and five fixed resistors, as shown.



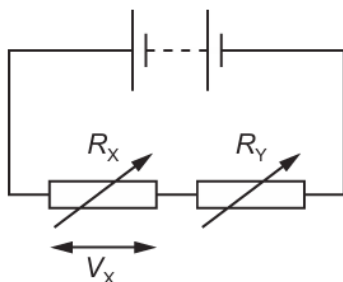
The voltmeter reading is zero.

What is the resistance of resistor R?

- A** $1.1\ \Omega$ **B** $2.1\ \Omega$ **C** $4.0\ \Omega$ **D** $8.0\ \Omega$

262. 9702/13/M/J/22/Q38

A potential divider circuit is formed by connecting a battery of negligible internal resistance in series with two variable resistors, as shown.



The variable resistors have resistances R_X and R_Y .

V_X is the potential difference (p.d.) across the variable resistor with resistance R_X .

R_X and R_Y are both changed at the same time.

Which combination of changes **must** cause V_X to increase?

	R_X	R_Y
A	larger	larger
B	larger	smaller
C	smaller	larger
D	smaller	smaller

263. 9702/11/O/N/22/Q35

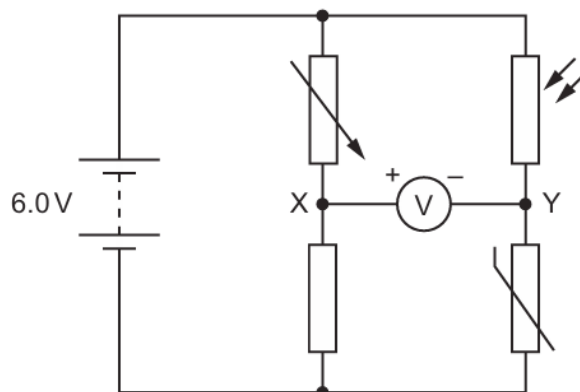
Which ratio has the same units as electromotive force (e.m.f.)?

- A** charge per unit energy transferred
- B** charge per unit time
- C** energy transferred per unit charge
- D** energy transferred per unit time

264. 9702/11/O/N/22/Q37

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected to a voltmeter and four other components, as shown.

The voltmeter is connected between points X and Y. The positive terminal of the voltmeter is connected to X and the negative terminal of the voltmeter is connected to Y.



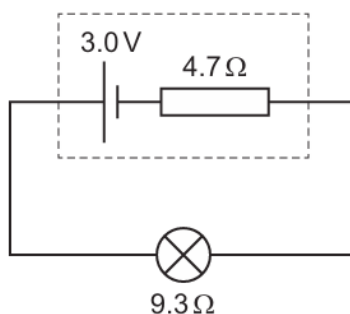
Initially, the resistance of each of the four components is $1.0\text{ k}\Omega$.

Which change, on its own, will cause the voltmeter to show a positive reading?

- A** Decrease the temperature of the thermistor.
- B** Increase the resistance of the variable resistor.
- C** Reduce the intensity of light incident on the light-dependent resistor (LDR).
- D** Replace the fixed resistor with a $500\text{ }\Omega$ resistor.

265. 9702/12/O/N/22/Q35

The diagram shows a cell of electromotive force (e.m.f.) 3.0 V and internal resistance $4.7\text{ }\Omega$ connected across a lamp. The lamp has a resistance of $9.3\text{ }\Omega$.

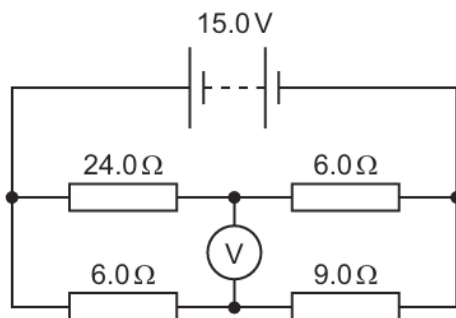


What is the power dissipated by the internal resistance of the cell?

- A** 0.22 W
- B** 0.43 W
- C** 0.64 W
- D** 1.0 W

266. 9702/12/O/N/22/Q36

A circuit consists of a battery, a high-resistance voltmeter and four fixed resistors, as shown. The battery has an electromotive force (e.m.f.) of 15.0 V and negligible internal resistance.

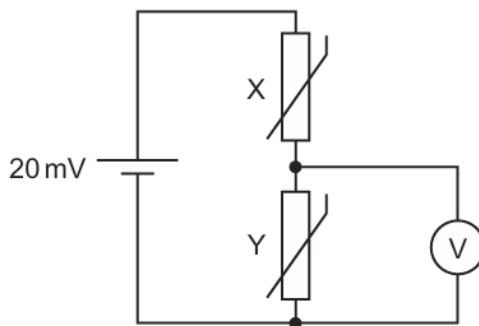


What is the reading on the voltmeter?

- A** 3.0 V **B** 6.0 V **C** 9.0 V **D** 12.0 V

267. 9702/12/O/N/22/Q37

A potential divider circuit is designed to detect the difference in temperature between two different places.



The cell has electromotive force (e.m.f.) 20 mV and negligible internal resistance.

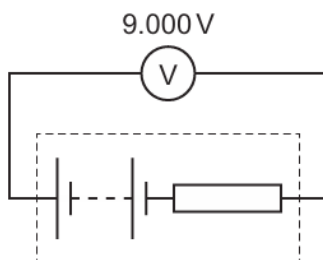
Initially, thermistors X and Y are at the same temperature and have the same resistance. The voltmeter reads 10 mV . X is then placed in a cold environment and its resistance doubles. Y is placed in a warm environment and its resistance halves.

What is the new reading on the voltmeter?

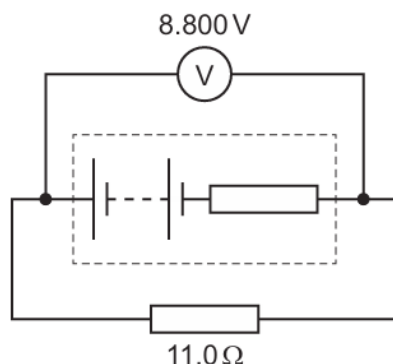
- A** 4 mV **B** 5 mV **C** 15 mV **D** 16 mV

268. 9702/13/O/N/22/Q34

A voltmeter reads 9.000 V when it is connected across the terminals of a battery.



When a resistor of resistance 11.0Ω is connected in parallel with the battery, the voltmeter reading changes to 8.800 V .



What is the internal resistance of the battery?

- A** 0.244Ω **B** 0.250Ω **C** 10.8Ω **D** 11.3Ω

269. 9702/13/O/N/22/Q35

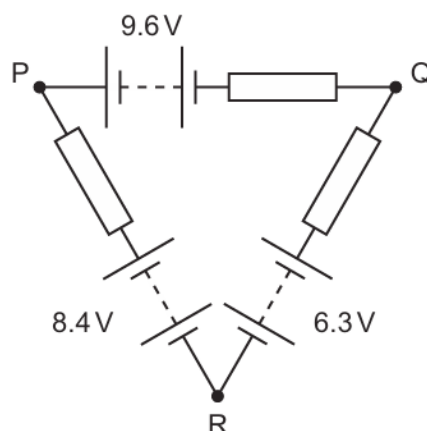
Each of Kirchhoff's laws is linked to the conservation of a physical quantity.

Which conserved physical quantities are used in the derivation of Kirchhoff's first law and of Kirchhoff's second law?

	Kirchhoff's first law	Kirchhoff's second law
A	energy	charge
B	energy	momentum
C	charge	energy
D	momentum	energy

270. 9702/13/O/N/22/Q36

Three batteries and three identical resistors are connected in a circuit PQR, as shown.



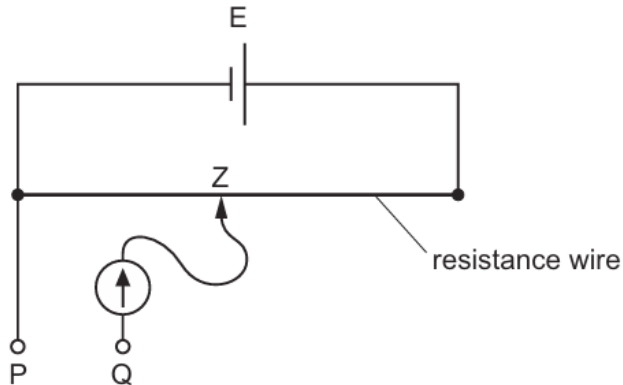
The batteries have negligible internal resistance.

What is the potential difference between points P and Q?

- A** 1.5 V **B** 2.1 V **C** 7.1 V **D** 12.1 V

271. 9702/13/O/N/22/Q37

A cell E, of electromotive force (e.m.f.) 2 V and negligible internal resistance, is connected to a uniform resistance wire of resistance $10\ \Omega$ and length 1.0 m.



Z is a connection that may be made at any position along the resistance wire. A galvanometer is connected between Z and a point Q.

A new source of e.m.f. of approximately 8 mV is connected between points P and Q. The e.m.f. of the new source is determined by changing the position of Z until the reading on the galvanometer is zero.

Which change to the circuit allows a much more precise value for the e.m.f. of the new source to be obtained?

- A** Add a resistor of resistance $0.1\ \Omega$ in series with cell E.
B Add a resistor of resistance $1000\ \Omega$ in series with cell E.
C Add a resistor of resistance $10\ \Omega$ in series with the new source.
D Add a resistor of resistance $800\ \Omega$ in series with the new source.

272. 9702/12/F/M/23/Q34

A cell has a constant electromotive force.

A variable resistor is connected between the terminals of the cell.

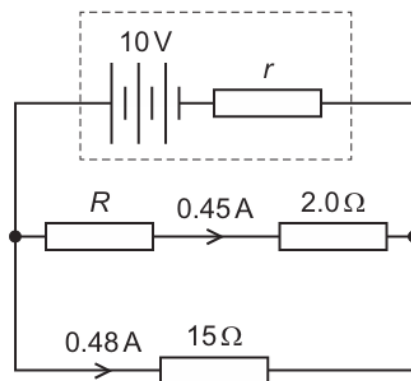
The resistance of the variable resistor is decreased.

Which statement about the change of the cell's terminal potential difference (p.d.) is correct?

- A** The terminal p.d. is decreased because more work is done moving unit charge through the internal resistance of the cell.
- B** The terminal p.d. is decreased because the current in the variable resistor is decreased.
- C** The terminal p.d. is increased because more work is done moving unit charge through the variable resistor.
- D** The terminal p.d. is increased because the current in the variable resistor is increased.

273. 9702/12/F/M/23/Q36

A battery of electromotive force (e.m.f.) 10V and internal resistance r is connected to three resistors of resistances R , 2.0Ω and 15Ω , as shown. A current of 0.45A is in the resistor of resistance 2.0Ω and a current of 0.48A is in the resistor of resistance 15Ω .

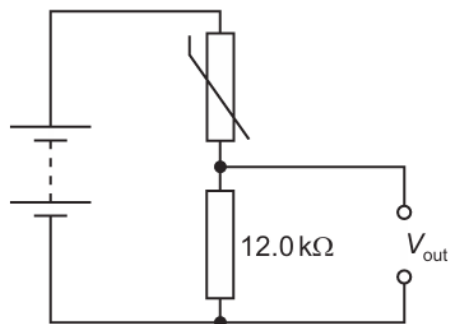


What are the values of r and R ?

	r/Ω	R/Ω
A	3.0	14
B	3.0	20
C	5.8	14
D	5.8	20

274. 9702/12/F/M/23/Q37

A battery of negligible internal resistance is connected in series with a thermistor and a fixed resistor of resistance $12.0\text{ k}\Omega$, as shown.



The table shows the resistance of the thermistor at two different temperatures.

temperature / $^{\circ}\text{C}$	resistance of thermistor / $\text{k}\Omega$
20.0	12.0
50.0	5.00

The potential difference V_{out} across the fixed resistor is 4.50 V when the thermistor is at a temperature of 20.0°C .

What is V_{out} when the thermistor is at a temperature of 50.0°C ?

- A** 2.65 V **B** 3.18 V **C** 6.35 V **D** 10.8 V

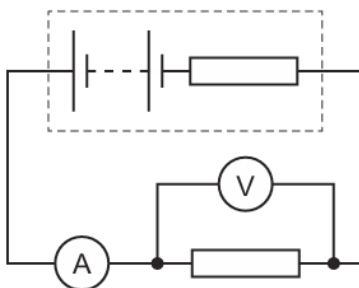
275. 9702/11/M/J/23/Q34

Which circuit symbol represents a microphone?



276. 9702/11/M/J/23/Q35

A battery with internal resistance is connected to a fixed resistor, an ammeter and a voltmeter, as shown.



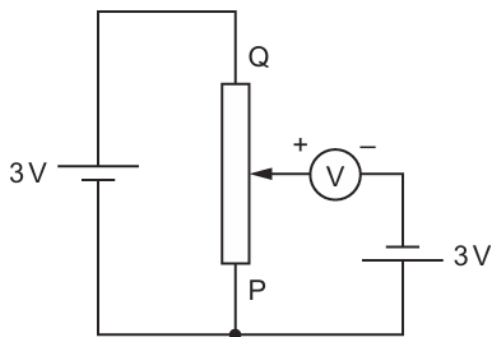
The battery is replaced by a different battery that has the same electromotive force (e.m.f.) but a greater internal resistance.

What happens to the readings on the ammeter and voltmeter?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

277. 9702/11/M/J/23/Q37

A voltmeter is connected into a circuit with the polarity shown.



The sliding contact is moved to end P of the potentiometer and then to end Q.

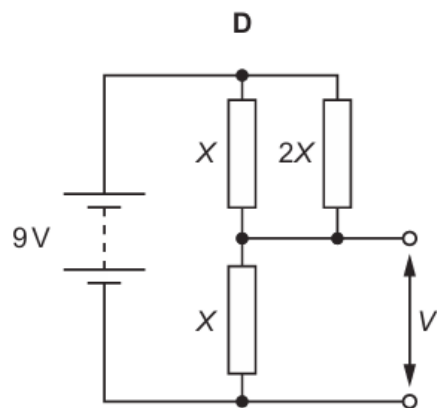
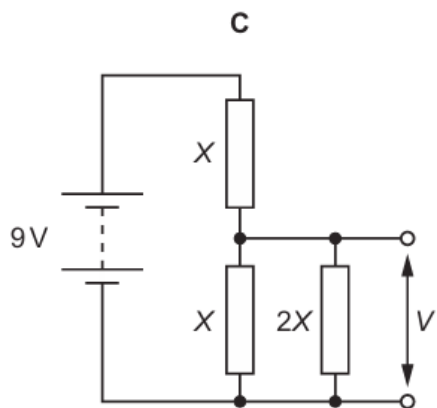
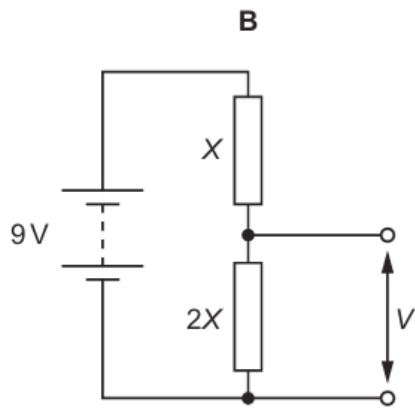
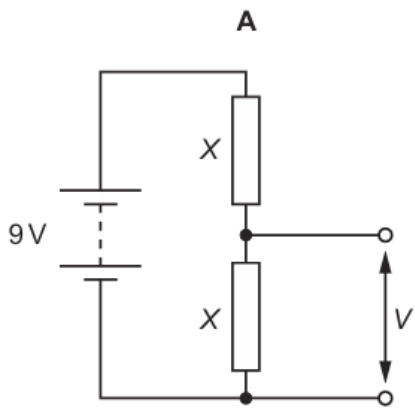
What are the two readings of the voltmeter?

	sliding contact at end P	sliding contact at end Q
A	0V	3V
B	0V	6V
C	3V	3V
D	3V	6V

278. 9702/11/M/J/23/Q36

Four potential divider circuits each consist of a battery of electromotive force (e.m.f.) 9 V and negligible internal resistance connected to a combination of resistors. Each of the resistors in the circuits has a resistance of X or $2X$.

Which circuit has the largest output voltage V ?



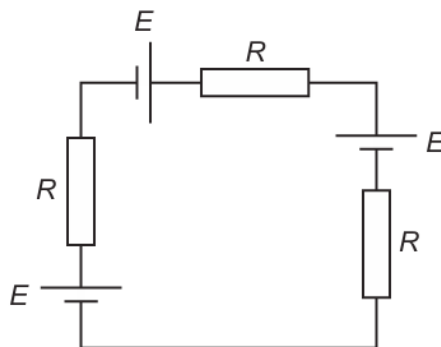
279. 9702/12/M/J/23/Q35

Kirchhoff's first law is a consequence of the conservation of which quantity?

- A** charge
- B** energy
- C** momentum
- D** potential difference

280. 9702/12/M/J/23/Q36

Three identical cells each have electromotive force (e.m.f.) E and negligible internal resistance. The cells are connected to three identical resistors, each of resistance R , as shown.

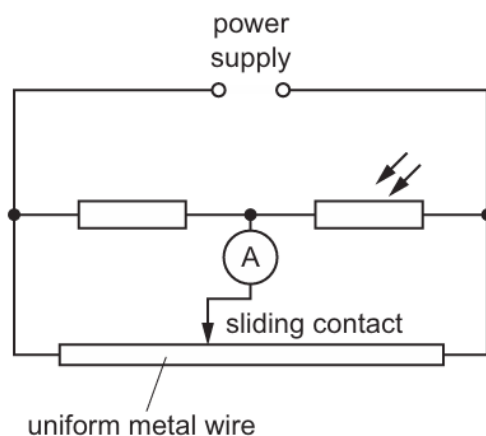


What is the potential difference across one of the resistors?

- A** 0 **B** $\frac{E}{3}$ **C** $\frac{2E}{3}$ **D** E

281. 9702/12/M/J/23/Q37

In the potentiometer circuit shown, the reading on the ammeter is zero.



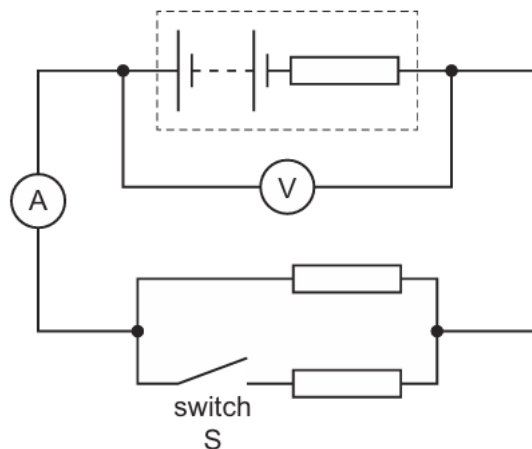
The light-dependent resistor (LDR) is then covered and the ammeter gives a non-zero reading.

Which change could return the ammeter reading to zero?

- A** decreasing the supply voltage
B increasing the supply voltage
C moving the sliding contact to the left
D moving the sliding contact to the right

282. 9702/13/M/J/23/Q34

A battery with internal resistance is connected to a parallel arrangement of two resistors and a switch S, as shown.



Initially, switch S is open.

What happens to the voltmeter and ammeter readings when switch S is closed?

	voltmeter reading	ammeter reading
A	decreases	increases
B	decreases	decreases
C	increases	increases
D	increases	decreases

283. 9702/13/M/J/23/Q36

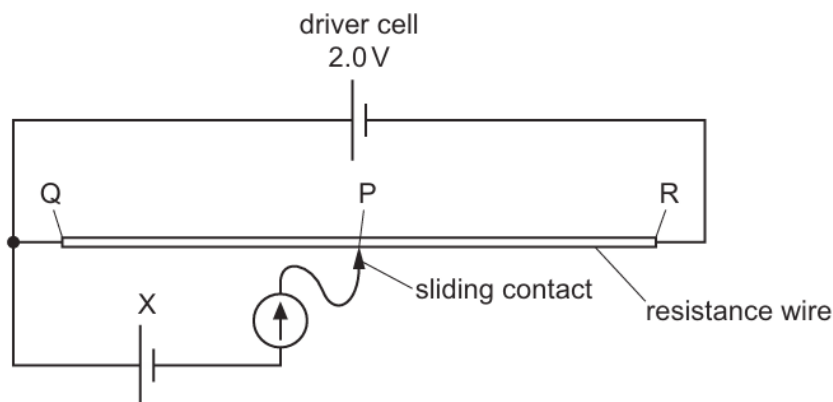
Two resistors have a combined resistance of $34\ \Omega$ when connected in series. The same resistors have a combined resistance of $7.4\ \Omega$ when connected in parallel.

What is the resistance of one of the resistors?

- A** $15\ \Omega$ **B** $17\ \Omega$ **C** $23\ \Omega$ **D** $27\ \Omega$

284. 9702/13/M/J/23/Q37

A potentiometer circuit is used to investigate the electromotive force (e.m.f.) of a cell X.



The e.m.f. of cell X is known to be approximately 0.50 V.

The driver cell has negligible internal resistance and an e.m.f. of 2.0 V. The sliding contact is moved along the uniform resistance wire between ends Q and R to a point P where the reading on the galvanometer is zero.

What is an expression for the approximate length QP?

- A $\frac{QR}{4}$ B $\frac{QR}{3}$ C $\frac{2QR}{3}$ D $\frac{3QR}{4}$

285. 9702/11/O/N/23/Q34

Which statement about the electromotive force (e.m.f.) of a cell is **always** correct?

- A The e.m.f. is the energy converted from electrical to other forms in the cell.
- B The e.m.f. is the energy provided by the cell per unit charge passing through it.
- C The e.m.f. is the potential difference across the internal resistance of the cell.
- D The e.m.f. is the potential difference across the terminals of the cell.

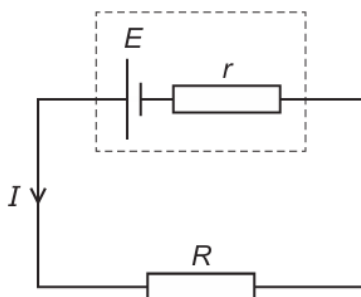
286. 9702/11/O/N/23/Q35

Kirchhoff's first and second laws are a consequence of the conservation of which quantities?

- A charge and energy
- B charge and resistance
- C mass and energy
- D mass and resistance

287. 9702/11/O/N/23/Q36

A circuit contains a cell of electromotive force E and internal resistance r connected to a resistor of resistance R . The current in the circuit is I .

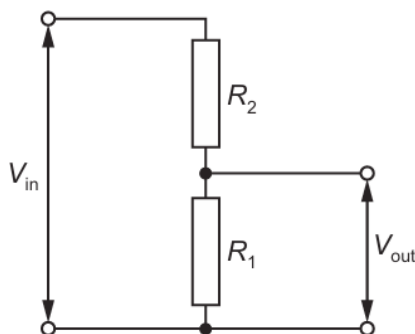


Which equation is correct?

- A** $E - Ir = IR$ **B** $E = Ir - IR$ **C** $E + Ir = IR$ **D** $E = IR$

288. 9702/11/O/N/23/Q37

A potential divider consists of two resistors of resistances R_1 and R_2 connected in series across a source of potential difference (p.d.) V_{in} . The p.d. across R_1 is V_{out} .

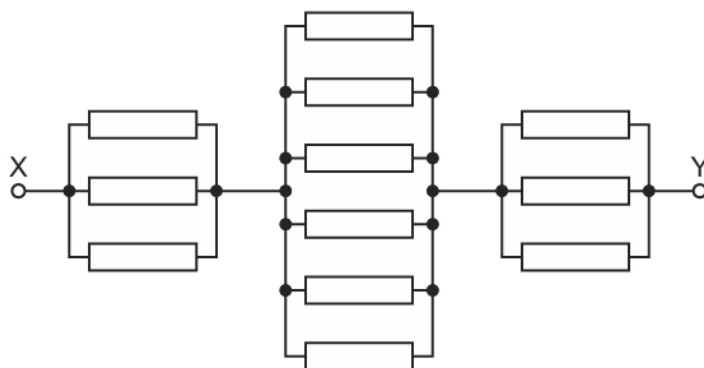


Which changes to R_1 and R_2 will increase the value of V_{out} ?

	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

289. 9702/13/O/N/23/Q36

The diagram shows a network of resistors. Each resistor has a resistance of $6.0\ \Omega$.

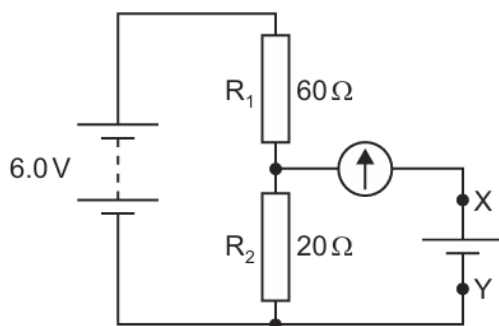


What is the total resistance of the network between points X and Y?

- A** $3.0\ \Omega$ **B** $5.0\ \Omega$ **C** $7.2\ \Omega$ **D** $18\ \Omega$

290. 9702/13/O/N/23/Q37

In the circuit shown, a battery of negligible internal resistance is connected in series with a pair of fixed resistors R_1 and R_2 .



The circuit is to be used to test whether the electromotive force (e.m.f.) of a particular cell is 1.5 V . The cell is connected between terminals X and Y in parallel with R_2 and in series with a galvanometer.

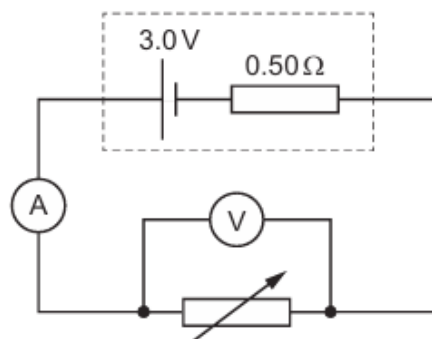
Which statement about the test is correct?

- A** Any non-zero reading on the galvanometer means the cell has an e.m.f. of 1.5 V .
B The battery does not need to have an e.m.f. of 6.0 V .
C The cell may be connected either way round between X and Y.
D The galvanometer does not need a scale calibrated in amperes.

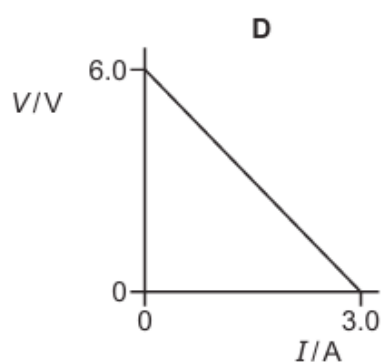
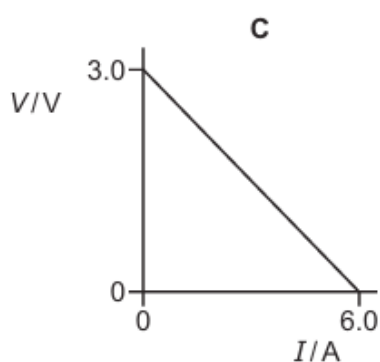
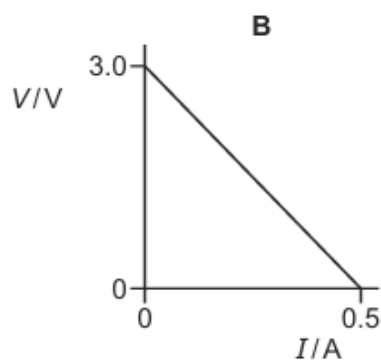
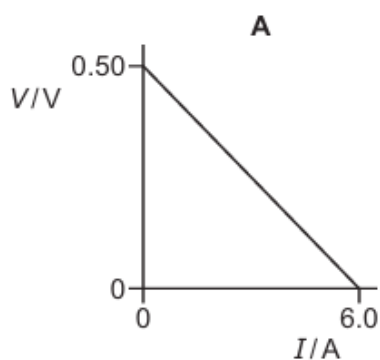
291. 9702/13/O/N/23/Q34

A cell of electromotive force (e.m.f.) 3.0 V and internal resistance 0.50Ω is connected to a variable resistor, a voltmeter and an ammeter, as shown. The resistance of the variable resistor is varied.

The reading on the ammeter I and the reading on the voltmeter V are recorded.

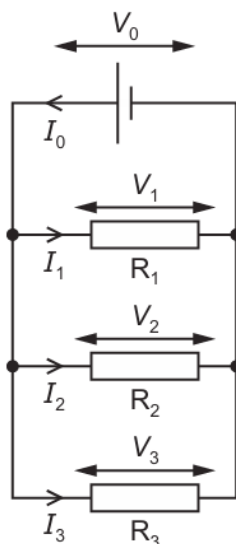


Which graph shows how V varies with I ?



292. 9702/13/O/N/23/Q35

Three resistors, R_1 , R_2 and R_3 , are connected in parallel to a cell. The currents in the resistors are I_1 , I_2 and I_3 . The potential differences across the resistors are V_1 , V_2 and V_3 . The current in the cell is I_0 . The potential difference across the cell is V_0 , as shown.

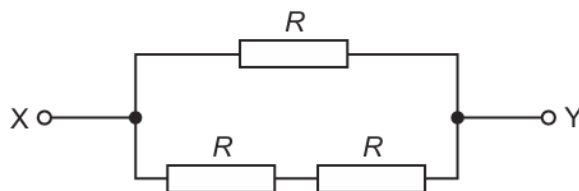


Which equation can be obtained by applying Kirchhoff's second law to the circuit?

- A** $I_0 = I_1 = I_2 = I_3$
- B** $I_0 = I_1 + I_2 + I_3$
- C** $V_0 = V_1 = V_2 = V_3$
- D** $V_0 = V_1 + V_2 + V_3$

293. 9702/13/O/N/23/Q36

Three resistors, each of resistance R , are connected in a network, as shown.



The total resistance between points X and Y is 8.0Ω .

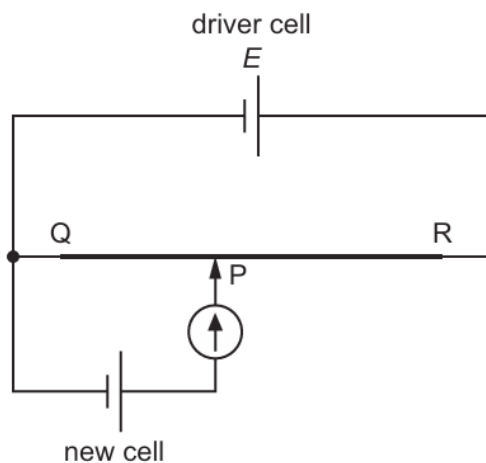
What is the value of R ?

- A** 2.7Ω
- B** 4.0Ω
- C** 5.3Ω
- D** 12Ω

294. 9702/13/O/N/23/Q37

A potentiometer and a driver cell of electromotive force (e.m.f.) E are used to measure the e.m.f. of a new cell.

A sliding contact at P is moved along a resistance wire QR until the reading on the galvanometer is zero.



What is an essential requirement for the e.m.f. of the new cell to be measured accurately?

- A** The e.m.f. of the driver cell must be less than the e.m.f. of the new cell.
- B** The galvanometer must have a large resistance.
- C** The internal resistance of the new cell must be zero.
- D** The resistance per unit length of the wire QR must be constant.